

Influence of Flow Pathway Geometry on Barite Scale Deposition in Marcellus Shale during Hydraulic Fracturing

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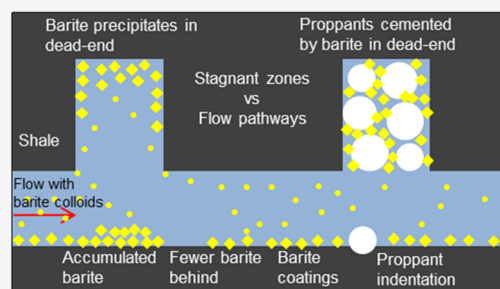
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ABSTRACT: Barite scaling is a common problem in the shale gas industry. Barite precipitation due to rock-fluid interactions has been studied intensively in static and flow-through experiments, but the impact of flow pathway geometry on barite scaling remains a question. The complex fracture passages can lead to local concentrated geochemical interactions, resulting in spatially variable scaling distribution. In this study, designed patterns with channels and holes were milled in two Marcellus shale cores to represent the main flow pathways, where the slow flow travels from the inlet to the outlet, and near-stagnant zones, where the fluid is trapped. Hydraulic fracturing fluid, created using synthetic produced water and Monongahela River water as base fluid, was injected at 0.02 mL/min into the cores at 66 °C, a core pressure of 12.4 MPa (1800 psi), and a confining pressure of 13.8 MPa (2000 psi) for 28 days. For the core with both channels and holes, barite coatings formed in the channels and the holes, with thicker barite accumulations occurring on the non-patterned half of the core adjacent to the holes on the milled half. For the core with only holes, proppants were added to prop up the main flow pathways. Barite formed on the propped fracture surface. The proppants in the holes were covered with and glued together by barite. Such barite-proppant conglomerates were not found in the main propped fracture. These barite-proppant conglomerates may block transport pathways as well as the barite coatings.



1. INTRODUCTION

Hydraulic fracturing is a technology that involves injecting high pressure water-based fluid into low-permeability formations to induce fractures, which enhances reservoir permeability and extractability of oil and gas.^{1–3} Natural gas production from shale formations has boomed over the past two decades due to the development of hydraulic fracturing technology in the United States.⁴ The main issues during shale gas production include low recovery rates, 20–30% compared to the estimated resource,^{5,6} and the fast decrease of gas production within several months to years after the initial extraction.^{6,7} The sharp decline in gas production is likely the result of many factors, including a reduced pressure gradient as the reservoir is depleted over time.⁶ Additionally, mineral scaling as a result of rock-fluid geochemical interactions may block pores and fractures in shale, as well as pipelines,⁸ thereby impacting flow. Among the shale plays in the United States, the Marcellus Shale is the most geographically extensive, stretching across Pennsylvania, West Virginia, and into eastern Ohio and western New York.⁹ Mineral scale precipitation, especially barite scaling, is a common problem in Marcellus shale gas production.^{10–14}

Monongahela River water is widely used as base fluid for hydraulic fracturing in shale reservoirs in Pennsylvania and West Virginia.¹⁵ Monongahela River water contains high concentrations of SO_4^{2-} (50–300 mg/L)¹⁶ compared to typical fresh natural water (20 mg/L)¹⁷ due to fuel extraction

activities.¹⁶ Produced water reuse is common in Marcellus shale gas production sites.¹⁸ The produced water contains high total dissolved solids (TDS) including Ba^{2+} , Sr^{2+} , Na^+ , Cl^- , Mg^{2+} , Fe^{2+} , and Ca^{2+} as major components.^{19–22} The hydraulic fracturing fluid (HFF) can contain 80% or more Monongahela River water that is blended with reused Marcellus produced water.^{23,24} Mixing the Monongahela River water with produced water can result in an injection fluid supersaturated with respect to barite, leading to barite scaling once injected into the reservoir. Some operators remove pre-precipitated barite by filtration in above-ground facilities, but a study showed that barite would still precipitate due to rock-fluid interactions with the supersaturated HFF after injection.²⁴ As long as produced water is reused with Monongahela River water for hydraulic fracturing in this region, barite scaling can be a potential problem.

Depending on the location of barite scaling, transport pathways may be inhibited or even blocked for shale gas extraction. Barite precipitation in shale due to rock-fluid

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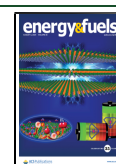


Table 1. Semi-Quantitative Mineralogical Composition (weight %) of the Marcellus Shale Core³¹

sample ID	location	quartz (%)	muscovite/illite (%)	chlorite (%)	calcite (%)	pyrite (%)	barite (%)
MIP3H-36	Monongalia CO, WV, USA	37	48	8	1	3	n/a

interactions also can affect other observed mineral reactions that impact transport pathways from the shale matrix to primary fractures. Reaction fronts containing an Fe(III) layer as a result of pyrite oxidative dissolution were observed at rock-fluid interfaces due to oxygen diffusion into the shale matrix.²⁵ Barite coatings observed on the Marcellus shale fracture surfaces^{15,26,27} may reduce oxygen transport, thereby inhibiting further pyrite oxidation.²⁸ Alternatively, shale mineral dissolution by the HFF may increase porosity and enhance hydrocarbon transport.²⁹ The presence of barite coatings could reduce direct contact between rock and fluid, inhibiting further dissolution reactions. In a diffusion experiment with barium and sulfate injected from opposite ends, barite precipitated in larger microfractures within the rock, causing localized clogging, and then further grew into nanopores and smaller fractures, which are diffusion-controlled regions.³⁰ During the hydraulic fracturing process, the main flow pathways are the places where the flow is laminar or turbulent and the stagnant zones in shale may be diffusion-dominant. How different flow pathways impact barite scaling in hydraulically fractured shale remains a question.

In this study, Marcellus shale cores with engineered patterns representing the main flow transport pathways and stagnant zones were used for flow-through experiments at representative reservoir temperature and pressure. The designed patterns enabled two pairs of comparison: (1) channeled fracture without proppants vs propped fracture and (2) main flow regions vs dead-end holes. In addition to experimental analysis, reactive transport modeling was conducted to illustrate barite scaling in the flow field of the designed pathways. The objectives of this study include (1) locating barite precipitation in the main flow-pathways and stagnant zones, (2) investigating the impact of flow pathways on barite extent and distribution, and (3) identifying implications for barite control and shale gas production.

2. EXPERIMENTAL METHODS

2.1. Marcellus Shale with Engineered Flow Pathways. Two Marcellus shale sub-cores (3.81 cm in diameter and 10.16 cm in length) from a 2279.4 m depth were extracted from a larger parent core taken from the Marcellus Shale Energy and Environment Laboratory (MSEEL). The mineral composition of this parent core was analyzed using a X-ray diffractometer (XRD) and reported previously (Table 1).³¹

The two cores were cut into half cylinders. Two patterns were milled on one half of each core (Figure 1). The M-core pattern

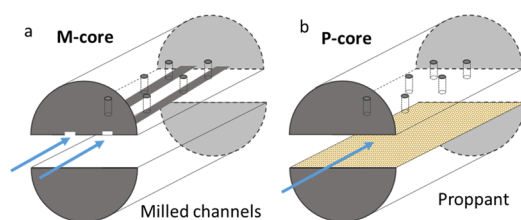


Figure 1. Patterns that simulate the main flow pathway and stagnant zones. M-core: the core with milled channels, P-core: the core that was propped with proppants.

geometry (Figure 1a) included two channels (~0.5 mm depth and ~4 mm width) and three dead-end holes (~2 mm diameter and ~3 mm depth) evenly distributed on each channel. The channels served as the main flow pathways, and no proppants were needed to keep the channels open. The flow in the channels can be viewed as simple laminar flow without obstacles. The P-core pattern geometry (Figure 1b) included the dead-end holes at the same locations but without the channels. A layer of Innoprop 40/60 mesh quartz proppants was added in between the half cores to maintain a propped fracture as the main flow pathway. The propped fracture width was in the range of 50–700 μm based on post-reaction X-ray computed tomography (CT) images. The presence of proppants added tortuosity to the main flow pathway and may impact the reactive transport of the fluid. The dead-end holes were the stagnant zones where fluid has very limited mobility. The flat surfaces of the half cylinders were polished with super fine (400 grit) sand paper after milling. The patterned half cylinders were placed on top the non-patterned half to minimize the gravity influence during the flow-through experiment.

2.2. Hydraulic Fracturing Fluid (HFF). The HFF used the same synthesized recipe as our previous studies¹⁵ with fresh Monongahela River water (Mon water) as base water. Mon water was collected from the near surface of the river on June 10, 2019, and kept at 4 °C. Major species concentrations of the Mon water are listed in Table 2. The hydraulic fracturing chemicals (Table 3)^{15,26,32} and salts (Table 4)^{15,26,33} were added to the Mon water 2 h prior to the start of the experiment. This formulation simulates the concentrated mixture where produced water is blended with fresh Mon water for hydraulic fracturing in the field.

2.3. Flow-Through Experimental Setup. The same flow-through experimental system (Figure 2)¹⁵ as our previous studies was used in the current study. The cores were placed in rubber sleeves in separate core holders with the patterned side on the top. A 13.8 MPa (2000 psi) confining pressure was added on the core holder via a syringe pump with deionized water, and 12.4 MPa (1800 psi) core pressure was set. The core holder temperature was maintained at 66 °C, which is representative of the downhole temperature. The HFF was purged with N₂ gas to limit O₂ in the fluid in a reservoir bottle. However, the system was not in a strictly anoxic environment, and the HFF contains oxidative chemicals such as persulfate. The fluid was pumped into each core at a slow rate of 0.02 mL/min. Because the experiment simulated the shut-in period, the flow rate was set as slow as possible while generating enough effluent for analysis every few days. The effluent was stored in the backpressure syringe pump and sampled every few days of the experiment. For the experimental initial setup, the HFF was pumped in quickly at 2 mL/min for 5 min to flush through the entire flow-through system. Then the flow rate was reduced to the 0.02 mL/min experimental rate while increasing the pressure and temperature to experimental conditions. Based on our previous experiments investigating the flow rate impact on barite precipitation using the same system,¹⁵ the dissolved mass amount from the initial 5 min flush-through was negligible compared to the entire reaction time. Both cores reacted for 28 days. The system took approximately 2 h to stabilize, and the effluent from this period was removed. The reaction time was considered as the time after the start-up period.

2.4. Analytical Methods. Inductively coupled plasma mass spectrometry (ICP-MS, PerkinElmer Nexion 350D), optical emission spectrometry (ICP-OES, PerkinElmer DV 7300), and ion chromatography (IC, DIONEX ICS-5000, Thermo Scientific) were used to measure the aqueous species concentrations. Inorganic carbon and non-purgeable organic carbon were measured by TOC analysis (TOC-L, Shimadzu). Mineral saturation indices were calculated using Geochemists' Workbench (GWB) 10.0 based on measured species concentrations and pH in the solution at 66 °C, with the GWB-provided thermo_pitzer database, which is suitable for high-salinity

Table 2. Mon River Major Cation and Anion Concentrations

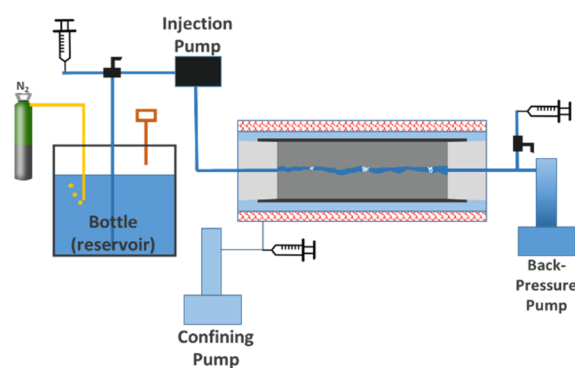
species	Mg	Ca	K	Na	chloride	nitrate	sulfate
concentration, mg/L	6.22	24.37	1.46	11.63	5.69	2.54	59.43

Table 3. Hydraulic Fracturing Fluid Chemical Additives^{15,26,32}

HFF ingredients	purpose	weight %
HCL (37%)	perforation cleaner	0.25
WGA-15 L	gelling agent	0.15
WCS-63ILC	clay stabilizer	0.106
WFR-6ILA	friction reducer	0.049
ammonium persulfate	breaker	0.02
glutaraldehyde	biocide	0.019
potassium hydroxide	pH adjuster	0.014
potassium carbonate	pH adjuster	0.012
revert flow	surfactant	0.0079
ethylene glycol	scale inhibitor	0.0045
citric acid	iron control	0.0034
boric acid	cross-linker	0.002
ethanolamine	cross-linker	0.0014
WAI-25ILC	corrosion inhibitor	0.0013

Table 4. Chemicals Added to Simulate Produced Water Composition^{15,26,33}

produced water ingredients	weight %
boric acid	0.002
potassium carbonate	0.024
barium chloride dihydrate	0.046
potassium chloride	0.022
strontium chloride	0.14
ammonium chloride	0.016
sodium bromide	0.018
calcium chloride dihydrate	0.74
magnesium chloride	0.19
sodium chloride	1.67
sodium sulfate	0.00003
sodium bicarbonate	0.015

Figure 2. Schematic of the high pressure and high temperature flow-through experimental system.¹⁵

solution. Pre-reaction and post-reaction maps of the cores were collected via a back-scatter detector (BSE) on a scanning electron microscope (SEM, Quanta 600F, FEI). Point analyses with energy dispersive X-ray spectroscopy (EDS, Oxford) were performed. The reacted surface and bottom of the holes were also imaged via digital microscopy (Olympus). X-ray computed tomography (CT, M5000 Industrial Computed Tomography System, North Star Imaging) was performed with the cores inside the sleeve at ambient conditions

before and after reaction. The voxel resolution of the CT images was 17 μm for pre-reaction scans and 18.8 μm for the post-reaction scans. Since the post-reaction scans were taken after depressurization, the fracture aperture derived from the CT data is likely slightly larger than it was during in situ conditions. CT images were processed using ImageJ.³⁴ Pre-reaction X-ray fluorescence (XRF) maps of selected elements on the core fracture surfaces were taken at the Stanford Synchrotron Radiation Lightsources (SSRL) with Beamline 10-2.

2.5. Reactive Transport Modeling. Reactive transport modeling was performed using a coupled CrunchTope-Comsol framework.³⁵ In this modeling framework, flow is solved within Comsol using the Brinkman solver. A 2D cross section (Figure 3) along the milled channel was selected, which considers both the main flow channel and the dead-ends.

The flow field is then imported into CrunchTope for solving the transport and reactions. A quasi-steady state approach is adopted: flow calculation is updated at times based on the geometry change caused by mineral reactions. The flow field simulations used an extremely fine finite element mesh generated by the internal mesh generator of Comsol with a constant average velocity ($\bar{u}$) of 8.3333×10^{-5} m/s (which is calculated based on the experimental flow rate and the fracture cross section area) at the inlet and fixed atmospheric pressure at the outlet. The results were mapped to the Cartesian grid cells in CrunchTope, which has a resolution of 400 μm along the flow direction, and a resolution of 50 and 100 μm perpendicular to the major flow direction in the channel and in the dead ends, respectively. The reactive transport simulations used mineral composition and influent chemistry reported in this study and the previously reported reaction networks³⁶ developed for Marcellus shale and hydraulic fracturing fluid interactions. Specifically, barite was allowed to precipitate on the fracture surface.

3. RESULTS AND DISCUSSION

3.1. Fluid Chemistry. The influent and effluent of the M-core (with milled channels) and the P-core (with proppants), pH, and concentrations are displayed in Figure 4, and other species are included in the Supporting Information (Figures S1 and S2). Sulfate concentration in Mon water was ~ 60 mg/L and decreased to ~ 40 mg/L in the injection fluid (Figure 4a), indicating that approximately 33% of the sulfate precipitated from solution during the preparation of HFF. The relatively high sulfate concentration (50–300 mg/L) in the Mon River is due to regional mining and fossil fuel extraction activities.¹⁶ Acid mine drainage is formed when pyrite, which is a common mineral in Marcellus shale, is exposed to air and water during extraction. The inorganic S can be converted to sulfate and H_2SO_4 with the overall reaction $2\text{FeS}_2 + 7\text{O}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{FeSO}_4 + 2\text{H}_2\text{SO}_4$.³⁷ Acid mine drainage from coal fields causes high concentrations of free sulfuric acid and sulfate in most streams in the west Pennsylvania region.³⁸ When sulfate-containing Mon water and Ba-rich produced water for HFF reuse are mixed, barite forms even with a scale inhibitor.¹⁵ Small round barite seeds in the injection fluid were observed from our previous study using the same HFF recipe.¹⁵ The presence of these barite seeds in the HFF can facilitate barite precipitation once injected into the reservoir.

After the fluid was pumped into the cores at elevated temperature and pressure, Ba concentration decreased drastically in both the effluents (Figure 4b), indicating barite precipitation inside the cores. $\text{Ba}^{2+}(\text{aq})$ in the HFF is a result of the reuse of the high TDS produced water. A recent study

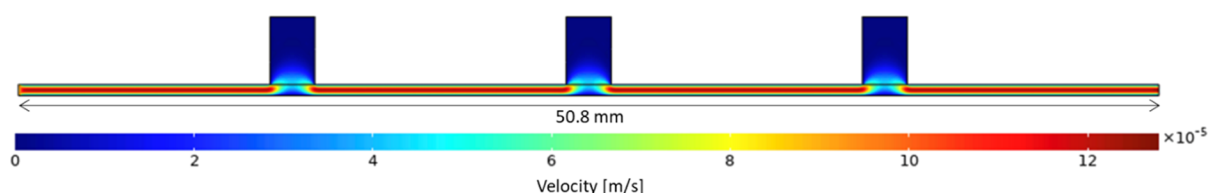


Figure 3. 2D cross section of the fracture used for the simulations, and the velocity magnitude of the original flow field from COMSOL.

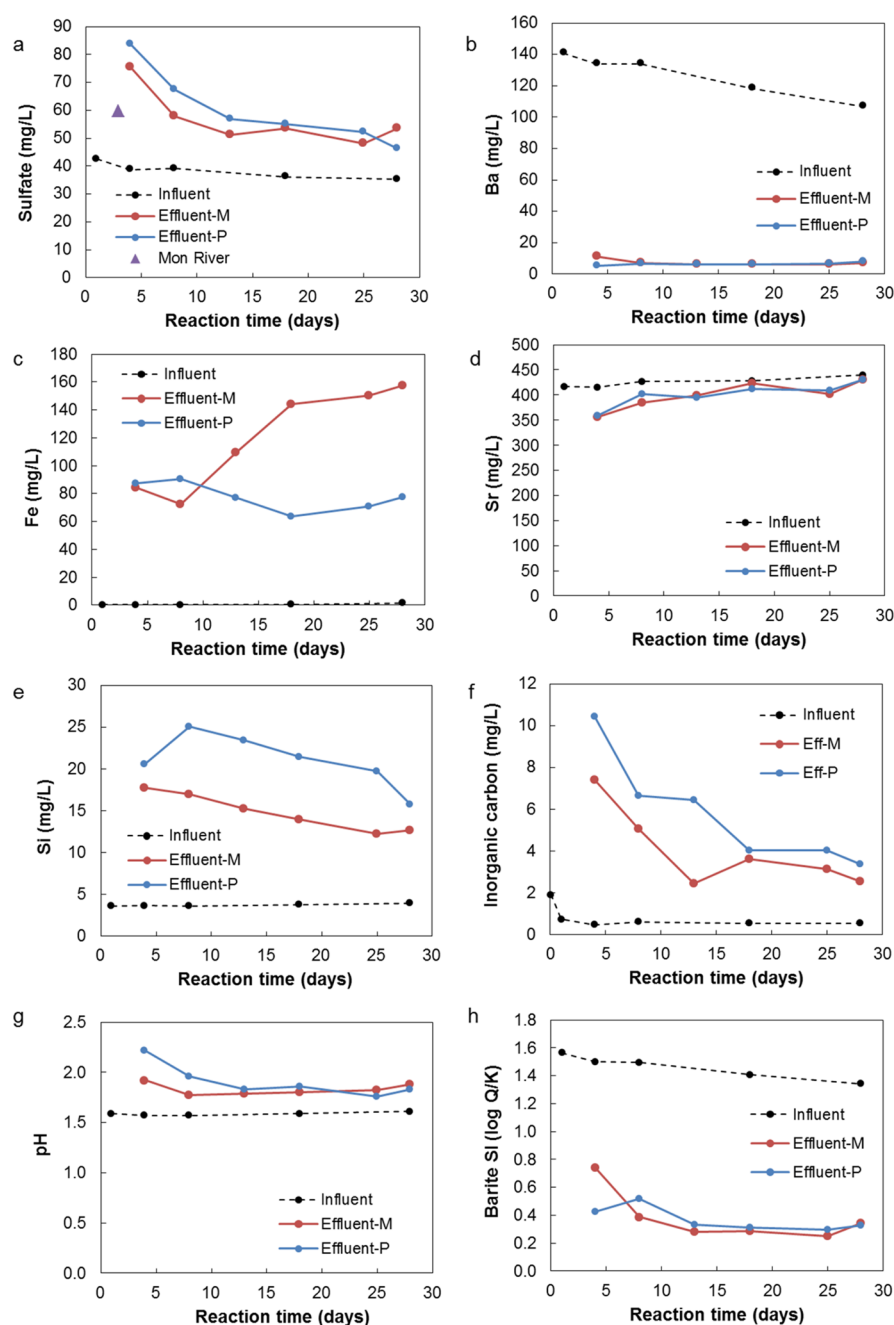


Figure 4. Selected concentrations (a–e), inorganic carbon (f), pH (g), and calculated barite saturation index (h) of the influent and effluent samples. M stands for the core with milled channels, and P stands for the core with proppants.

shows that the high Ba in the produced water may come from rock-fluid interactions by exchanging Ba^{2+} in clays with Na^+ and Ca^{2+} in the reused produced water, and barite solubility and dissolution kinetics are increased under high ionic strength conditions.³⁹ Another study suggests that drilling mud, which

contains two orders of magnitude more barite than Ba-rich oil/gas shales, can be the major source of both barium and sulfate in hydraulic fracturing systems.¹¹ Sr concentration decreased a little in the effluent (Figure 4d). In this study, the Ba in barite precipitations primarily came from the injecting fluid. No

detectable barite or other Ba-containing minerals were found in the Marcellus shale cores before reaction (Table 1). Very limited Ba was observed in XRF maps of the pre-reaction fracture surfaces (Figure S3).

Sulfate concentrations in the effluents were higher than that in the influent, meaning more sulfate was released than consumed by barite precipitation. Fe also increased in the effluent (Figure 4c). Pyrite oxidative dissolution contributed to the increase of sulfate and Fe. Oxidized pyrite grains were found after reaction in both our previous study¹⁵ and the current study, while the pyrite in pre-reaction shale was unaltered without detectable oxygen in SEM–EDX (see the solid analysis below). The decomposition of persulfate was another source of sulfate. At elevated temperature (>50 °C), persulfate decomposes to a sulfate radical ($\text{SO}_4^{\cdot-}$), a strong one-electron oxidant,⁴⁰ which can rapidly degrade the guar-based polymer by defragmenting the polymer into shorter molecules, thus reducing the viscosity.⁴¹

The TDS in the effluent remained at similar levels present in the injection fluid (in the range of 25,000–27,000 mg/L). An aqueous Si increase (Figure 4e) indicated clay and quartz dissolution from the shale matrix. Clay dissolution is also supported by the Al increase (Figure S1). The Si concentration from the P-core was larger than that from the M-core. This could be due to larger reactive surface area, thus the Si-rich shale matrix, in the propped fracture than the milled channels exposed to the HFF. However, the Fe concentration was larger in the M-core than in the P-core, and it had an increasing trend over time. This may be because pyrite was not homogeneously distributed in the shale. As the surface minerals dissolved in the acidic HFF, it was possible that more pyrite grains were exposed on the reactive surface in the M-core than the P-core. The increase of inorganic carbon (Figure 4f) in the effluent reflects carbonate mineral dissolution when the acidic HFF flowed through the cores.

Most of the aqueous species concentrations in the effluents decreased over time. This may result from limited reactivity of fluids in the fracture with deeper fresh shale minerals below the fluid-rock interface as the reaction time increased. The pH in both effluents increased (Figure 4g) as a result of mineral dissolution to buffer the acidic HFF exposure. The injection HFF was supersaturated with respect to barite (Figure 4h), and small barite seeds were present. The barite saturation index decreased significantly in the effluent. Elevated temperature and pressure could promote barite precipitation by increasing nucleation and crystal growth rate⁴² once the fluid was injected into the shale cores.

3.1.1. Barite Precipitation and Pyrite Dissolution. To quantify barite precipitation and pyrite oxidative dissolution, mass balance calculations were conducted based on the fluid chemistry data and the flow rate. Because very limited Ba would be released from the core as stated above, it is reasonable to assume that the decreased barium in the effluent turned into barite (barite Ba^{2+}). Sulfate concentrations in the effluent resulted from the completing processes. Reactions happened inside the cores, including pyrite oxidative dissolution, other minor sulfate mineral dissolution, and persulfate thermal decomposition, provided new sulfate. Since it was difficult to estimate how much persulfate was decomposed, the sulfate species generated inside the cores were grouped together as core SO_4^{2-} and barite precipitation consumed sulfate (barite SO_4^{2-}). The molar mass balance for the entire reaction duration is listed below.

$$\text{influent Ba}^{2+} + \text{barite Ba}^{2+} = \text{effluent Ba}^{2+}$$

$$\text{barite Ba}^{2+} = \text{barite SO}_4^{2-}$$

$$\begin{aligned} \text{influent SO}_4^{2-} + \text{core SO}_4^{2-} + \text{barite SO}_4^{2-} \\ = \text{effluent SO}_4^{2-} \end{aligned}$$

The mass balance calculation results are listed in Table 5. Influent and effluent Ba^{2+} and SO_4^{2-} were calculated based on

Table 5. Mass Balance Calculation for Barium and Sulfate for 28 Days of Reaction

mass (mmol)	sulfate	sulfate net change	barium	barium net change	core SO_4^{2-}
influent	0.32		0.73		
inside core-M		0.52		−0.68	1.20
effluent-M	0.84		0.05		
inside core-P		0.61		−0.69	1.30
effluent-P	0.93		0.04		

the measured concentrations. The sulfate and barium ratio was not stoichiometric with respect to barite in the influent, with sulfate being the limiting species for barite precipitation. Sulfate and barium net changes were the difference between the effluent and influent. Sulfate net change was positive, meaning more sulfate was released inside the core than consumed. Barium net change was negative, turning into solid barite inside the core. Comparing the influent and effluent Ba mass, more than 93% barium precipitated as barite inside the cores. Compared to the influent sulfate, an appreciable amount of additional sulfate was generated inside the core (core SO_4^{2-}) from mineral dissolution and chemical decomposition. During the slow-flow shut-in period, pyrite in shale could be the primary sulfate source, other than sulfate directly from the fluid, which plays a more important role during the fast-flow stimulation period.¹⁵ Barium became the limiting species for barite precipitation inside the cores with an ample sulfate supply during rock-fluid interactions. Limiting the barium concentration in the injection HFF can be an important step to control barite formation during the slow-flow shut-in period of hydraulic fracturing in shale.

3.2. Barite Distribution in Flow Pathways. Barite precipitates were found in both the main flow pathways and stagnant zones (Figure 5). In the SEM/BSE images, silicate minerals in the shale matrix appeared to be darker than barite and pyrite. Barite contains heavier elements than pyrite and backscatters electrons more strongly, so barite appears to be brighter than pyrite in a grayscale. In the post-reaction cores, barite size and amount are much larger than pre-existing pyrite in shale. Barite precipitates stand out obviously (the brighter regions) in the SEM images. In the P-core, large new cracks were observed inside the core in the CT images before and after reaction. These fractures may be a result of uneven pressure on the rock surface in the presence of proppants when loading the core into the sleeve and applying confining pressure around the core holder (Figure 6a). The proppants created and expanded a new reactive surface for rock-fluid interactions in shale. No obvious new fractures formed in the M-core.

3.2.1. Barite Coating on Main Fluid Pathways. Barite coating was found in the channels of the M-core after reaction, with more barite on the bottom non-patterned side due to

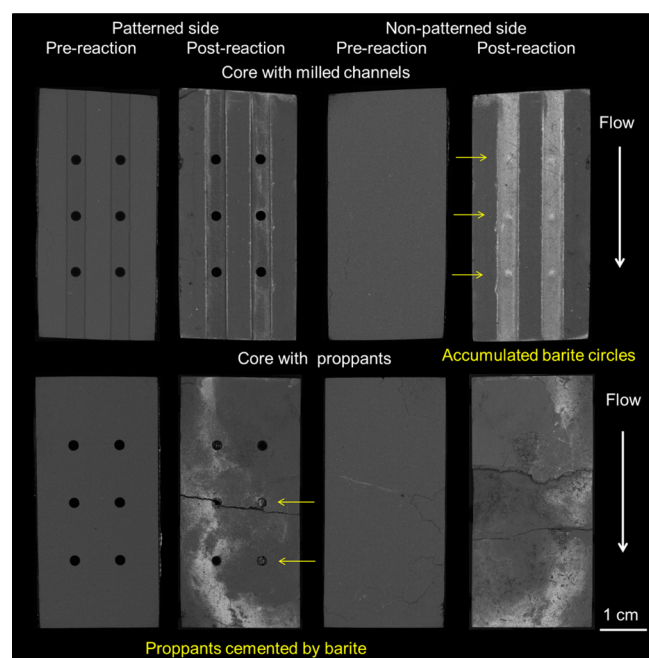


Figure 5. SEM-BSE maps of pre-reaction and post-reaction shale surfaces.

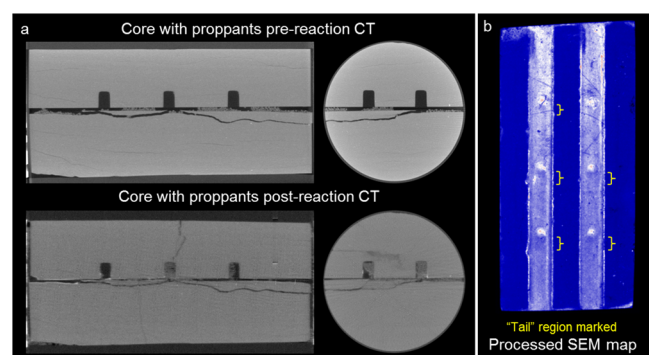


Figure 6. (a) CT images of the core with proppants before and after reaction including a side view of the left three holes and a top view of the middle two holes. Bright coating on the post-reaction fracture surface is barite. (b) Processed SEM map of the core with milled patterns, taken from the non-patterned surface, where a lighter color indicates the presence of barite. Gaussian blur and Unionjack LUT were applied on the original image using ImageJ to reduce noise and highlight gradients.

gravity. Barite seeds settled and continued to grow as the supersaturated fluid went through the channels. Figure 6b shows a processed SEM image to better show the barite distribution. The barite particle extent showed a decreasing trend in the flow direction on the surface of the channels, with more barite near the flow inlet and fewer barite near the outlet, which may be due to reduced barite seeds and Ba^{2+} supply from the fluid as the fluid-rock interactions occur along the channels. On the bottom non-patterned surface, accumulated barite spots were found at the same locations of the dead-end holes on the other side. These accumulated barite circles have a “tail” region of fewer barite behind at downstream (marked with yellow braces in Figure 6b). Some of the tail regions were more obvious than others. For laminar flow, the velocity would decrease in the hole regions. The slowed supersaturated fluid allowed more barite to settle and grow on the non-patterned

bottom surface. Because barite accumulated in these circles, when the fluid came out of the hole regions, the concentration of barite seeds and Ba/sulfate supply in the fluid were lowered compared to the fluid when it came into the hole regions. Fewer barite would precipitate in the “tail” regions. When the “tail” fluid joined the main flow away from this region again, the barite seed concentration and Ba/sulfate supply increased, and the barite precipitation distribution went back to behave like in the main flow pathways with their own decreasing trend along the channel.

Barite precipitation, location, and amount are directly related to the flow access and magnitude.¹⁵ The barite distribution on the fracture surface can in turn affect flow pathways in the propped fracture. Barite coatings were found on the surface of both sides of the P-core (Figure 5). The flow was more turbulent in the presence of proppants compared to the milled channels. The location of barite precipitation was symmetric on the top and bottom halves of the core. Barite precipitated mostly on one side of the fracture (left or right), indicating that more fluid went through that side. It is unclear as to whether this is because the core was slightly rotated inside the core holder (fracture surface was not exactly horizontal) and/or if more proppant propped the fracture wider on that side.

The initially flat surface was cracked and roughened by the proppants under confining pressure (Figure 5). The uneven pressure might trigger the pre-existing tiny fractures in the core to crack. The formation of fresh cracks was not observed in the milled channelled core without proppants.

3.2.2. Mineral Chemistry and Morphology. The pyrite grains and framboids in the shale before reaction were mostly intact with only Fe and S in the EDS spectra (Figure S4). After reaction, broken and scattered pyrite framboids were found on the fracture surface (Figure 7a,e). Much of the pyrite in the flow paths was oxidized with an appreciable amount of oxygen, as shown in the EDS spectra (Figure S4). Pyrite oxidative dissolution would provide additional sulfate to the reacted fluids. Barite precipitates were primarily BaSO_4 , with some Sr substitutions in barite grains (Figure S5). Sr is used as a geochemical tracer of fluid-rock reactions in unconventional reservoirs.⁴³ Some barite contained more Sr, and some contained less. This is consistent with findings from a previous study that the distribution of Sr in barite grains was not homogeneous, even in the same barite grain.⁴⁴ This variance in Sr substitution is likely due to a shift of the crystal growth mechanism when the aqueous Sr/Ba ratio is above 5.⁴⁵ Considering that the Sr/Ba ratio went from ~ 3 in the influent to ~ 70 in the effluent, the localized fluid Sr/Ba ratio in the fluid pathways may be different enough to introduce heterogeneous Sr substitutions.

Barite precipitates on the shale fracture surface varied in size and morphology along the flow pathways in the experimental results. Many of the barite precipitates were small ($< 5 \mu\text{m}$) and irregularly shaped. These may be the barite seeds that settled on the fracture surface from supersaturation of the injection fluid. Some barite seeds were clustered with the gelling agents in the HFF and settled together on the surface (Figure 7b). Alternatively, euhedral barite crystals (mostly $10\text{--}30 \mu\text{m}$) were found in barite coated regions (Figure 7c). These barite most likely formed due to continued crystal growth during the course of the experimental reaction. Some barite even grew into tiny, pre-existing fractures on the shale surface (Figure 7d). The barite seeds and euhedral barite crystals were consistent

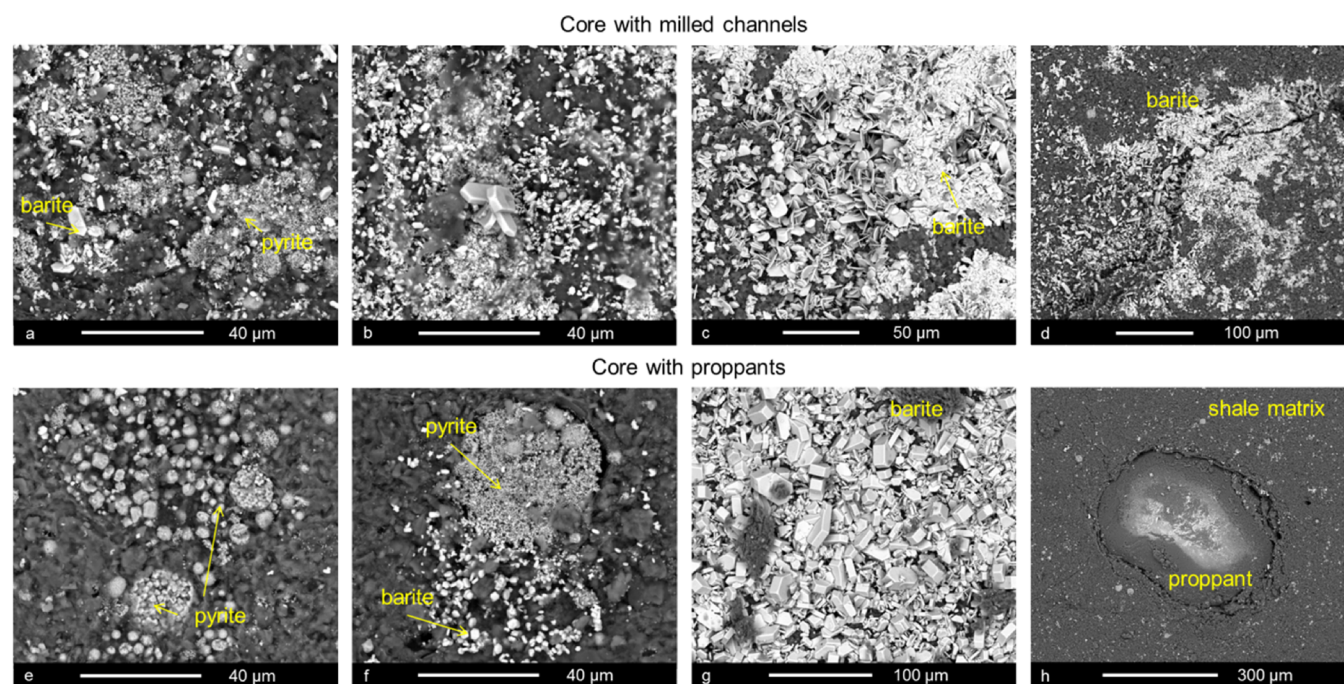


Figure 7. SEM-BSE images of minerals on the reacted core surfaces. (a–d) Core with milled channels. (e–h) Core with proppants.

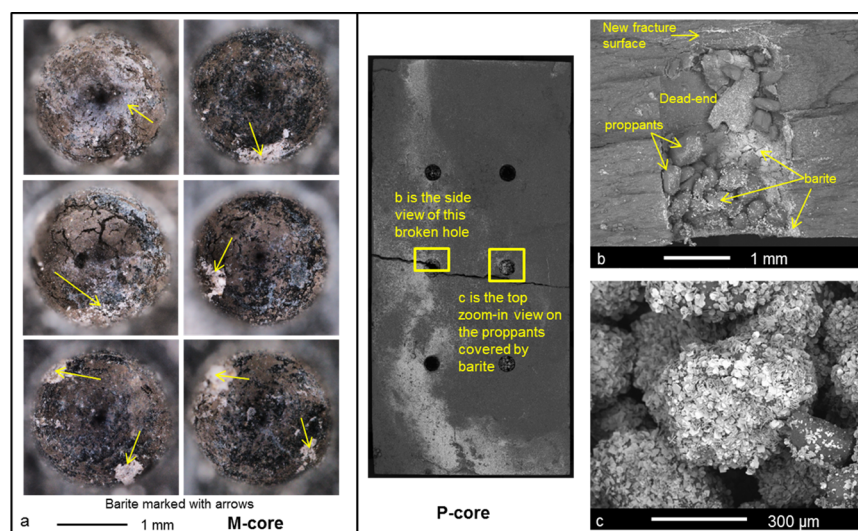


Figure 8. (a) Microscope images of the holes of the M-core, hole locations corresponding to Figure 4. White precipitates are barite; barite growth connecting proppants in dead-end holes in the P-core, (b) side view of the parted hole, (c) proppants cemented together by barite in the hole.

with the observations in our previous study.¹⁵ Another study pointed out that euhedral barite predominantly formed at high temperature ($>40\text{ }^{\circ}\text{C}$) and higher solution Sr/Ba ratio (>15).⁴⁶ Euhedral barite growth was expected in this experiment because the temperature was $66\text{ }^{\circ}\text{C}$ and the Sr/Ba ratio in the solution was very high especially when considering Ba uptake by barite precipitation.

Oxidized pyrite, barite seeds, and euhedral barite were also observed in the P-core (Figure 7e–g). No clear association between the locations of barite and pyrite was observed in both cores. In some places, barite formed around the pyrite or covered a large reactive surface containing pyrite minerals. In some places, some scattered barite was found around pyrite. In some places, no barite was around oxidized pyrite. The sulfate released from pyrite oxidative dissolution may travel with the

flow and participate in barite precipitation reactions away from the origin. Sulfate from persulfate decomposition could also directly react with Ba in the fluid to form barite. Indented proppants were found on the fracture surface (Figure 7h). The proppant indentation occurred probably because radial confining pressure was applied on the core or because the shale matrix near the proppants was weakened due to dissolution in HFF exposure.⁴⁷

3.2.3. Barite Growth in Stagnant Zones. Barite precipitation accumulated in circular patterns on the flat fracture surface opposite to the holes on the milled side of the M-core (Figure 5). These circles of barite appear to be a result of barite seeds settling directly from the HFF and continued barite growth in the slowed supersaturated flow in the hole zones. Barite precipitates were also found at the bottom of the

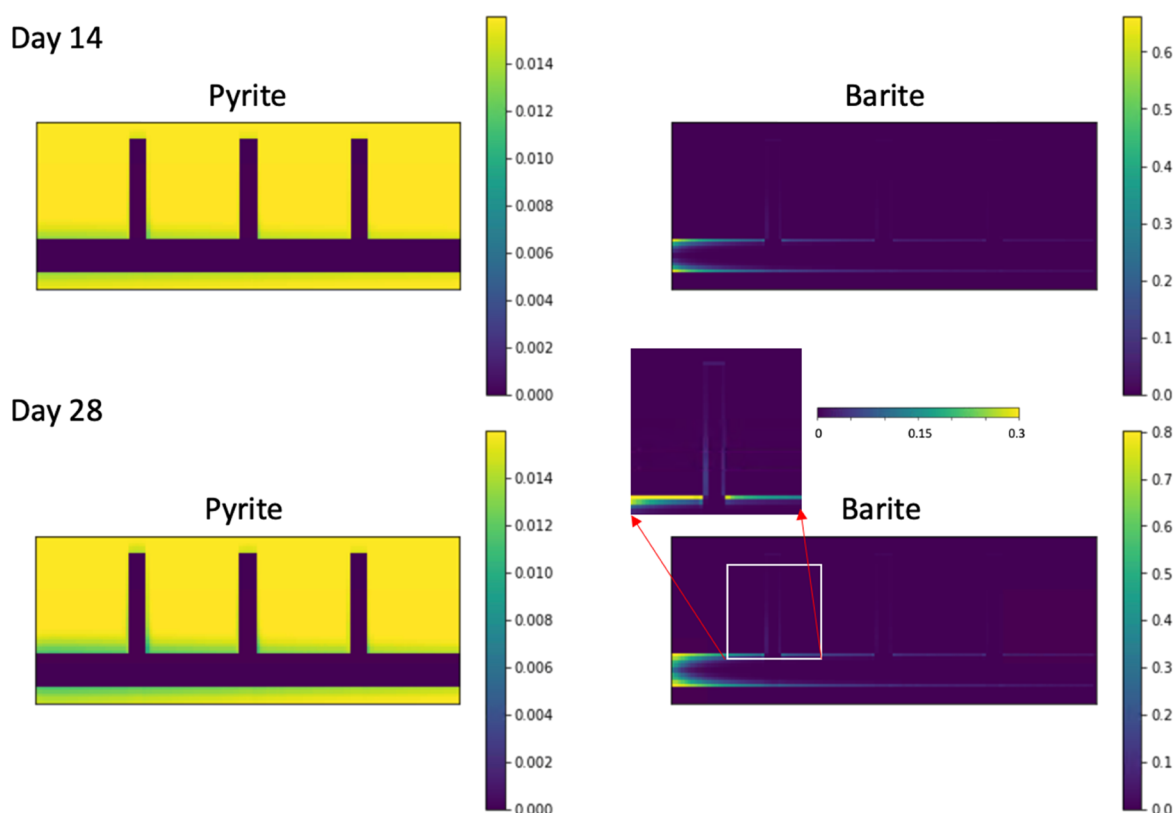


Figure 9. Volume fraction of pyrite (left) and barite (right) at day 14 (top panel) and day 28 (bottom panel).

holes on the patterned side of the M-core (Figure 8a). The patterned side was placed on top of the non-patterned side in the core holder during the course of the experiments, so these are not due to settling. Barite growth from the stagnant fluid led to the scaling on the inner surface of the holes. Continuous Ba and sulfate supply from the hole opening from the laminar flow also contributed to barite growth in the stagnant zones.

In the P-core, the proppants inside the dead-end holes were cemented together into larger accretions by barite (Figure 8b,c). The cemented proppants remained stable inside these holes even with the non-patterned side facing down. The patterned side was on top of the non-patterned side, so fewer small barite seeds were settled there due to gravity. These euhedral barite crystals formed due to geochemical reactions within the core. In a flow-through study using glass beads as porous media, barite precipitations were found to cover the bead surfaces and fill void space in between the beads.⁴⁸ In contrast, proppants located in the main flow pathways were rarely found to be covered with barite (Figure 7h). Barite also grew further into new crack surfaces (Figure 8b).

The stagnant zones can be viewed as porous media after the proppants were flushed into the dead-end holes. The minimal flow velocity in the stagnant zones allowed barite to precipitate with less turbulent supersaturated fluid and the continuous supply of Ba and sulfate from the main flow at the opening of the hole in diffusion-controlled transport zones. Barite scaling and proppant cementation can be an issue for shale gas production if they block the transport pathways in shale and/or pipelines.

3.3. Reactive Transport Modeling. Reactive transport modeling showed a decreased pyrite volume fraction within the shale matrix over time (Figure 9). Sulfate was predicted to be present in the fluid due to pyrite oxidative dissolution (Figure

S6), which was consistent with the presence of sulfate in the effluent from the experiment. The amount of pyrite dissolution was not as significant as observed in the experiment because only dissolved oxygen under atmospheric conditions was included in the simulation, while the actual fluid contained other oxidants such as persulfate. The amount of barite precipitation in the fracture increased over time, and the extent of barite coating on the channel surface decreased along the flow direction (Figure 9). The decreasing barite gradient along the flow direction was because of decreasing supersaturation following the consumption of Ba^{2+} from the fluid. A similar gradient along the channels was observed in the experiment (Figure 6b).

The total barite precipitation within each column of grid cells is shown in Figure 10, assuming a thickness of 1 mm for the 2D cross section. The barite precipitation amount close to the inlet was one order of magnitude higher than that at the outlet. Barite precipitation hotspots (the spikes) were also

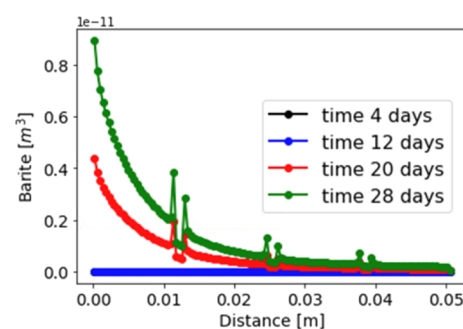


Figure 10. Amount of barite precipitation along the flow direction. The spikes are in the dead-end regions.

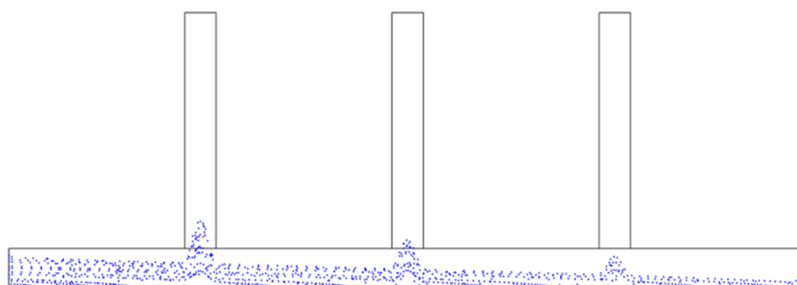


Figure 11. Particle (blue dots) transport in the simulated domain. The y axis is exaggerated by 5 to better show the particles.

observed around the dead-ends. The walls of the dead-ends provided additional surface area for the precipitation reaction in the simulation.

Based on the pressure difference (ΔP) recorded in the reactive transport modeling, fracture permeability (k) can be evaluated using $k = \frac{\mu \mu L}{\Delta P}$. Due to barite precipitation, the fracture permeability decreased by $\sim 16\%$ from the initial value of 22 Darcy to ~ 18 Darcy. Another common practice is to estimate fracture permeability based on average aperture ($\bar{b}$) using the cubic law, $k = \frac{\bar{b}^2}{12}$. The amount of barite precipitation corresponded to $\sim 5\%$ reduction in the average aperture and thus $\sim 10\%$ in the permeability. This difference is because precipitation and thus aperture change were not uniform along the flow direction. More precipitation was observed at the inlet and created a stronger flow restriction. It further highlights the importance of the spatial distribution of the precipitates in controlling fracture permeability change.

3.3.1. The Transport of Colloidal Barite. The reactive transport modeling did not consider particle transport and deposition and thus the impacts of colloidal barite. Colloidal barite formed in the injection fluid before entering into the fracture domain. Once the fluid was injected, the colloidal barite would act as the seed for further crystal growth. Because of gravity, the colloidal barite settled on the bottom fracture surface, providing the additional surface area for continuous barite growth. In the reactive transport simulation, all fracture surface area was considered equal for precipitation, whereas in reality, precipitation on existing barite surfaces was more energetically favorable. In the experiment, barite precipitation was primarily observed on the fracture surface of the bottom half core, on which barite settled primarily.

Figure 11 illustrates particle settling distribution within the fracture domain for the flow field given in Figure 3. The simulation was performed in the COMSOL Multiphysics code using the “particle tracing for fluid flow” application. The particle size was set to $5 \mu\text{m}$, and the density was 1050 kg/m^3 , and 10 particles were released every second at the inlet. The particles settled on the bottom of the channels with a decreasing extent along the flow direction. Accumulated barite particles were found at the openings of the dead-end regions, which was consistent with the experimental observations (Figure 6b).

3.3.2. Impact of Reactive Flow on Scaling in Different Pathways. Barite distribution in channeled fracture vs propped fracture and stagnant zones vs main flow regions was investigated. In the channeled fracture, the flow was close to laminar flow. Barite settled and grew along the channel surface with a decreasing extent trend. Such a barite distribution trend was not obvious in the propped fracture, in which the flow was

turbulent. Barite distribution was associated with the flow access and magnitude in the propped fracture.¹⁵ The distribution of barite in the main flow regions and stagnant zones was a result of the coupling of reactive transport and particle settling. The barite formed in the injection fluid settled in the main flow pathways and served as seeds for continuous barite growth in the reactive flow. In the particle transport simulation without geochemical reactions, no barite was predicted on the wall of the dead-ends, which were placed on the top (Figure 11). In reactive transport modeling, stagnant fluids in the dead-ends were supersaturated and barite could form on the surface in these regions (Figure 9), as observed in the experiment (Figure 8). This indicates that even without settled barite seeds to facilitate crystal growth, barite can still form in regions where the supersaturated fluid is stagnant. Our previous study with Marcellus shale powders showed continued barite precipitation due to rock-fluid interactions after removing the pre-precipitated barite in the initial HFF, even with a scale inhibitor.²⁴ Apart from the main flow pathways for barite to precipitate, these dead-end regions also allow barite precipitation once the supersaturated fluid was injected into the reservoir. More importantly, barite could connect and cement the proppants that were trapped in such stagnant zones and form larger conglomerates. Such barite-proppant conglomerates were not found in the main flow pathways in the P-core. Shale mineral dissolution may increase porosity and permeability for gas extraction. However, the barite coatings on the reactive surfaces and barite-proppant conglomerates could inhibit mass transport and mineral dissolution in the rock-fluid interface.

4. CONCLUSIONS

Colloidal barite in the injection HFF can settle in the main flow pathways and serve as a seed for continuous barite growth during the entire fracturing process. Barite coatings can cover a large portion of the reactive surface in shale, blocking transport pathways in the rock-fluid interface. Removing pre-precipitated barite before injection may help control barite scaling once the fluid is injected into the reservoir. Barite may still continue to form due to rock-fluid interactions in the reservoir. Supersaturated fluids allow barite to precipitate in stagnant zones. With the trapped proppants in the stagnant zones, individual proppants can be cemented together by barite precipitations and become bulk barite-proppant conglomerates, inhibiting or even block the transport pathways. Proppants conglomerating via barite scaling are a potential issue associated with scale control in the shale gas industry.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.energyfuels.1c01374>.

Other major cation and anion concentrations; XRF elemental maps of the fracture surfaces of the pre-reaction cores; SEM–EDX of pyrite in pre- and post-reaction shale; SEM image of barite with Sr substitution; model predicted effluent concentrations (PDF)

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Notes

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RESEARCH ARTICLE

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Key Points:

- Pore-scale reactive transport modeling of diffusion controlled mineral precipitation
- Examined the evolution of pore structures and averaged bulk properties
- Developed modified porosity-diffusivity relationship that accounts for pore-scale mineral precipitation dynamics

Supporting Information:

Supporting Information may be found in the online version of this article.

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A Pore-Scale Investigation of Mineral Precipitation Driven Diffusivity Change at the Column-Scale

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Abstract Mineral precipitation affects the pore structure and thus transport properties of porous media. In this study, we investigated the pore-scale dynamics of precipitation in diffusion controlled systems and the resulting impacts on the effective diffusivity, using a micro-continuum reactive transport model. Forty two-dimensional pore structures representing both idealized and realistic geometries were simulated with consideration of different precipitation scenarios and rates. A homogeneous nucleation scenario reproduced patterns observed in previous experimental study showing mixing-induced precipitation, and a surface growth scenario captured the pattern for mineral precipitation on a substrate with the same or similar mineral structures. In all cases, local precipitation resulted in the reduction in the average porosity of the domain (Φ) until the cessation of diffusive transport and the termination of precipitation. The minimum porosity reached was referred to as the critical porosity (Φ_c). The effective diffusivity (D_{eff}) decreased with Φ and dropped sharply to effectively zero, that is, the critical effective diffusivity ($D_{\text{eff},c}$), as Φ_c was reached. These pore-scale dynamics can be captured by a revised $\Phi - D_{\text{eff}}$ relationship that explicitly considers the critical porosity and the corresponding effective diffusivity, and the pre-exponential coefficient and the exponent of the relationship varied with initial pore structure and the precipitation kinetics. Overall, the homogeneous nucleation scenario results in systematically larger Φ_c and coefficients that give rise to a sharper decrease in diffusivity as Φ_c is approached, compared to the surface growth scenario. The revised relationship was also implemented at continuum scale and used to examine column scale diffusivity change and reactions.

Plain Language Summary Mineral precipitation alters the internal structures of geo-materials, and thus affects how fluid and solutes migrate in these porous media. Typically, this alteration has been modeled by assuming that the changes in transport properties are proportional to the amount of mineral precipitation, following a simple power law function. However, previous experimental studies have pointed out that in addition to the amount of mineral precipitation, how the structures of porous media are altered by the mineral precipitates can be important as well. In our study, we used numerical modeling that resolves the geometry of porous media explicitly to investigate how the structures evolve as a result of mineral precipitation. In these simulations, we consider initial geometry of different complexity and a range of precipitation rates that have been reported in previous studies. All the simulations showed that a small amount of precipitation can result in plugging of the porous media, by clogging a small portion of the transport pathways. Based on the simulation results, we demonstrated that the simple power law function can be modified to account for these structural changes. We have also implemented this modified power law relationship to simulate a column experiment published previously and showed good agreements.

1. Introduction

Mineral dissolution and precipitation can change the structures of fractured porous media, and thus control their mechanical and hydrodynamic properties. Mineral precipitation is widely observed and occurs when the fluid is over-saturated with respect to one or multiple mineral phases. Some conditions that can be encountered in natural and engineered systems and lead to mineral precipitation include (i) solubility change, for example, due to the presence of temperature gradients (Griffiths et al., 2016; Steefel & Lasaga, 1994), (ii) mixing of fluids (Emmanuel & Berkowitz, 2005; Tartakovsky et al., 2008), and (iii) mineral dissolution that increases the concentration of key chemical species (Garcia-Rios et al., 2015; Li et al., 2017; Seigneur et al., 2019; Steefel & Lichtner, 1994; Xie et al., 2015). Understanding and predicting the evolution of Earth's

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systems resulting from mineral precipitation are important. For instance, mineralization of CO₂ injected into the underground as part of geologic carbon storage is an important permanent trapping mechanism (Matter et al., 2016), and mineral precipitation can affect the efficiency of hydraulic fracturing and the performance of other subsurface energy applications (Vankeuren et al., 2017). In addition, salt precipitation alters the hydraulic properties of soils (Nachshon et al., 2011) and the porosity in coastal environments, especially in the saltwater-freshwater mixing zones (Singurindy et al., 2004). Mineral precipitation is one mechanism used to scavenge contaminants in aquifers (Wright et al., 2011), a by-product of remediation technologies such as *in situ* chemical oxidation (Chen and Park, 2017), and can reduce the efficiency of membranes used for water desalination (Rahardianto et al., 2008). Therefore, evaluating and predicting mineral precipitation and its impacts on porous media are needed for improving management practices of soils and groundwater resources.

In continuum scale modeling (Lichtner, 1996; Steefel & Lichtner, 1994; 1998; Steefel et al., 2005; Steefel, Appelo, et al., 2015), the changes in flow and transport properties resulting from mineral precipitation, and mineral reactions in general, are typically described by power law functions of porosity (ϕ). Commonly used functional relations include the Kozeny-Carman relationship between porosity and permeability and Archie's law that relates effective diffusivity to porosity. These relationships are easy to implement and allow some considerations of system specificity such as textural heterogeneity through the empirical coefficients (Cai et al., 2017; Luquot & Gouze, 2009; Menke et al., 2017; Peng et al., 2012; Smith et al., 2013; Steinwinder & Beckingham, 2019).

However, adjusting the parameters alone in some cases fails to produce flow and transport properties that are directly measured or inferred from observations. In flow-through experiments in carbonate rocks with induced precipitation (Luquot & Gouze, 2009), it is observed that porosity-permeability relationship shifted away from a power law relationship as precipitation progresses. Navarre-Sitchler et al. (2009) performed diffusion experiments on weathered basalt samples and the corresponding parent materials and showed that the effective diffusivity (D_{eff}) evaluated using micro-x-ray fluorescence spectroscopy cannot be fitted with a simple form of Archie's law given by $D_{\text{eff}} = D_m a \phi^m$, where D_m is the molecular diffusion coefficient and a and m are the empirical coefficients. In Chagneau et al. (2015), a set of column experiments were carried out to assess the changes in the effective diffusivity of the system as a result of mineral precipitation. In the study of Chagneau et al. (2015), a one-dimensional continuum reactive transport model using Archie's law either over-estimated the amount of precipitation or over-estimated the tracer transport.

These studies highlighted the need of further development of constitutive relations or other means to account for pore-scale dynamics that are not captured by bulk averaged parameters used in continuum scale modeling. For this purpose, pore-scale experiments and modeling have been used to investigate microscopic scale dynamics that arise from geometric and mineralogical heterogeneity and that can propagate to larger scale processes (Cil et al., 2017; Deng et al., 2018; Chen et al., 2014; Molins et al., 2020; Zhang, Kang, et al., 2010). For system evolution driven by mineral precipitation, the complexity of pore-scale dynamics is compounded by the processes of nucleation and crystal growth. It was shown that the morphology and texture of the precipitates depend on the substrates (Noiriel et al., 2016), supersaturation (Kim et al., 2020; Rajyaguru et al., 2019; Zhang, Dehoff, et al., 2010), and local hydrodynamics (Poonosamy et al., 2019). Parallel pore-scale modeling studies were used to simulate precipitation patterns (Yoon et al., 2012), and their dependence on fluid saturation (Prasianakis et al., 2017). Pore-scale models were also used to examine system evolution and their dependence on different nucleation mechanisms, for example, epitaxial crystal growth versus crystal growth along the direction of maximum concentration gradient or probabilistic nucleation versus deterministic nucleation (Fazeli et al., 2020; Chen et al., 2014).

These studies with well controlled and characterized experiments and high-fidelity modeling have provided valuable data and important mechanistic understanding of precipitation-driven pore-scale dynamics. While a few studies have examined permeability evolution following precipitation and its relationship with respect to porosity change (Fazeli et al., 2020; Noiriel et al., 2016), how pore-scale dynamics translate into continuum scale transport properties remains an open question. In our study, we aim to use carefully designed pore-scale reactive transport simulations to address this gap. The complexity and uncertainties of precipitation kinetics were considered by including different precipitation scenarios and rates. In the following sections, we first present the model and simulation approach and then follow this with pore-scale

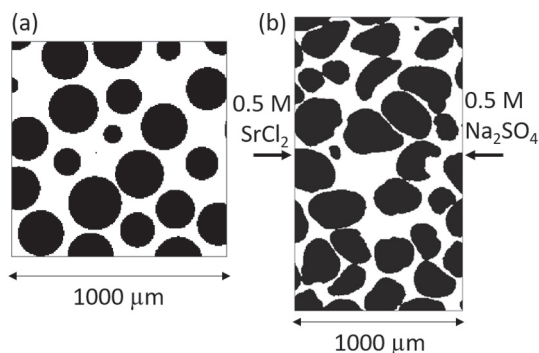


Figure 1. Illustration of the 2D domains of (a) the Finney pack geometry and (b) the quartz sand pack geometry, and of the fluid chemistry boundary conditions used in the simulations.

simulation results and discussions of the revised mathematical relation that integrates the pore-scale dynamics into column/continuum scale simulations.

2. Methods

The approach taken in this work is the simulation of diffusion-reaction processes in 2D domains derived from openly available digital rock data. The simulations were performed systematically under different model assumptions and over a range of parameters. The simulations used the solution chemistry and major physical bulk properties of the experimental work of Chagneau et al. (2015), in which celestite precipitation was induced in the quartz sand packed column by the mixing of a Sr-rich fluid and a sulfate-rich fluid introduced via diffusion from the two ends, respectively. This allowed us to leverage the experimental data for comparison and discussion. The precipitation reaction writes as follows.

All aqueous reactions and the corresponding equilibrium constants are summarized in the supporting information.



2.1. Pore Geometry

Two pore geometries were considered, using data archived at the digital rock portal (<https://www.digital-rocksporal.org/>). One data set represents an idealized geometry from a random packing of hard spheres with uniform radii (Finney, 2016). A three-dimensional (3D) volume image was generated using the reported coordinates of the sphere centers, using a radius of 150 μm and a voxel size of 5 μm. The voxel size was representative of the resolution of micro-tomography images, and the diameter was comparable to the grain size used in the experiments of Chagneau et al. (2015). The other data set included grayscale synchrotron x-ray computed tomography images of a column that was packed with quartz sands of grain sizes of 420–600 μm (Molnar, 2016). The images had a resolution of 9.87 μm and were rescaled to a resolution of 5 μm such that the grain sizes were comparable with the Finney pack geometries. For the simulations, the CT images from Molnar (2016) were segmented in MATLAB using Otsu's method (Otsu, 1979) to generate a binary volume image. The average porosity of the middle section of the Finney pack was 48%, whereas the average porosity of the quartz pack was 33%. These values are on the higher and lower end of the values reported for the experimental columns (42% ± 9%), respectively.

For our simulations, twenty two-dimensional (2D) domains were randomly selected from each 3D volume, as illustrated in Figure 1. The domain size was 1 by 1 mm (200 by 200 grid cells) for the Finney pack geometries (numbered as FGeo1-20) and was 1 by 1.75 mm (200 by 350 grid cells) for the quartz pack geometries (numbered as QGeo1-20). The width was larger for the quartz pack geometries to ensure initial connectedness of the domain.

2.2. Reactive Transport Model for Precipitation Simulations

In our study, the code CrunchTope (Steefel, Appelo, et al., 2015) was used for all the reactive transport simulations. For pore-scale simulations, modifications were made to implement the micro-continuum approach: the pore geometry was explicitly represented, whereas porosity was used to keep track of partial volume in individual grid cells due to precipitation. As the system considered was diffusion controlled, the main governing equation written with respect to the total concentration of a chemical component Ψ_i is given as follows:

$$\frac{\partial \phi \Psi_i}{\partial t} = \nabla \cdot (\phi D \nabla \Psi_i) - \mathcal{R}_i R_i, \quad (2)$$

where the local porosity (ϕ) is 1.0 in the pore space, 0.0 inside the mineral grains and is a fractional value as precipitation progresses and occupies part of the grid cell. D is the diffusion coefficient, which is equal to the molecular diffusion coefficient in the pore space, and is $D_m\phi$ in grid cells occupied by the precipitates to account for the tortuosity caused by the presence of the solid phase. Two additional non-linear tortuosity models ($D = D_m\phi^{n_0}$, $n_0 = 2$ or 3) were also tested for a subset of the simulation conditions to examine the impacts of this assumption. For simplicity, we assumed that the molecular diffusion for all aqueous species is the same, with $D_m = 2.24 \times 10^{-9}$ m²/s following that of the tracer in Chagneau et al. (2015). The total concentration is the sum of the concentration of the primary species and those of other species that can be expressed in terms of the primary species based on mass action laws. The concentrations of all primary species were calculated by solving the mass balance equation (Equation 2). Spatial discretization for numerically solving Equation 2 followed the image resolution.

The reaction term (last term on the right hand side) accounts for the concentration change caused by mineral reactions, with ϑ_i being the stoichiometric coefficient and one for celestite. Two precipitation conceptualizations were considered to bracket different reaction mechanisms. In one scenario, homogeneous nucleation was enabled in the pore space and provided the initial surface area for subsequent crystal growth. This is referred to as the homogeneous nucleation (HN) scenario. In the other scenario, referred to as the surface growth (SG) scenario, homogeneous nucleation in the pore space was not enabled. This is also equivalent of assuming that heterogeneous nucleation is fast and crystal growth picks up immediately.

$$R_i = \begin{cases} R_{i,nuc} + R_{i,tst}, & \text{for HN} \\ R_{i,tst}, & \text{for SG} \end{cases} \quad (3)$$

Crystal growth rate was calculated from the transition state theory rate law ($R_{i,tst}$ [mol / m³s]) and is positive for the precipitation reaction.

$$R_{i,tst} = Ak_{rxn} \prod a_j^{n_j} \exp\left(-\frac{E_a}{RT}\right) \left(1 - \left(\frac{IAP}{K_{sp}}\right)^{m_1}\right)^{m_2} \quad (4)$$

where A (m² / m³) is the bulk surface area of the mineral, k_{rxn} (mol / m²s) is the kinetic factor, a_j is the activity of any catalytic and inhibitory species, and n_j is the corresponding reaction order. The Arrhenius term accounts for the temperature (T [K]) effect, where E_a (J / mol) is the activation energy, and R (J/mol K) is the ideal gas constant. The thermodynamic driving force is measured by the saturation index, which is the ratio of ionic activity product (IAP) to solubility (K_{sp}), and m_1 and m_2 are constants dependent on the reaction mechanisms.

Homogeneous nucleation rate ($R_{i,nuc}$ [mol / m³s]) was described by the classical nucleation theory and was implemented in CrunchTope as follows (Li et al., 2017):

$$R_{i,nuc} = A_0 J_0 \exp - \left(\frac{16\pi v^2 \alpha^3}{3k_B^3 T^3 \left(\ln \left(\frac{IAP}{K_{sp}} \right) \right)^2} \right) \quad (5)$$

where A_0 is unity (m² / m³) and J_0 is the kinetic factor with a value of 1.0×10^{-8} mol/m²s for celestite, which is within the range inferred from the experimental study of Poonosamy et al. (2019). v is the molecular volume (8.21×10^{-29} m³ for celestite), α is the effective interfacial tension (0.092 J/m² for celestite, Nielsen & Söhnel, 1971), and k_B is the Boltzmann constant. It is assumed that the nuclei formed via homogeneous nucleation stay in the same grid cell, where the crystal growth continues.

For the homogeneous nucleation scenario, at a given time (t), the surface area used in Equation 4 was calculated as

Table 1
The Kinetic Data and Surface Area Setup for Different Precipitation Scenarios Simulated in This Study

Scenario	Nucleation	Crystal growth kinetic data	Surface area	Reference
Homogeneous nucleation	$\alpha = 0.092 \text{ J / m}^2 *$	$k_{\text{rxn}} = k_{\text{Marty}}$	Specific surface area 0.44 m ² /g	ssa0.44
	$J_0 = 1.0 \times 10^{-8} \text{ mol / m}^2 \text{ s}^{**}$	$m_1 = 0.5, m_2 = 2$		
		$\log k_{\text{Marty}} = -7.23 \text{ ***}$		
	$\alpha = 0.092 \text{ J / m}^2$	$k_{\text{rxn}} = k_{\text{usgs}} a_{\text{H}^+}^{0.109}$	Specific surface area 0.44 m ² /g	ssa0.44_usgs
	$J_0 = 1.0 \times 10^{-8} \text{ mol / m}^2 \text{ s}$	$m_1 = m_2 = 1$		
		$\log k_{\text{usgs}} = -5.66 \text{ ****}$		
Surface growth	$\alpha = 0.092 \text{ J / m}^2$	$k_{\text{rxn}} = k_{\text{Marty}}$	Specific surface area 44 m ² /g	ssa44
	$J_0 = 1.0 \times 10^{-8} \text{ mol / m}^2 \text{ s}$	$m_1 = 0.5, m_2 = 2$		
		$\log k_{\text{Marty}} = -7.23$		
	N/A	$k_{\text{rxn}} = k_{\text{Marty}}$	Porosity gradient area	pga
		$m_1 = 2, m_2 = 0.5$		
		$\log k_{\text{Marty}} = -7.23$		
	N/A	$k_{\text{rxn}} = k_{\text{usgs}} a_{\text{H}^+}^{0.109}$	Porosity gradient area	pga_usgs
		$m_1 = m_2 = 1$		
		$\log k_{\text{usgs}} = -5.66$		

The data are from *Nielsen & Söhnel (1971), **Poonosamy et al. (2019), ***Marty et al. (2015), and **** Palandri (2004).

$$A = MW \cdot \text{ssa} \cdot \int_0^t R_i, \quad (6)$$

where MW (g / mol) is the molecular weight and ssa is the specific surface area (m²/g).

For the surface growth scenario, the surface area used in Equation 4 was the geometric surface area of the solid phase and was calculated based on the pore geometry as the gradient of the local porosity using the micro-continuum formulation derived from the volume averaging theorem (Soulaine et al., 2017 and references therein):

$$A = |\nabla \phi|. \quad (7)$$

In order to account for uncertainties introduced by the kinetic data, two sets of kinetic coefficients and two specific surface areas were examined in the simulations. One kinetic formulation followed Palandri (2004) and included a non-linear dependence on pH. The other kinetic formulation used was from Marty et al. (2015) and was independent of any other species. The specific surface area of celestite varies substantially, as reported in Chagneau et al. (2015). Accordingly, two values, 0.44 and 44 m²/g, were chosen as the bounding cases. Overall, higher specific surface area and the kinetic coefficient from Marty et al. (2015) would result in higher reaction rate. The parameters and kinetic scenarios are summarized in Table 1.

Mineral precipitation was used to update local porosity:

$$\phi^{t+dt} = \phi^t - R_i V_m dt, \quad (8)$$

where V_m is the molar volume of celestite and dt is the time step.

The precipitation simulations used boundary conditions similar to the experiment. The two ends of the computational domain were assigned a Dirichlet boundary condition, in contact with a solution of 0.5 M SrCl_2 and a Na_2SO_4 solution of 0.5 M, respectively (Figure 1). The other two sides were assigned no flux boundary conditions.

2.3. Numerical Tracer Experiments

During the precipitation simulations, pore structures were recorded at multiple time points. Numerical tracer experiments were performed to evaluate the corresponding effective diffusivity, similar in concept and execution to those considered by Navarre-Sitchler et al. (2009) and Steefel, Beckingham, & Landrot (2015). The governing equation of the numerical tracer experiments was similar to the precipitation simulations but without the reaction term.

$$\frac{\partial \phi C_{\text{tracer}}}{\partial t} = \nabla \cdot (\phi D \nabla C_{\text{tracer}}). \quad (9)$$

In the tracer simulations, the Dirichlet boundary on the left was considered as the inlet and was exposed to a solution with a non-reactive tracer of 1 M. The pairing boundary, that is, the outlet, had a zero-gradient boundary condition and is where the tracer break-through curve ($C_{\text{tracer}}(t)|_{x=L}$) was monitored. Other boundary conditions, spatial discretization, and diffusion coefficients were set up in the same way as the precipitation simulations.

The effective diffusivity (D_{eff}) is the bulk transport property of the entire computational domain and is evaluated by fitting the tracer break-through curve from the numerical tracer experiments using the one-dimensional (1D) diffusion equation.

$$\frac{\partial C_{\text{tracer}}}{\partial t} = D_{\text{eff}} \frac{\partial^2 C_{\text{tracer}}}{\partial x^2}. \quad (10)$$

Its relation with respect to the domain porosity (Φ), which is the average of the local porosity in all grid cells in the domain ($\Phi = \sum_x \sum_y \phi(x, y)$), was analyzed. In the following sections, porosity refers to the domain porosity unless specified otherwise.

Numerical tracer experiments were performed for all Finney pack geometries and a subset of four quartz sand pack geometries.

3. Simulation Results

3.1. Pore Geometry Evolution and Porosity Reduction

In the homogeneous nucleation scenario, a precipitation band formed in the pore space (Figure 2a–2c). This type of pattern was observed in the microfluidic experiments by Zhang, Dehoff, et al. (2010), in which calcium-carbonate precipitation was induced by the mixing of the calcium-rich and carbonate-rich fluids in the analog porous media, and the precipitation band tracked the mixing line of the two fluids. Their experimental observation was also confirmed by the companion modeling work (Yoon et al., 2012), and the initial surface area was assumed to be provided by the top and bottom surfaces of the microfluidic cell. In our study, the initial surface area was created by the homogeneous nucleation in the fluid that was locally supersaturated.

In contrast, under the surface growth scenario, the precipitates coated the mineral grains, and the precipitation pattern was thus largely influenced by the initial pore structure (Figures 2d and 2e). This precipitation pattern was consistent with the scenario when the substrate had a similar crystal structure to the precipitating mineral. Similar patterns have been observed in the experimental study of Noiri et al. (2012) when the injected fluid over-saturated with respect to calcium carbonate interacted with the calcite spar substrate. In these cases, heterogeneous nucleation, which was not considered in our model, is fast, and the following crystal growth is the dominant process.

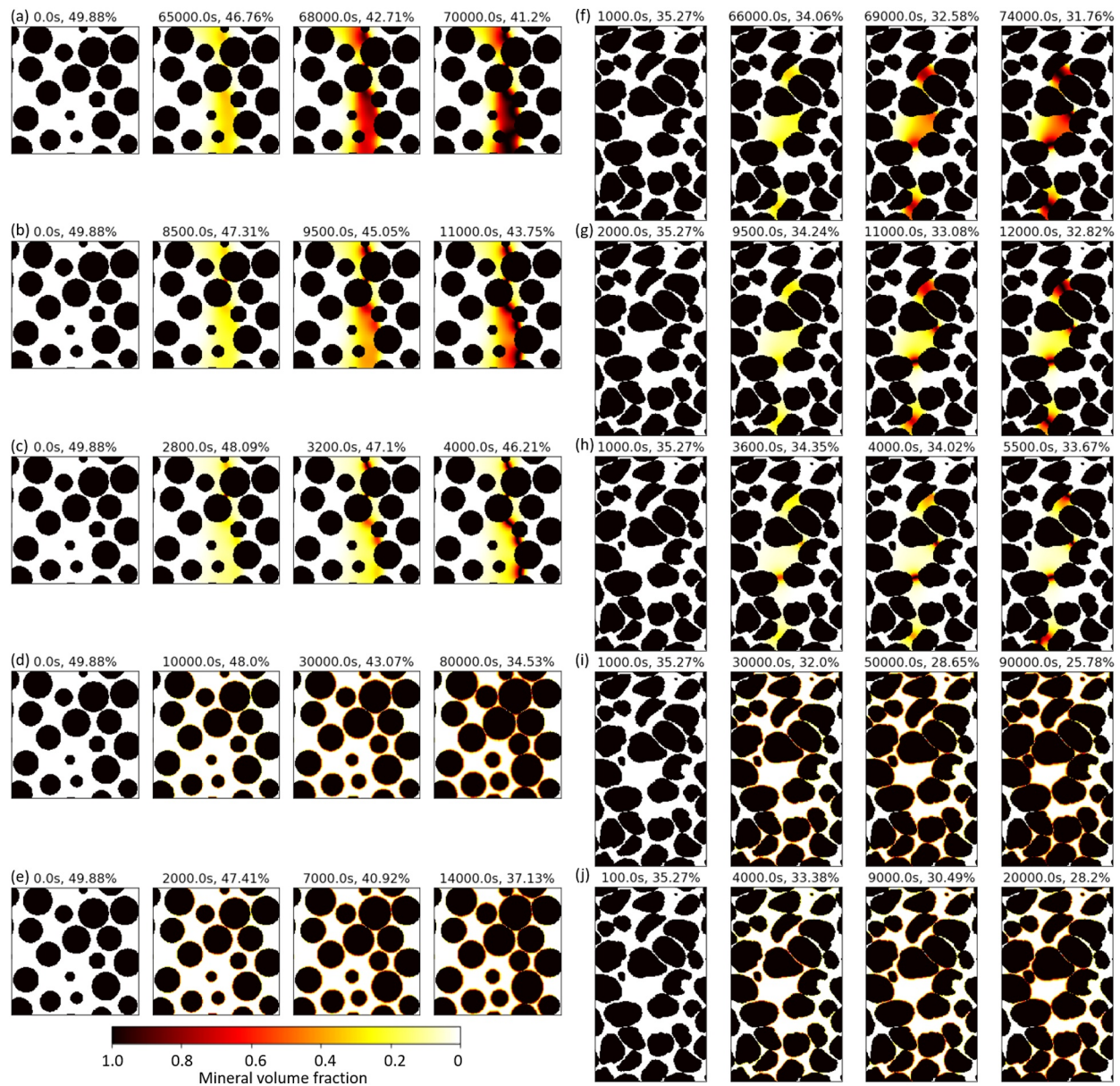


Figure 2. Snapshots of pore geometry evolution for a Finney pack geometry (a)–(e) and a quartz sand pack geometry (f)–(j) for different precipitation setups: (a) (f)—ssa0.44, (b) (g)—ssa0.44_usgs, (c) (h)—ssa44, (d) (i)—pga, (e) (j)—pga_usgs. Note that due to the difference in reaction rate and geometry, each case reached the clogging point at different time steps, and the snapshots shown are thus at different times.

Across different precipitation model setups and initial pore geometries, the domain porosity (Φ) decreased as a result of precipitation and reached a non-zero constant value afterward (Figure 3), indicating that the precipitation reaction became inhibited. As the precipitates blocked the transport pathways, mixing of the two fluids was suppressed, and thus, the precipitation reaction was no longer sustained. Here, the average porosity within the 2D computational domain at the end of the precipitation reaction is referred to as the critical porosity (Φ_c).

Under the surface growth scenario, domain porosity decreased continuously from the beginning of the simulation until the critical porosity was reached. In comparison, in the homogeneous nucleation scenario, the domain porosity remained unchanged initially as the nucleation dominated over the crystal growth pathway. The decrease in Φ became evident once enough surface area had been created by nucleation for the crystal growth to pick up. The critical porosity tended to be lower under the surface growth scenario.

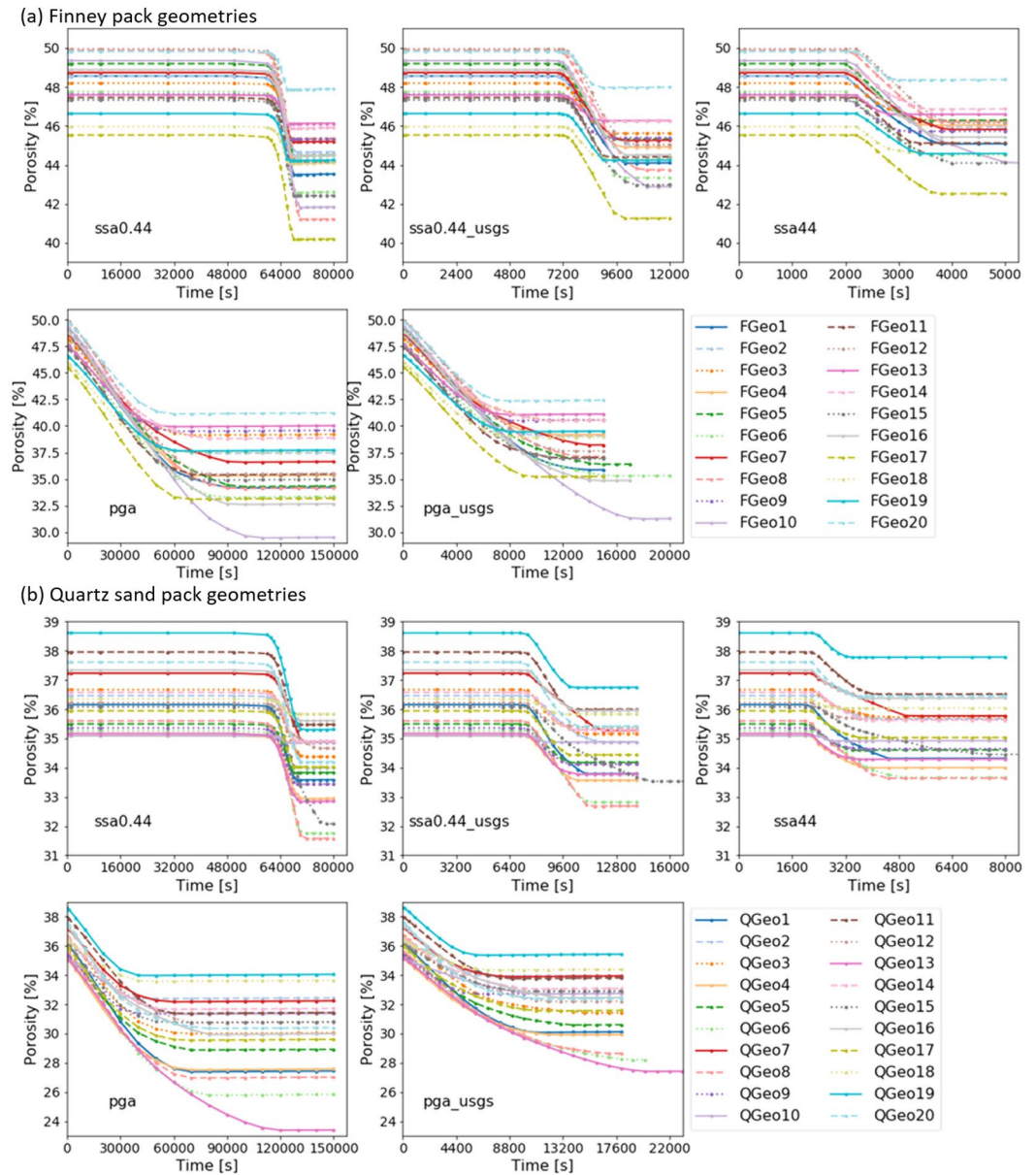


Figure 3. Domain porosity (Φ) evolution for (a) 20 Finney pack geometries (FGeo1-20) and (b) 20 quartz sand pack geometries (QGeo1-20) with different precipitation setups. Note that some cases had longer simulations as they required longer to reach the clogging time point than other cases.

For instance, for the quartz pack geometries, surface growth simulations had critical porosity as low as 24%, whereas the critical porosity of homogeneous nucleation simulations was always above 31%. This is because in the surface growth scenario, precipitation occurred on the grain surfaces outside of the pore throats were less effective in clogging the transport pathway (Figures 2d, 2e, 2i and 2j).

For the same precipitation conceptualization and geometry, a lower critical porosity was observed at a slower reaction rate. For example, in the Finney pack simulations, the critical porosity for the ssa0.44 (specific surface area = $0.44 \text{ m}^2/\text{g}$) case was lower than the ssa44 (specific surface area = $44 \text{ m}^2/\text{g}$) case by 3 percentage points. This is a result of the interplay between precipitation and diffusion, which is typically measured by the Damköhler number (Da). At the lower precipitation rate (lower Da), in grid cells that are over-saturated, less chemicals (Sr^{2+} and SO_4^{2-}) are removed from the solution by precipitation and more are transported downgradient. As a result, more grid cells become oversaturated, and by the time when the transport

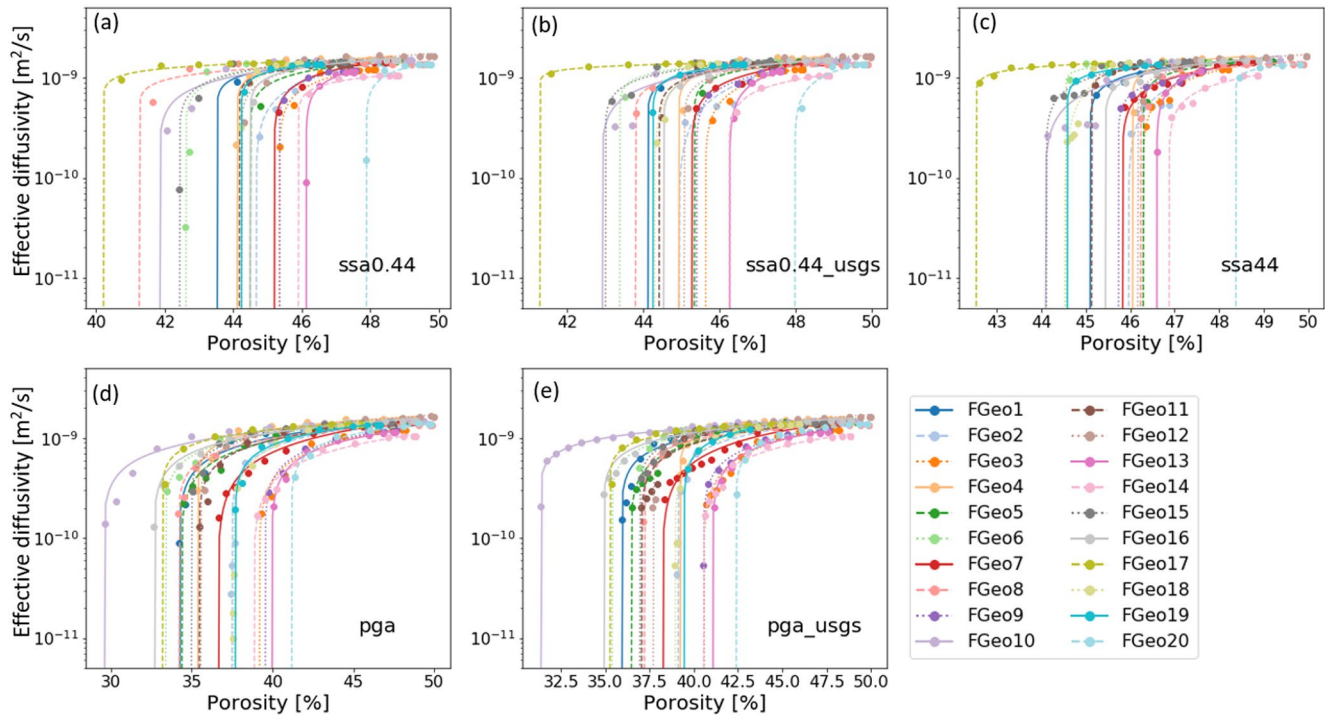


Figure 4. Effective diffusivity in relation to the porosity from the tracer simulations and the fitting line using Equation 11 for the five precipitation kinetic setups of all the Finney pack geometries (FGeo1-20). The goodness-of-fitting R^2 parameter is 0.97 on average, and larger than 0.91 for all 100 cases except for two.

paths become clogged, more precipitation would have happened. As shown in Figure 2, the precipitation band formed in the homogeneous nucleation scenario was thicker at a slower reaction rate, and the growth of the grain surfaces was observable in a wider region in the surface growth cases.

The overall reduction in the domain porosity varied between less than 1%–12% for the quartz pack geometries and between ~1% and 20% for the Finney pack geometries (Figure 3). The reduction in porosity is positively correlated with the initial porosity in the Finney pack geometries; whereas there is no clear correlation between the porosity reduction and the initial porosity for the quartz pack geometries (Figure S1). This implies that for the idealized hard sphere packing of Finney geometries, initial porosity is a good indicator of the pore structure and connectivity. In contrast, for the quartz pack geometries, there is more spatial heterogeneity in the pore structure, and porosity does not accurately reflect the connectivity of the pore structures. For instance, it was reported that permeability of sandstones from different basins does not vary monotonically with porosity and samples with comparable porosity can have different permeability and thus connectivity (Zou et al., 2012).

3.2. Effective Diffusivity Evolution

The effective diffusivity decreased with the porosity (Figure 4). The initial effective diffusivity ranged between $1.1\text{e}-9$ and $1.6\text{e}-9$ m^2/s in the 20 Finney pack geometries given a molecular diffusion coefficient of $2.24\text{e}-9$ m^2/s for the tracer. The relatively large initial effective diffusivity was indicative of the presence of open transport pathways. As the porosity approached the clogging value, the effective diffusivity decreased significantly. At the critical porosity, the effective diffusivity became effectively zero, as evidenced by the termination of the mixing and precipitation. This was also expected in our simulations, as precipitation was allowed to proceed until local porosity (ϕ) was effectively zero (a non-zero but negligible value was used for numerical reasons). Such clogging has also been observed experimentally. In the experiments of Zhang, Dehoff, et al. (2010), dissolution of the precipitated calcium carbonate was observed at the late stage,

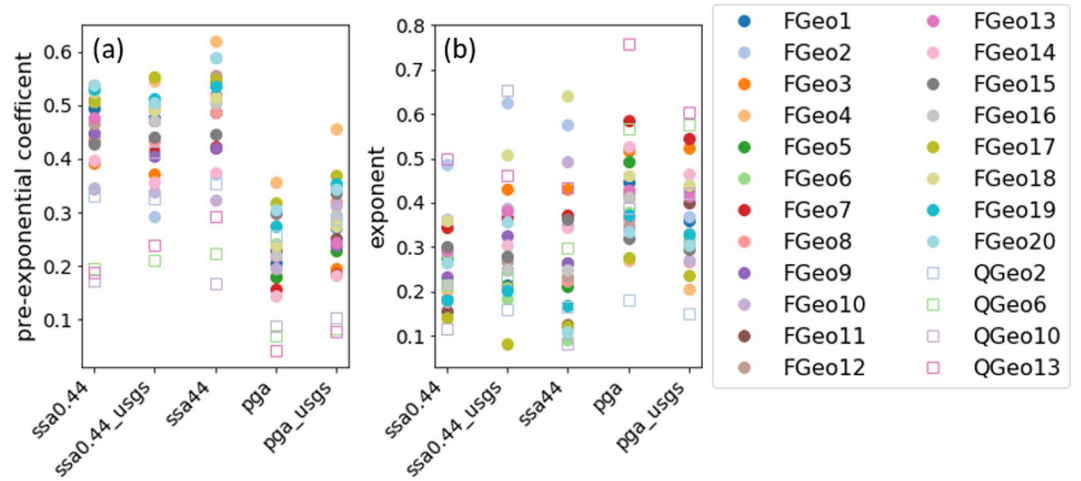


Figure 5. (a) Pre-exponential coefficient and (b) exponent of Equation 11 for all Finney pack geometries (filled circle) and the subset of quartz sand pack geometries simulated (open square) for different precipitation kinetic setups.

confirming negligible transport through the precipitates and thus the termination of mixing and further precipitation as observed in the simulations here.

These trends are consistent with what was observed in Navarre-Sitchler et al. (2009). Therefore, we accounted for the clogging effect of the precipitates on effective diffusivity by adopting the following mathematical formulation:

$$D_{\text{eff}} = D_{\text{effc}} + aD_m(\phi - \phi_c)^n, \quad (11)$$

where a and n are empirical coefficients. Theoretically, the value of D_{effc} can be determined by running the numerical tracer experiment on the pore structure at the critical porosity. Here, because $D_m\phi \rightarrow 0$ when $\phi \rightarrow 0$ in the pore space that is clogged by the precipitates, $D_{\text{effc}} \rightarrow 0$, as also confirmed by the numerical tracer experiments, which showed no break-through of the tracer after 10 million seconds. Therefore, a value of $1\text{e}-15\text{ m}^2/\text{s}$ was assumed for D_{effc} across all geometries and precipitation scenarios. This value is on the same order of magnitude as the value used for unweathered basalt rocks in Navarre-Sitchler et al. (2009) and provides a reasonable estimate for low diffusivity systems that can be encountered. Using a value of 0 or $1\text{e}-15\text{ m}^2/\text{s}$ for D_{effc} did not affect the evaluation of the parameters a and n . Because each simulation evolves differently and it was impossible to predict the clogging time a priori and thus set the output time steps accordingly, data points are sparse in the region where the diffusivity drops dramatically within a small porosity change and a relatively short time period.

For the Finney pack geometries (Figure 5), the pre-exponential coefficients (a) for the two surface growth cases (i.e., “pga” and “pga_usgs”) were 0.24 ± 0.06 (mean $\pm$ standard deviation) and 0.29 ± 0.07 , respectively, and were smaller than those for the three homogeneous nucleation cases (i.e., “ssa0.44”, “ssa0.44_usgs”, and “ssa44”), which were 0.46 ± 0.06 , 0.45 ± 0.07 , and 0.49 ± 0.08 , respectively. Under the same precipitation scenario, the reaction rate only had a minor impact on the pre-exponential coefficient. The exponent (n) in the surface growth scenario, which was 0.41 ± 0.08 and 0.36 ± 0.09 for the two cases, was slightly larger than in the homogeneous nucleation scenario. For the three homogeneous nucleation cases, n showed a larger variation across different geometries, with values of 0.26 ± 0.09 , 0.30 ± 0.13 , and 0.29 ± 0.16 . For FGeo8, simulations were performed with larger domain sizes for “ssa0.44_usgs” and “pga_usgs”. The simulations showed similar evolution in the pore structures. The pre-exponential coefficient and exponent values were within the ranges observed for different geometries. More information is provided in the supporting information.

Diffusion simulations of the quartz pack geometries resulted in similar observations (Figure 5). The pre-exponential coefficient was lower than that of the Finney pack geometries. It was 0.22 ± 0.07 , 0.3 ± 0.09 , 0.26 ± 0.08 for “ssa0.44”, “ssa0.44_usgs”, and “ssa44”, respectively, which was larger than the two surface growth

cases (0.11 ± 0.1 and 0.14 ± 0.1). The exponent values were 0.21 ± 0.2 , 0.38 ± 0.22 , 0.24 ± 0.15 for the three homogeneous nucleation cases, and were comparable to the average values of the Finney pack geometries. The exponent for the two surface growth cases were on the higher end of the range observed for the Finney pack geometries, with values of 0.47 ± 0.25 and 0.44 ± 0.21 .

Everything else being equal, a smaller pre-exponential coefficient leads to a lower effective diffusivity. This is consistent with the observation that the pre-exponential coefficients of the quartz pack geometries, for which the initial diffusivity ranges between 3.2 and $7.2 \times 10^{-10} \text{ m}^2/\text{s}$, are systematically lower than those of the Finney pack geometries, for which the initial diffusivities are 1.1 – $1.6 \times 10^{-9} \text{ m}^2/\text{s}$. A weak positive correlation was observed between the pre-exponential coefficient and the initial diffusivity (supporting information). The exponent controls the shape of the curve described by Equation 11. A small exponent corresponds to a limited reduction in diffusivity at early stages of porosity reduction and a sharp decrease toward the critical porosity. For instance, in the homogeneous nucleation scenario (for which the exponent tends to be lower), the initial development of the precipitation bands did not necessarily translate to transport pathway clogging and thus reduction in the diffusivity. In contrast, in the surface growth scenario, precipitation resulted in grain dilation very early on, the pore throats and thus diffusivity decreased continuously from the beginning of the simulations, corresponding to higher exponents. The trajectory of the diffusivity evolution depends on not only the precipitation mechanism but also the pore structures. This is why large standard deviations in the exponent were observed among different geometries. However, no clear correlation was observed between the exponent and the initial porosity or diffusivity in our simulations. Additional pore structure characteristics need to be examined in the future.

4. Discussion

Our work advances our understanding and predictive capability of precipitation driven alteration of porous media in two ways. The modeling workflow presented enables the prediction of precipitation as a result of the chemical gradients developed from pore structure heterogeneity and various chemical interactions, in comparison with assumed uniform precipitation or other prescribed precipitation patterns. It therefore gives a more truthful representation of the pore-scale dynamics. In addition, while considerable variations were observed in the precipitation patterns among different geometries and precipitation kinetic setups, the macroscopic $\Phi - D_{\text{eff}}$ relationship was fairly consistent across all the simulations. This observation confirms the possibility of using this mathematical relationship for upscaling.

Our study is nevertheless based on 2D simulations (given the prohibitively high computational cost for hundreds of full 3D simulations) of pore structures with fairly large pore sizes, whereas real systems involve three dimensional processes and smaller pore sizes, especially ones in which diffusion is the dominant transport mechanism. Regardless of these differences, we expect the major observations of our study to hold in a more general sense. For instance, in the experimental study of (Rajyaguru et al., 2019), a chalk was used as a proxy material to examine diffusion-reaction dynamics in low-permeability systems. The sample used in the experiment had a porosity of 45%, comparable to the ones used in our study. Although the median pore throat size was $\sim 0.66 \text{ }\mu\text{m}$, one order of magnitude lower than the resolution of our model, a thin precipitation disk was observed due to homogeneous nucleation of barite, similar to the precipitation band predicted in our simulations. More importantly, the experimental study also observed that the same amount of precipitates resulted in a larger reduction of the diffusivity when homogeneous nucleation was dominant. This was in contrast to when heterogeneous nucleation on the substrate (similar to the surface growth scenario in our simulations) was dominant, consistent with our observation that homogeneous nucleation results in higher Φ_c and lower porosity reduction. The traditional Archie's law has been extensively evaluated experimentally and numerically, and the same formulation has been recovered for 2D and 3D domains. But the exponent of Archie's law was found to be lower in 2D, approximately equal to 1, than in 3D, approximately equal to 2, for various 3D natural media (Hunt, 2004). Following these observations, we expect that Equation 11 is broadly applicable, but the coefficients may be different in 3D due to the different number of degrees of freedom or different level of disorder, in addition to varying with the precipitation scenario and pore geometries as shown in our simulations. In this section, we discuss some of the assumptions and limitations, with the objective of increasing clarity of the applicability of our results and highlighting some future research needs.

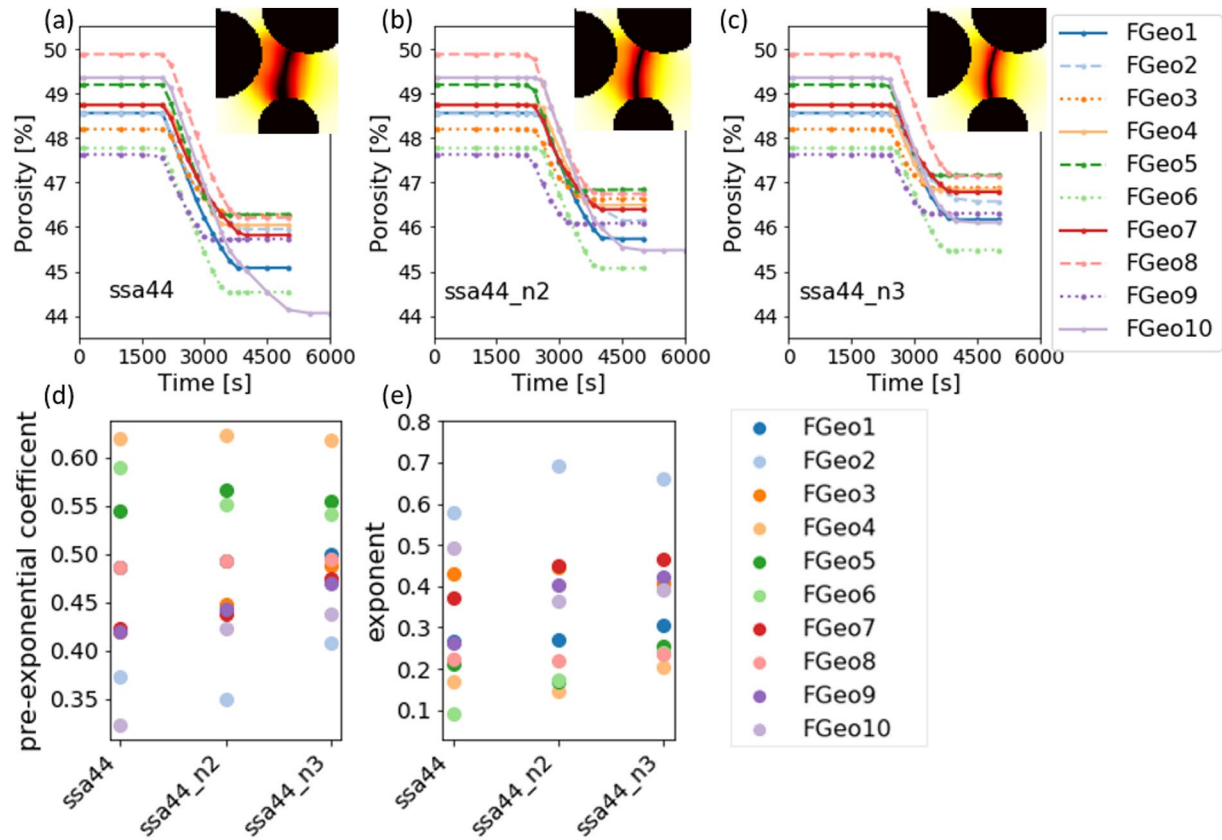


Figure 6. Comparison of the porosity evolution of 10 Finney pack geometries using the default linear grid-cell level tortuosity model (a), the nonlinear tortuosity model with exponent of 2 (b) and 3 (c), and the pre-exponential coefficient (d) and exponent (e) of the $\Phi - D_{\text{eff}}$ relationship of the three cases.

4.1. Impacts of Mineral Precipitation Mechanisms and Texture

Our simulations accounted for the impacts of pore geometry and precipitation kinetics of homogeneous nucleation and surface growth but did not resolve single crystals or the textures of the precipitates that can vary with nucleation and crystal growth mechanisms (Chen et al., 2014; Fazeli et al., 2020).

Additional simulations for 10 Finney pack geometries and the “ssa44” case were performed to evaluate how the texture of and thus tortuosity in the developing precipitates affect the macroscopic porosity-diffusivity evolution. These simulations used the non-linear relationships, $D_m \phi^2$ and $D_m \phi^3$, at the grid cell level, and showed the same trend as the linear tortuosity model simulations (Figure 6). However, for the higher exponents, the critical porosity (Φ_c) was larger, that is, the porosity reduction was smaller. Across the 10 geometries simulated, the reduction was 1.6%–3.9% and 1.3%–3.3% for the exponent of 2 and 3, respectively; compared to 1.9%–5.3% in the linear tortuosity model simulations. This is consistent with the conceptualization that a larger exponent results in increased loss of connectivity for a given amount of porosity reduction. Accordingly, the transport pathways are clogged at a faster speed even though the precipitation rate is the same, as illustrated by the thinner precipitation bands (Figures 6a–6c). The non-linear tortuosity model did not affect the pre-exponential coefficient of the $\Phi - D_{\text{eff}}$ relationship noticeably, which was 0.48 ± 0.08 and 0.50 ± 0.06 for the exponent of 2 and 3, respectively, or the exponent (n), which was 0.33 ± 0.17 and 0.36 ± 0.14 for the exponent of 2 and 3, respectively.

Another textural feature that may affect the macroscopic $\Phi - D_{\text{eff}}$ relationship is the “microporosity”, that is, nanopores that remain in the precipitates due to reduced nucleation and precipitation rate under confinement. It has been reported that nucleation and precipitation in nanopores are prohibited or significantly reduced (Stack et al., 2014) because of the pore-size controlled solubility (PCS) effect (Emmanuel & Ague, 2009) or local transport limitations. If the PCS effect is dominant, and the diffusive

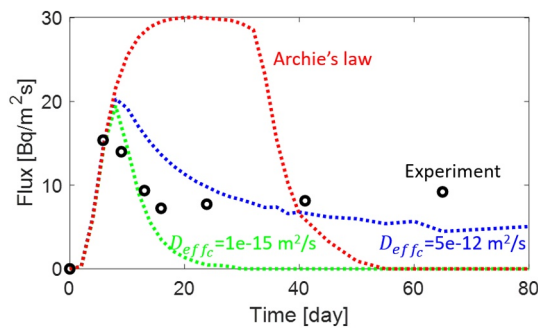


Figure 7. Tracer flux at the outlet. The experimental observations are shown in the open black circles, in comparison with simulation results using the traditional form of Archie's law (red dotted line) and using the revised relationship with $D_{\text{effc}} = 1\text{e-}15\text{ m}^2/\text{s}$ (green dotted line) and $D_{\text{effc}} = 5\text{e-}12\text{ m}^2/\text{s}$ (blue dotted line). The traditional Archie's law is given by $D_{\text{eff}} = 1.3149\phi^2$, the exponent is the same as the revised relationship based on literature data, and the pre-exponential factor is determined based on the initial porosity and diffusivity reported in the experimental study.

the column experiments. The 1D continuum model solves Equations 2, 4, 6, and 8 with equal discretization of 0.1 mm. Each grid cell is a combination of mineral grains and pore space with an initial porosity (Φ_{ini}) of 39% according to the CT image data. The precipitation rate follows the transition state theory rate law seeded with a tiny surface area initially. More details are documented in the supporting information. For the four parameters in Equation 11, the critical porosity (Φ_c) was set at 31% based on the post-reaction CT images. The exponent was set to be 2 according to prior studies of 3D natural rocks (Hunt, 2004; Navarre-Sitchler et al., 2009). The pre-exponential factor was constrained by the initial effective diffusivity (D_{effini}) of the column, that is, $4.48\text{e-}10\text{ m}^2/\text{s}$:

$$a(\Phi_{\text{ini}} - \Phi_c)^n = D_{\text{effini}}. \quad (12)$$

There was no direct measurement of D_{effc} , and thus two values $1\text{e-}15\text{ m}^2/\text{s}$ and $5\text{e-}12\text{ m}^2/\text{s}$ were tested. The simulation results were compared with the effluent tracer flux and the amount of precipitates from the experiment.

In the experiment, tritiated water (HTO) was used as the tracer. The effluent flux peaked at $\sim 15\text{ Bq/m}^2\text{s}$ around day 6 and stayed around $8\text{ Bq/m}^2\text{s}$ after day 15. The precipitates were observed in a $\sim 6\text{ mm}$ zone and were largely located in a 0.35 mm thick disk, and amounted to $20\text{--}35\text{ mg}$ in different parallel experiments. Compared to the simple form of Archie's law, the revised relationship provides a better agreement with the experimental observations (Figure 7 and Table S1). The case of $D_{\text{effc}} = 1\text{e-}15\text{ m}^2/\text{s}$ implicitly assumes that diffusion limitation is the only mechanism for precipitation termination as in the pore scale simulations. In this simulation, the precipitates were present in a $\sim 7\text{ mm}$ zone and amounted to 20 mg by day 80. However, the flux went to zero after about 20 days. A D_{effc} of $5\text{e-}12\text{ m}^2/\text{s}$ effectively assumes the microporosity scenario due to PCS effect as discussed above. The presence of microporosity was implied by the variable gray values in the CT images collected in the experimental study (Chagneau et al., 2015), even though the individual pores constitute the microporosity were not resolved given the resolution of the images. In this case, the flux remained non-zero after the initial decrease and agreed well with the experimental observations. As the suppression of precipitation in microporosity was not explicitly modeled, the simulation adopts a simple treatment of setting the precipitation rate to zero when Φ_c was approached. The total amount of precipitates was $\sim 113\text{ mg}$ within a $\sim 9.5\text{ mm}$ zone. Both are larger than the low D_{effc} case given the higher diffusive flux and mixing. This over-estimation in precipitation may be attributed to sample heterogeneity along the column that was not considered, and that different precipitation mechanisms may be present in microporosity, that is, nano-pores.

transport through the nanopores is non-negligible, the mixing region and the precipitation bands will continue to grow until precipitation becomes undetectable, leading to phenomenological plugging of the system. The corresponding Φ_c is expected to be lower than what would be expected from our pore-scale simulations, whereas the corresponding D_{effc} will be constrained by the tortuosity in the precipitates and the thickness of the precipitation band, and is likely to be larger. In contrast, if the predominant mechanism for the remaining nanopores is local transport limitation, we may expect slight increase in Φ_c , and D_{effc} is by definition close to zero. However, in order to quantitatively evaluate the impacts of microporosity, information on the amount and tortuosity of the remaining nanopores is required. Thus, future studies on mechanistic understanding of various nucleation and crystal growth processes that contribute to the creation of microporosity and proper conceptualization of these processes in pore-scale models are needed.

4.2. Application of the Revised Relationship

The revised formulation (Equation 11) was implemented in 1D continuum simulations similar to Chagneau et al. (2015), for comparison with

5. Conclusions

In this study, we performed numerical investigation of pore-scale precipitation dynamics and their impacts on effective diffusivity (D_{eff}). The modeling results reproduced precipitation patterns that were observed in previous experiments. A precipitation band formed in the scenario of homogeneous nucleation, whereas the precipitates coated the grain surfaces in the surface growth scenario. For all geometries and precipitation rate setups simulated, the progress of the precipitation reaction led to the decrease of domain porosity (Φ) and D_{eff} until transport in the system was completely clogged. The surface growth scenario required more precipitation than the homogeneous nucleation scenario to clog the system and was largely influenced by the initial pore geometry. At a higher precipitation rate, the system was clogged with a smaller amount of the precipitate, that is, the critical porosity (Φ_c) was higher. Based on these observations, the $\Phi - D_{\text{eff}}$ relationship (traditional Archie's law) was revised to explicitly account for the clogging effect, that is, Φ_c and the critical effective diffusivity (D_{effc}) were included in the relationship. The two fitting parameters, the pre-exponential coefficient, and the exponent were between 0–1 for all simulations and showed large variations across different geometries. For the surface growth scenario, the exponent was slightly higher than for the homogeneous nucleation scenario. Application of this revised relationship in the 1D continuum scale model produced improved results that agree better with the column experiment compared to the traditional Archie's law. The amount of precipitation is still over-estimated, which indicates that other precipitation mechanisms may be present that reduced mineral precipitation. Future studies are needed to better understand processes such as precipitation under confinement and to conceptualize these processes for implementation in reactive transport models.

Data Availability Statement

All geometry data can be accessed through the digital rock portal (<https://www.digitalrockportal.org/>). The fluid chemistry and thermodynamic and kinetic data are available in the cited references: Chagneau et al. (2015), Marty et al. (2015), Nielsen & Söhnel, (1971), Palandri (2004), and Poonosamy et al. (2019). The code of CrunchTope is available at <https://bitbucket.org/crunchflow/crunchtope-dev>. Example input files and database for the CrunchTope simulation can be accessed at <https://github.com/denghangPU/diffusion-dominated-pore-scale-precipitation.git>.

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Key Points:

- Three-dimensional simulations of reactive transport processes using the micro-continuum approach based on a fractured dolostone experiment
- The altered layer (AL) as a diffusion barrier limits reactions of the fast-reacting mineral and thus the matrix alteration
- Advection in the AL affects spatial patterns of mineral dissolution that controls subsequent AL development and fracture enlargement

Supporting Information:

Supporting Information may be found in the online version of this article.

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Investigation of Coupled Processes in Fractures and the Bordering Matrix via a Micro-Continuum Reactive Transport Model

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Abstract In multi-mineral fractured rocks, the altered porous layer on the fracture surface resulting from preferential dissolution of the fast-reacting minerals can have profound impacts on subsequent chemical-physical alteration of the fractures. This study adopts the micro-continuum approach to provide further understanding of reactive transport processes in the altered layer (AL), and mass exchanges with the bordering matrix and fracture. The modeling framework couples the Darcy-Brinkman-Stokes (DBS) solver in COMSOL Multiphysics and the geochemical modeling capability of CrunchFlow. Three-dimensional steady state simulations with systematically varied chemical-physical parameters of the AL were performed to examine the impacts of individual factors and processes. Our simulation results confirm previous observations that dissolution of the fast-reacting mineral (i.e., calcite) is largely controlled by diffusion across the AL. We also show that dissolution of the slow-reacting mineral (i.e., dolomite), which controls AL development and fracture enlargement, increases with surface area and has a complex dependence on different local rate-limiting processes. In particular, advection can result in evident spatial variations in the local dissolution rates of dolomite, although it does not affect the bulk chemistry significantly. The difference in the spatial patterns between simulations with and without advection in the AL is more noticeable in the locations with smaller apertures, with up to 20% difference in local reaction rates. Therefore, it is important to include a full depiction of advection, diffusion, and reactions for accurately capturing local dynamics that control long-term fracture evolution.

1. Introduction

Fractures act as preferential flow pathways that govern fluid migration and solute transport in many natural and engineered systems, and are subject to alteration driven by geochemical reactions (Berkowitz et al., 2016; Deng & Spycher, 2019; Steefel, Beckingham, & Landrot, 2015). Large contrasts in morphological and mineralogical properties between fractures and bordering rock media can result in non-linear feedback between reactions, fluid flow, and transport (Molins et al., 2017; Schädle et al., 2019). Improved fundamental understanding of these coupled processes in fractured rocks is critical for accurate assessment of the long-term performance of different energy and environmental systems (Mehmani et al., 2020; Noiriél & Daval, 2017).

Experimental observations and numerical simulations have explored the alteration of fracture morphology and thus hydrophysical properties induced by changes in fluid composition, temperature, and pressure (Al-Khulaifi et al., 2017; Davila et al., 2016; Deng & Spycher, 2019). Dimensionless numbers, including Péclet and Damköhler numbers, have been used to investigate and predict dissolution behaviors in different flow regimes (Brunet et al., 2016; Detwiler et al., 2003; Szymczak & Ladd, 2009). These studies, however, typically focus on single-mineral systems in which mineral dissolution/precipitation mainly occur at the fracture surface, whereas mineral heterogeneity that is widely observed can lead to complex alteration in the properties of the bordering rock matrix (Upadhyay et al., 2015). In multi-mineral systems with varied reactivities, preferential dissolution of the fast-reacting minerals results in the development of an altered porous layer bordering the fracture, which affects the subsequent reactions (Laubach et al., 2019; Wen et al., 2016), and thus the evolution of fracture morphology and hydraulic conductivity (Ellis et al., 2011; Noiriél et al., 2013).

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Continuum scale models have been developed to demonstrate and quantify the role of the altered layer (AL) as a diffusion barrier, which limits mass transfer between the fracture and rock matrix (Deng et al., 2016; Deng, Molins, et al., 2018; Noiriél & Daval, 2017). While these modeling studies have enabled predictions of AL development and their impacts on reactions using simplified treatments, they are limited to specific conditions and they only partially represent the coupled (flow-)transport-reaction processes in the AL. For instance, while the effective diffusion coefficient that was introduced to describe the transport limitation on reaction rate of the fast-reacting minerals (Deng et al., 2016) can potentially be adjusted to account for the effects of advection in the AL, it would be purely empirical unless physically based understanding of such effects is taken into consideration (Scheibe et al., 2015). In this regard, the relative importance of different processes within the AL in controlling the overall fracture evolution is still an open question.

Recent advances in microscopic characterization techniques and computational power have enabled considerations of reactive transport processes in heterogeneous fractured porous media at the pore-scale (Deng et al., 2020; Molins, 2015; Singh et al., 2017). In pore-scale models, such as the mesh based pore-scale reactive transport model (Bultreys et al., 2016; Molins et al., 2014; Starchenko & Ladd, 2018), lattice Boltzmann method (LBM; Curti et al., 2019; Kang et al., 2010; Prasianakis et al., 2017), and smooth particle hydrodynamics (SPH; Tartakovsky et al., 2007, 2016), the interfaces between fluid and solid phases are explicitly resolved and the flow field reflects fine-scale morphology. Chemical gradients arise from the interplay between local advective transport, diffusion, and chemical reactions can be evaluated to reveal surfaces versus transport limitations on reactivity (Deng, Steefel, Molins, & DePaolo, 2018; Fazeli et al., 2019). As such, pore-scale models can be a valuable tool to connect microscopic geometric heterogeneity and the macroscopic properties (e.g., permeability and reaction rate; Molins & Knabner, 2019; Starchenko & Ladd, 2018).

However, the morphological and mineralogical heterogeneity in the fracture and the rock matrix can vary across a wide range of scales, resulting in a scale dependence of coupled processes and parameters (Al-Khulaifi et al., 2019; Jung & Navarre-Sitchler, 2018). In addition to the coupling of multiple processes, the controlling processes with different characteristic length scales and their interactions need to be identified and accounted for. Pore-scale simulations have been performed in two-dimensional regions or in a synthetic binary mineral system to investigate mass transfer between fractures and altering rock matrix, they, however, did not consider the dissolution of slow-reacting minerals (Chen et al., 2014; Fazeli et al., 2019). The application of pore-scale models in three-dimensional (3D) cm-scale experimental or natural systems is even more rare.

In light of the scale differences and computational challenges of the pore-scale approach, reactive transport modeling in fractured media can be conceptualized as a multiscale problem that couples a pore-scale component in the fracture, and a Darcy-scale component in the porous matrix (Molins et al., 2019), which is commonly referred to as multiscale modeling (Molins & Knabner, 2019; Yang et al., 2021). One approach of multiscale modeling involves the use of hybrid algorithms, which divide the domain into subdomains according to their respective characteristic length scales. Different governing equations are solved for each subdomain and the subdomains are coupled by enforcing mass and momentum continuity at the interface (Mehmani & Balhoff, 2015; Tartakovsky et al., 2008; Tomin & Lunati, 2013; Yousefzadeh & Battiato, 2017). However, implementation of different discretization across the interface can be complicated, especially when the interface moves as a result of reactions, as different resolutions are required to capture features in each individual subdomain (Molins & Knabner, 2019).

Another multiscale modeling approach involves the implementation of the Darcy-Brinkman-Stokes (DBS) equation in a single micro-continuum domain (Carrillo et al., 2020; Gulbransen et al., 2010; Soulaïne et al., 2017; Soulaïne & Tchelepi, 2016). The DBS equation solves for the Stokes (or Navier-Stokes) flow and Darcy flow simultaneously and has been used across spatial scales under changing flow regimes (Brinkman, 1947; Neale & Nader, 1974). Reactive transport models using the DBS approach have been developed for various applications. Golfier et al. (2002) applied a DBS-based model at Darcy-scale to capture the wormhole development of different dissolution regimes. Soulaïne and Tchelepi (2016) proposed a DBS micro-continuum model for pore-scale simulations of evolving porous media. Although the fluid-solid interfaces are not explicitly resolved as in the pore-scale models, the micro-continuum approach captured pore structure evolution and has the advantage of accommodating complex geometries without requiring extremely high resolution or adding significant computational costs. Compared to the hybrid methods, the micro-continuum approach unites the mathematical description of features at different scales using one set of governing equations and enables computationally efficient representation of heterogeneity and all relevant processes at both pore- and continuum scale. It is especially useful

to capture the reactive transport processes through porous regions and solid-free regions (Carrillo et al., 2020; Molins et al., 2021). In spite of these advantages, this approach has not been applied to simulate reactive transport in fractured media, a prime example of multiscale media in that there is a sharp contrast—with a clear separation of scales—between the porosity in the fracture opening and in the rock matrix (Deng & Spycher, 2019).

Our study focuses on reactive transport processes in fractured media, particularly in the porous AL bordering the fracture and their impacts on fractured alteration. For this purpose, we perform 3D multiscale simulations of a fractured porous medium with a spatially heterogeneous AL bordering the fracture using the micro-continuum approach, including a full representation of advection, diffusion, and reactions (dissolution of both the fast-reacting mineral and the slow-reacting mineral) in the entire domain. This is enabled by a framework that couples a multiphysics code (COMSOL Multiphysics, COMSOL) with a reactive transport code (CrunchFlow). The mathematical model and numerical implementation of this coupling framework are described in Section 2. In Section 3, we present the setup of the 3D simulations, and we report the results and discuss the impacts of different physical-chemical parameters of the AL on the reactive transport dynamics in the fractured multiscale domain in Section 4. We conclude in Section 5.

2. Materials and Methods

2.1. Governing Equations

The micro-continuum approach is derived by volume averaging governing equations over each control volume (V ; Beavers & Joseph, 1967; Beckermann & Viskanta, 1988) instead of a full pore-scale representation. The control volumes may contain both a solid and a fluid phase or one of them. The volume fraction of fluid (ϵ_f) is 0 and 1 in the cells containing only solid or fluid, respectively. Cells with intermediate values ($0 < \epsilon_f < 1$) are identified as the fluid/solid interface when bordering the fluid cells, and are otherwise part of the porous matrix. In each cell or control volume, all the physical variables are volume averages.

2.1.1. Flow

Flow in a medium composed of porous and non-porous regions can be described by the single field momentum conservation equation arising from the integration of the Navier-Stokes (N-S) momentum equation (Bennon & Incropera, 1987; Bousquet-Melou et al., 2002; Brinkman, 1947; Ochoa-Tapia & Whitaker, 1995; Quintard & Whitaker, 1999):

$$\frac{1}{\epsilon_f} \left(\rho_f \frac{\partial \bar{\mathbf{u}}_f}{\partial t} + \rho_f \bar{\mathbf{u}}_f \cdot \nabla \left(\frac{\bar{\mathbf{u}}_f}{\epsilon_f} \right) \right) = -\nabla \bar{p}_f + \frac{\mu_f}{\epsilon_f} \nabla^2 \bar{\mathbf{u}}_f - \frac{\mu_f}{\mathbf{k}} \bar{\mathbf{u}}_f \quad (1)$$

where $\mathbf{k}$ is the permeability tensor of the medium, and the velocity is defined as

$$\bar{\mathbf{u}}_f = \frac{1}{V} \int_{V_f} \mathbf{u}_f dV, \quad (2)$$

physical properties of the fluid, that is, fluid density (ρ_f) and fluid viscosity (μ_f) and the pressure field are defined as intrinsic volume averages that correspond to the volume occupied by the fluid (V_f).

The left-hand side in Equation 1 describes the inertia, and the right-hand side terms represent the pressure gradient, the diffusive viscous contribution, the momentum exchange, respectively. The momentum exchange subject to the mutual friction between fluids and solids vanishes in solid-free regions (large k values) while tends to be dominant in porous regions where permeability is low. As such, this equation enables us to recover the NS flow field in the pore space when local porosity and permeability fields in the solid phase are set to arbitrarily small values (Molins et al., 2021). Moreover, Soulaïne and Tchelepi (2016) demonstrated that no-slip boundary conditions at the solid/fluid interfaces can be ensured when k within the mineral is at least four orders of magnitude below what is in the fluid zone.

2.1.2. Reactive Solute Transport

Reactive transport is solved by the averaged multicomponent advection-diffusion-reaction equation:

$$\frac{\partial(\epsilon_f \bar{\psi}_i)}{\partial t} = -\nabla \cdot (\bar{\mathbf{u}}_f \epsilon_f \bar{\psi}_i) + \nabla \cdot (\epsilon_f \mathbf{D}_i \nabla \bar{\psi}_i) + R_i \quad (3)$$

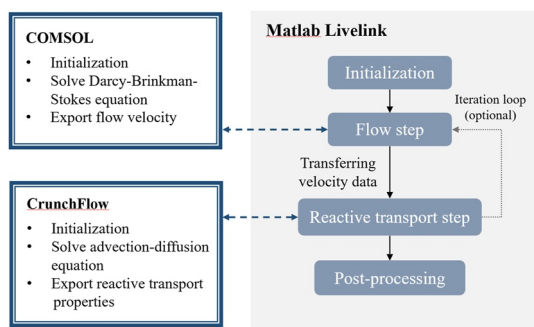


Figure 1. Schematic diagram of the structure of the COMSOL-CrunchFlow framework.

where $\bar{\psi}_i$ is the molal total concentration of a component, $\mathbf{D}_i$ is the combined dispersion-diffusion tensor, and R_i is the changes in the concentration of the species due to mineral reactions. For a multi-mineral system, $R_i = \sum_s v_{i,s} r_s$, where $v_{i,s}$ is the stoichiometric coefficient of reaction s , and the mineral reaction rate r_s for a given mineral is given by the transition state theory rate law (Lasaga, 2018):

$$r_s = -A_s \left\{ \sum_{l=1}^{N_r} k_l \left(\prod_{i=1}^{N_c+N_x} a_i^{p_{il}} \right) \right\} \left[1 - \left(\frac{Q_s}{K_s} \right)^M \right]^n, \quad (4)$$

where A_s refers to the surface area of individual minerals, k_l is the far from equilibrium dissolution rate constant for the l th parallel reaction, Q_s is the ion activity product, and K_s is the equilibrium constant. The term $\prod_{i=1}^{N_c+N_x} a_i^{p_{il}}$ incorporates the effects of catalytic/inhibitory ions in the solution. The exponents n and M allow for non-linear dependencies on the saturation state and are typically fitted from experimental studies. Both are one in our simulations.

2.2. Numerical Implementation

This framework is implemented by coupling the COMSOL, that is, a multi-physics and multi-purpose software, with CrunchFlow, that is, a software package with comprehensive geochemical modeling capability (Steefel, Appelo, et al., 2015; Steefel & Lasaga, 1994). The DBS equation for single phase flow (Equation 1) is solved in COMSOL, and the advection-diffusion-reaction equation is solved in CrunchFlow for the transport of the reactive species and the resulting evolution of pore structure when geometry update is considered.

The program structure of this coupling framework is illustrated in Figure 1. At the beginning of the simulation, domain geometry, discretization, properties, boundaries are initialized in COMSOL and CrunchFlow separately. After the initialization procedure, Matlab LiveLink is used as the application interface to step through COMSOL and CrunchFlow for the calculation of flow and reactive transport, respectively. All data, that is, fluid chemistry, porosity, and permeability, are stored and updated in CrunchFlow. The flow model in COMSOL discretizes the domain into an unstructured mesh, and computed using the finite element method. The velocity field is then mapped to the Cartesian mesh using the module of data export in COMSOL and transferred into CrunchFlow. The mesh resolution is presented later in the simulation setup section. In this study, we focus on steady state simulations, and thus no iteration between COMSOL and CrunchFlow is needed. But the framework can be used for cases with geometry evolution by adding the iteration loops. Both COMSOL and CrunchFlow have been extensively validated against known solutions and benchmark problems (Azad et al., 2016; Molins et al., 2021; Nardi et al., 2014).

3. Setup of the Multiscale Model in 3D Fractured Dolostone Experiment

A 3D multi-scale model is built using the coupled COMSOL-CrunchFlow framework based on a previous fracture-flow experiment to investigate the reactive transport processes in a fractured medium with complex geometries and mineralogy. In the experiment, CO_2 -saturated brine was injected into a fractured Duperow Dolostone, which resulted in the formation of a preferential channel and the development of an AL (Ajo-Franklin et al., 2018). This experiment has served as a test case for several modeling studies, and has been simulated by Deng et al. (2016) using a reduced-dimension Darcy-scale model and by Molins et al. (2019) using a hybrid reactive transport model in which flow and reactions were considered in the fracture domain and diffusion and reactions were included in the matrix domain. Here, similar to Molins et al. (2019), we simulate the steady state of a time snapshot from the experiment, but with full coupling of flow, transport, and reactions in both the fractures and the altered/unreacted matrix.

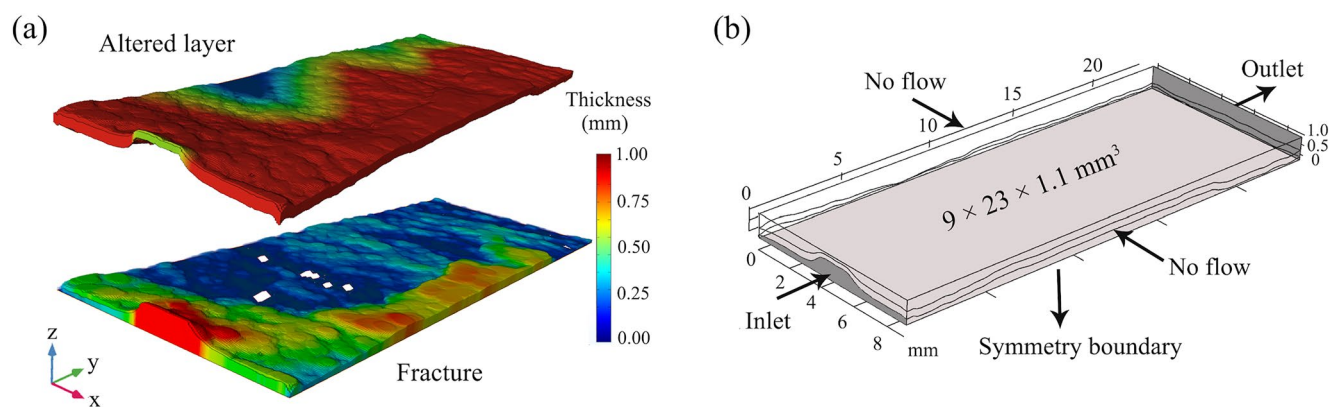


Figure 2. The (a) 3D domains and (b) boundary conditions for multi-scale models. The altered layer and the fracture were reconstructed from the thickness and aperture maps, respectively (fracture contact points are displayed in white). The domain bottom, that is, the central plane of the fracture was set as a symmetry boundary.

3.1. Multi-Scale Simulation Setup

The 3D domain is reconstructed from the fracture aperture and AL thickness maps at 80 hr from Deng et al. (2016). Assuming the core is symmetric with respect to the fracture plane, only half of the domain (Figure 2) was used for simulation. The aperture of the fracture ranges from 0 to 0.805 mm, and the thickness of the AL ranges from 0.028 to 1.009 mm. The entire domain is $9 \times 23 \times 1.1 \text{ mm}^3$ in size.

The multi-scale medium includes three zones, that is, the open fracture region, the AL characterized by the absence of calcite, and the unreacted rock matrix, with a volume fraction of 15.68%, 24.88%, and 59.44%, respectively. At the inlet, the reactive fluid was injected into the fracture with a constant rate of $2.5 \mu\text{L}/\text{min}$ (half of the experimental rate because half of the domain was simulated), and a constant pressure was used at the outlet. The symmetry boundary condition was applied at the bottom surface of the domain representing the central plane of the fracture, and other boundaries are without flow or mass transfer. The flow model in COMSOL used an unstructured mesh with $4.26\text{e}6$ elements and the finite element method to solve for the steady state velocity field. The velocity was mapped to the Cartesian mesh ($60 \times 200 \times 110$, $1.32\text{e}6$ cells) used in CrunchFlow to solve the advection-diffusion-reaction equation using the integrated finite difference method. The resolution of the CrunchFlow mesh used in this work is similar to the initial setup of the 3D hybrid model (Molins et al., 2019), which has $64 \times 160 \times 32$ cells. The vertical resolution in the hybrid model is coarser because adaptive mesh refinement was implemented. In our model, the resolution is $10 \mu\text{m}$ for the z -axis, which is close to the data resolution of the fracture aperture and thus able to capture the roughness. A mesh convergence study was performed in a small region—one tenth of the entire domain—to evaluate the effect of mesh resolutions on both flow and reactive transport simulations. Results of the pressure difference, and the effluent Ca and Mg concentrations differ by $<0.02\%$ when the number of grid cells increases by a factor of 2. It confirms that the mesh used in our simulations is fine enough. Furthermore, details regarding the mesh convergence study are presented in the Supporting Information. The computational time is 4–6.5 CPU hours for the COMSOL simulations and 100–250 CPU hours for the CrunchFlow simulations depending on the scenarios tested.

Following the experimental characterization, dolomite and calcite were assumed to be well mixed in the unreacted matrix. The volume fraction of dolomite is 81% in both the AL and matrix, and calcite is only present in the unreacted matrix with a volume fraction of 9%. Therefore, the porosities are 1, 0.19, and 0.10 in the fracture, the AL, and the matrix, respectively. For the steady state simulations, the porosities and volume fractions were not updated. Initial and boundary conditions are summarized in Table 1. The values of permeability were determined based on permeability measurements at different porosities reported in previous experimental studies (Ehrenberg et al., 2006; Kaminskaite et al., 2019). The rate equations for calcite and dolomite are given by (Chou et al., 1989; Deng et al., 2016) as follows

$$r_{\text{CaCO}_3} = (k_1 \gamma_{\text{H}^+} C_{\text{H}^+} + k_2 \gamma_{\text{H}_2\text{CO}_3^*} C_{\text{H}_2\text{CO}_3^*} + k_3) \left(1 - \frac{Q_{\text{CaCO}_3}}{K_{\text{CaCO}_3}} \right) \quad (5)$$

Table 1
Chemical and Physical Parameters Used in the Base Case Simulation

Domain	Porosity	Permeability (m ²)	Effective diffusion coefficient (m ² /s)	Reactive surface area (m ² /m ³)	Inlet flow rate (μL/min)	Solution parameters for initial ^a /inlet conditions
Fracture		Free flow in the fracture zone			2.5	pH 4.8/3.13
Altered layer	0.19	1e−14	2e−10	Dolomite 2,555		pCO ₂ 1.2/1.2 (mol/kg)
Matrix	0.10	5e−13	4e−12	Calcite 1e6		Mg 5e−3/0 (mol/kg)
				Dolomite 2,555		Ca 2e−2/0 (mol/kg)

^aInitial conditions were set to be at equilibrium with the mineral phases in order to reach steady state quickly.

$$r_{CaMg(CO_3)_2} = (k_4 \gamma_{H^+}^{0.5} C_{H^+}^{0.5}) \left(1 - \frac{Q_{CaMg(CO_3)_2}}{K_{CaMg(CO_3)_2}} \right) \quad (6)$$

where K_{CaCO_3} , $K_{CaMg(CO_3)_2}$ are the solubility constants and Q_{CaCO_3} , $Q_{CaMg(CO_3)_2}$ are the ion activity products of the reaction for calcite and dolomite, respectively. C_i and γ_i are the concentrations and activity coefficients of species i . In this single phase system, all aqueous reactions are treated as equilibrium reactions. The kinetic coefficients follow Molins et al. (2019), that is, $k_1 = 0.083$ mol/m²/s, $k_2 = 1.5e-4$ mol/m²/s, $k_3 = 1.1e-4$ mol/m²/s, and $k_4 = 4.5e-4$ mol/m²/s. The effective diffusion coefficients and reactive surface areas are specified following the continuum approach in the porous regions and are also consistent with previous modeling work (Molins et al., 2019).

3.2. Scenario Setup for Analysis of Altered Layer Properties and Processes

Several scenarios were designed for comparison to isolate the impact of flow, transport and reaction in the AL on the spatial patterns of reaction rate and the overall behaviors, and thus to identify the controlling factor on the subsequent evolution. Key parameters of the AL, that is, porosity, permeability, effective diffusion coefficient, and reactive surface area, were varied. Table 2 summarizes the scenarios and the respective input parameters. The DBS_ε0.19 scenario is the base case with the best available parameter values for the experiment and considers all processes. In the DBS_ε0.5 scenario, the AL is made more permeable by increasing the calcite volume fraction and AL porosity to 0.5. N-S_ε0.19 and N-S_ε0.5 scenarios have the same setup as DBS_ε0.19 and DBS_ε0.5,

Table 2
Reactive Transport Parameters for Simulation Scenarios

Parameters ^a		Transport metrics				Reaction metrics	
		Base case	Advection-diffusion effects		Diffusion effects		Surface area effects
		DBS_ε0.19 scenario (base case)	DBS_ε0.5 scenario	N-S_ε0.19 scenario	N-S_ε0.5 scenario	DBS_ε0.19 scenario	DBS_ε0.5 scenario
Porosity_ε	Altered layer	0.19	0.50	0.19	0.50	0.19	0.19
	Matrix	0.10	0.10	0.10	0.10	0.10	0.10
Permeability (m ²) ^c	Altered layer	5e−13	1e−10	5e−13	1e−10	5e−13	5e−13
	Matrix	1e−14	1e−14	1e−14	1e−14	1e−14	1e−14
Effective diffusion coefficient_D (m ² /s) ^c	Altered layer	2e−10	1e−9	2e−10	1e−9	4e−11	2e−10
	Matrix	4e−12	4e−12	4e−12	4e−12	4e−12	4e−12
Surface area_A (m ² /m ³)	Calcite	1e6	1e6	1e6	1e6	1e6	1e6
	Dolomite	2,555	2,555	2,555	2,555	2,555	12,775

^aThe bold fonts highlight varied parameters compared to the base case. ^bVelocity fields for N-S_ε0.19 and N-S_ε0.5 scenarios were calculated by the N-S equation, while others were given by the Darcy-Brinkman-Stokes equation. ^cPorosity-permeability relationship was determined by experimental measurements (Ehrenberg et al., 2006; Kaminskaite et al., 2019) and diffusion coefficients were calculated by Archie's law (Deng et al., 2016).

Table 3

Comparison of Effluent Concentrations of Calcium and Magnesium From Experiments and Different Models

Data concentration	Experiment (Ajo-Franklin et al., 2018)	Diffusion coefficient for the altered layer (m ² /s)	Reduced-dimension model (Deng et al., 2016)	Hybrid model (Molins et al., 2019)	Micro-continuum model (this work)
Calcium concentration/magnesium concentration (mol/L)	0.027/0.0087	2e−10 4e−11	0.027/0.0094 0.021/0.0111	0.031/0.0034 0.022/0.0049	0.025/0.0033 0.014/0.0031

respectively, except that they neglect flow in the AL and the remainder of the rock matrix. This is accomplished by obtaining the velocity field using the Navier-Stokes equation in the fracture opening, rather than the DBS equation, which is also the approach used in Molins et al. (2019). The DBS_ D_{low} scenario uses a lower diffusion coefficient for the AL, as also tested in Deng et al. (2016) and Molins et al. (2019), and the DBS_ A_{high} scenario examines the effect of reactive surface area of dolomite.

4. Results and Discussion

Overall, results from the model presented here compare well with the experimental measurements (Table 3). Discrepancies are of the same order as those observed in previous modeling efforts (Deng et al., 2016; Molins et al., 2019; Table 3). Although assumptions, solution, and meshing approaches and codes are different, this indicates the ability of the model to capture the main processes in the fractured domain. A detailed description of the differences between results from our model and previous models result is presented in the Supporting Information.

4.1. Reactive Transport Regimes in the Fractured Dolomite

The results show spatial variations in velocities and concentrations that are strongly influenced by the structural heterogeneity (Figure 3). The velocity maps highlight a fast flow path on the right side of the fracture domain, whereas the left side is characterized by low velocities due to the presence of contact areas or small apertures. The downstream velocities are more uniform. Overall, regions with higher flow rates in the fracture correlate

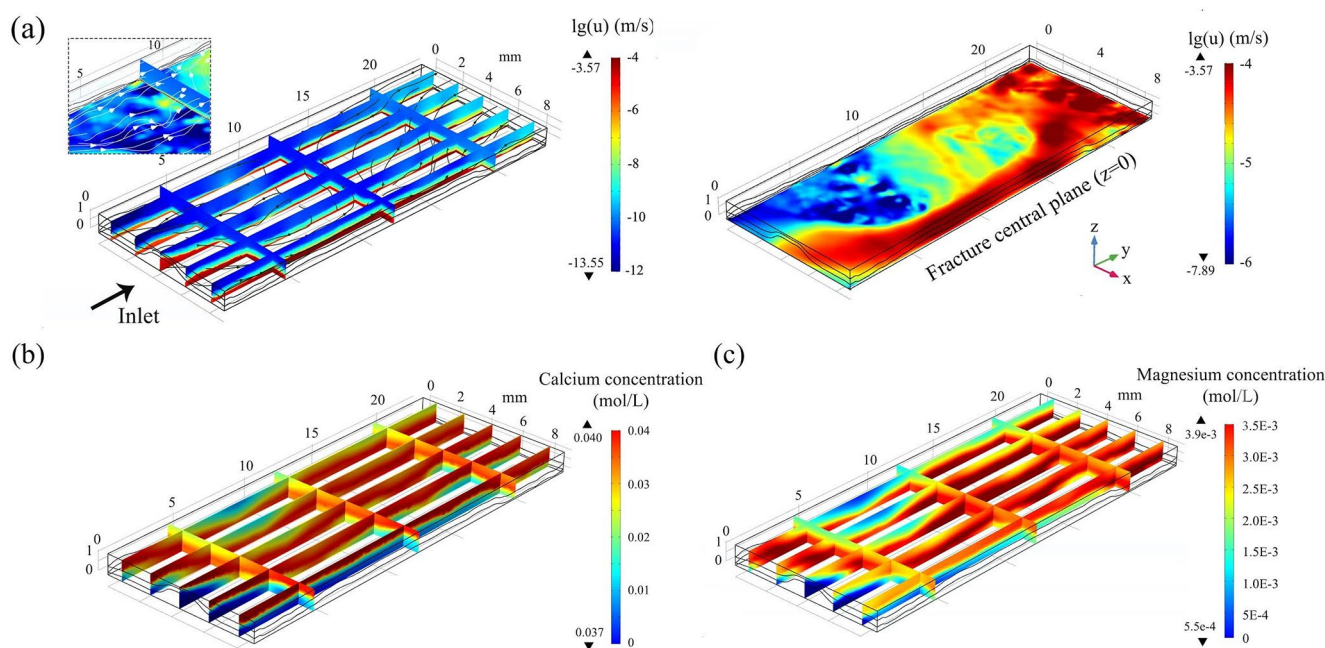


Figure 3. Visualizations of the (a) velocity field (including a close-up view of flow across the altered layer), (b) calcium concentrations, and (c) magnesium concentrations in the 3D domain.

to regions with lower concentrations of calcium (Ca) and magnesium (Mg). The average Darcy velocities are $4.52\text{e}-6$, $1.02\text{e}-9$, and $4.68\text{e}-11$ m/s in the fracture, AL, and unreacted matrix, respectively. In the AL and matrix, the velocities are several orders of magnitude slower and is transport limited, and thus higher concentrations were observed. However, the streamlines shown in Figure 3a also highlight flow through the AL, especially at regions where there are contacts or fracture apertures are small. These features can have important implications regarding local reactions and thus subsequent geochemical evolution, as will be analyzed and discussed in detail below.

The dimensionless Péclet (Pe) and diffusive Damköhler (Da_{II}) numbers may be used to characterize the interplay between flow, transport, and reactions. For a given characteristic length L_c (fracture aperture here), Pe and Da are estimated in this work by $\frac{u_f L_c}{D}$ and $\frac{k L_c^2}{D}$, respectively. The reaction rate constant k (s^{-1}) is calculated as $k = \pi r_s n L$, where S is the specific surface area, r_s is the intrinsic reaction rate without transport limitation, and n is the moles of mineral per unit volume, that is, $n = \frac{\rho_{\text{mineral}} f_{\text{mineral}}}{M_{\text{mineral}}}$. While the cut-off value for determining the dominance of diffusion versus advection is not necessarily 1, the evident increase in the $\log_{10}(Pe)$ values (from -5.64 to -1.78 and 1.42) suggests a decreasing control of the diffusive transport process moving from the matrix to the AL and to the fracture. The rate-limiting process for different mineral reactions differs in different zones. With the shielding effect of the AL, calcite dissolution in the matrix is diffusion-limited as also confirmed by the large Da_{II} , 365. For dolomite, Da_{II} is 0.389 and 0.006 in the matrix and AL, respectively, indicating an increased control of surface reaction. The advective Damköhler (Da_I) number, as given by Da_{II}/Pe , is on the order of 0.1 and suggests a more comparable control of surface reaction and advective transport on the reaction rate in the AL.

Figure 4 shows two-dimensional (2D) cross sections of concentrations and reaction rates, to illustrate the chemical gradients across the three regions. Two representative slices were chosen. One cuts through the flow channel in the fracture (yz_Slice 2), and the other is located in the region where the fracture is tight (yz_Slice 1). Generally, the dissolution of calcite only happens at the interface between the unreacted matrix and the AL, and Ca concentration is much higher in the matrix. The narrow region of reactions and the low effective reaction rates, even with a large surface area, highlight the diffusion limitations on calcite dissolution. For dolomite, dissolution is localized within the AL with higher reaction rate closer to the fracture surface. The Mg concentration is comparable in the AL and the unreacted matrix, and is much lower in the fracture. Comparison of the chemical profiles in the two locations provide further insights regarding the impacts of fracture geometry heterogeneity. In particular, fast flow in the wide fracture (yz_Slice 2) corresponds to lower Ca and Mg concentrations in the matrix, which enhances the geochemical disequilibrium and thus the extent of dissolution for both calcite and dolomite. This indicates a positive feedback, that is, porous media close to larger fracture apertures are the preferred regions of geochemical reactions (Fazeli et al., 2019). It leads to the increasingly heterogeneous pattern in both matrix alteration induced by calcite dissolution, and AL development and fracture enlargement from dolomite dissolution (Al-Khulaifi et al., 2019).

4.2. The Impacts of AL Processes on Bulk Chemistry

Variations in reactive transport parameters of the AL were implemented in different scenarios to examine the controlling factor on dissolution behaviors of both calcite and dolomite. Probability distribution functions (PDFs) of velocities (Figure 5) provide an overview of changes in the advective transport. For the DBS simulations, three peaks are observed, representing the flow in the matrix, the AL, and the fracture, from left to right. The velocity distributions in the fracture in the N-S simulations (Figure 5) overlap with the fracture peak of the DBS velocity distribution. The higher porosity and permeability used in the DBS_ε0.5 scenario enhanced both advective and diffusive transport in the AL compared to the base case. The middle peak shifts significantly to the right as flow velocities in the AL are two orders of magnitude higher than those in the base case. The shape of PDFs also shows a smoother transition of velocity distributions near fracture surfaces. As a result, the relative contribution of advective transport in the AL increases, which is confirmed by a more than one order of magnitude increase in the Pe number ($\log_{10}(Pe)$ is -0.55). This implies that advection in the AL could play a more important role when the AL is more permeable.

Table 4 summarizes the effluent concentrations and bulk reaction rates from all six scenarios to quantify the impacts of different properties of the AL on bulk chemistry. Compared to the base case, the DBS_ε0.5 scenario results in the largest decrease (73%) in the effluent Ca concentration and the DBS_Ahigh scenario leads to the

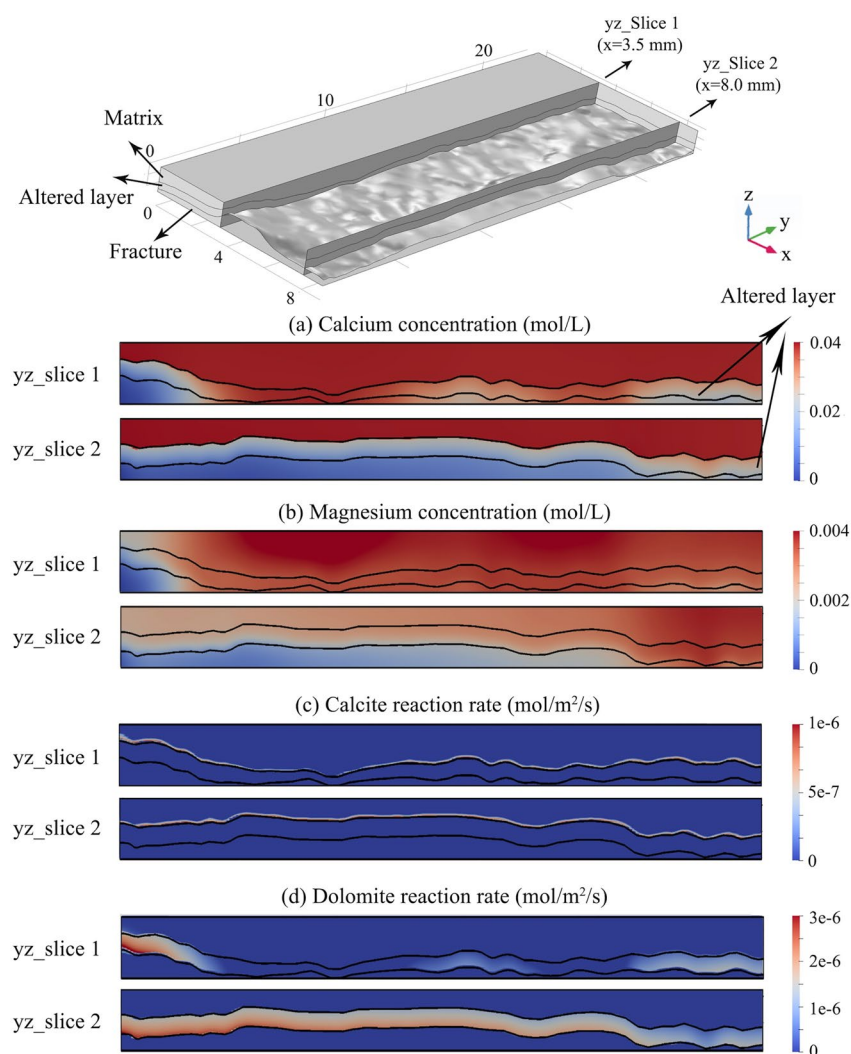


Figure 4. Chemical properties, that is, concentrations of (a) calcium and (b) magnesium, and reaction rates of (c) calcite and (d) dolomite from specific 2D profiles, yz_Slice 1 and yz_Slice 2 cut through the slow flow region and the fast flow path along the flow direction, respectively. The z-axis of slices is exaggerated by a factor of 2 to highlight variations through different domains.

largest increase (113%) in the effluent Mg concentration. The significant increase in the effluent Mg concentration observed in the DBS_ε_{high} scenario supports that the reactive surface area for dolomite used here which is sourced from the literature is low compared to the dolostone core.

As a result of the enhanced advection and diffusion across the AL in the DBS_ε0.5 scenario, overall dissolution of both calcite and dolomite are increased compared to the DBS_ε0.19 scenario. The effluent concentrations of Ca and Mg increase by 21% and 6%, respectively. Calcite dissolution increases with diffusivity as expected, but the level of the increase depends on advection in the AL. The increase in calcite dissolution when there is no advection in the AL (N-S_ε0.19 and N-S_ε0.5 scenarios) is 22%, which is less than the extent of increase when there is advection in the AL (DBS_ε0.19 and DBS_ε_{low}), that is, 73%. For dolomite, higher diffusivity results in slightly less dissolution without advection. It indicates that the impact of the pH-dependent reaction pathway of dolomite is important as more calcite dissolution results in more buffering of the pH and lower kinetic rate for dolomite. With advection, however, dolomite dissolution increases with the diffusivity due to the improved transport. The bulk chemistry results from N-S_ε0.19 and N-S_ε0.5 scenarios are comparable to the ones using the DBS velocity field. The lack of difference in the bulk chemistry with and without considering advection in the AL is a result of the spatial rearrangement of reactions, rather than an indicator that advection in the AL is

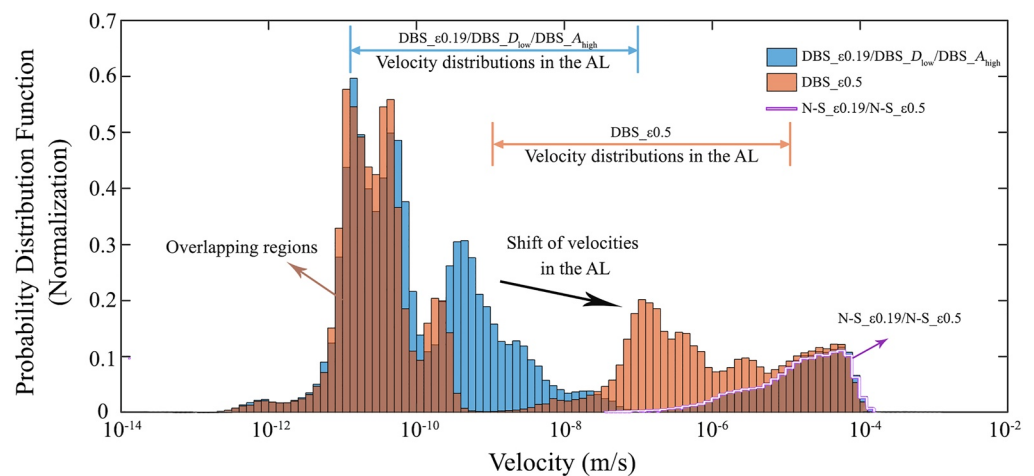


Figure 5. Comparison of probability distribution functions (PDFs) of velocities from six scenarios, grouping into three pairs. For N-S_ε0.19 and N-S_ε0.5 scenarios, the PDFs are calculated with zero values for the porous media while flow is only resolved in the fracture.

unimportant. As will be shown in Section 4.3, consideration of the advection in the AL results in spatial variations in local reaction rates.

In all the cases, bulk reaction rates of both calcite and dolomite are several orders of magnitude lower than the intrinsic reaction rates. Moreover, the effective reaction rates of dolomite are higher than those of calcite even though the intrinsic reaction rates are expected to be one order of magnitude lower. A similar phenomenon was observed in the previous experimental work (Al-Khulaifi et al., 2017) indicating limited calcite dissolution due to shielding provided by the remaining dolomite. These observations suggest that the slow-reacting mineral, that is, dolomite, can play a more important role in the later-stage alteration of fractured media when the AL becomes well established in a multi-mineral system, since the dissolution of the remaining minerals in the AL will control the subsequent fracture enlargement in addition to the AL development.

Overall, variations in the bulk reaction rates are similar to those in effluent concentrations, except for the case with the high surface area. In the DBS_A_high scenario, the higher reactive surface area results in more dolomite dissolution—with a significant increase (113%) in the effluent Mg concentration—but a lower surface area normalized rate due to more accumulation of the reaction products. This implies that dolomite reaction rates in the AL are strongly influenced by the surface reaction. Calcite dissolution is limited by both inefficient transport and exhaustion of fresh reactants. The net effect of the reduced calcite dissolution rate and increased dolomite dissolution rate is a higher effluent Ca concentration in this scenario.

Table 4
Comparison of Effluent Concentrations and Bulk Reaction Rates From Different Scenarios

	Base case	Transport metrics				Reaction metrics
		Advection-diffusion effects	Advection effects		Diffusion effects	Surface area effects
Effluent concentrations (mol/L)	DBS_ε0.19 scenario (base case)	DBS_ε0.5 scenario	N-S_ε0.19 scenario	N-S_ε0.5 scenario	DBS_D_low scenario	DBS_A_high scenario
Calcium	0.0248	0.0300	0.0249	0.0306	0.0143	0.0253
Magnesium	0.00326	0.00345	0.00325	0.00322	0.00309	0.00694
Bulk reaction rate (mol/m ² /s)						
Calcite	5.05e−9	6.29e−9	5.32e−9	6.74e−9	2.62e−9	4.30e−9
Dolomite	2.65e−7	2.81e−7	2.63e−7	2.60e−7	2.53e−7	1.12e−7

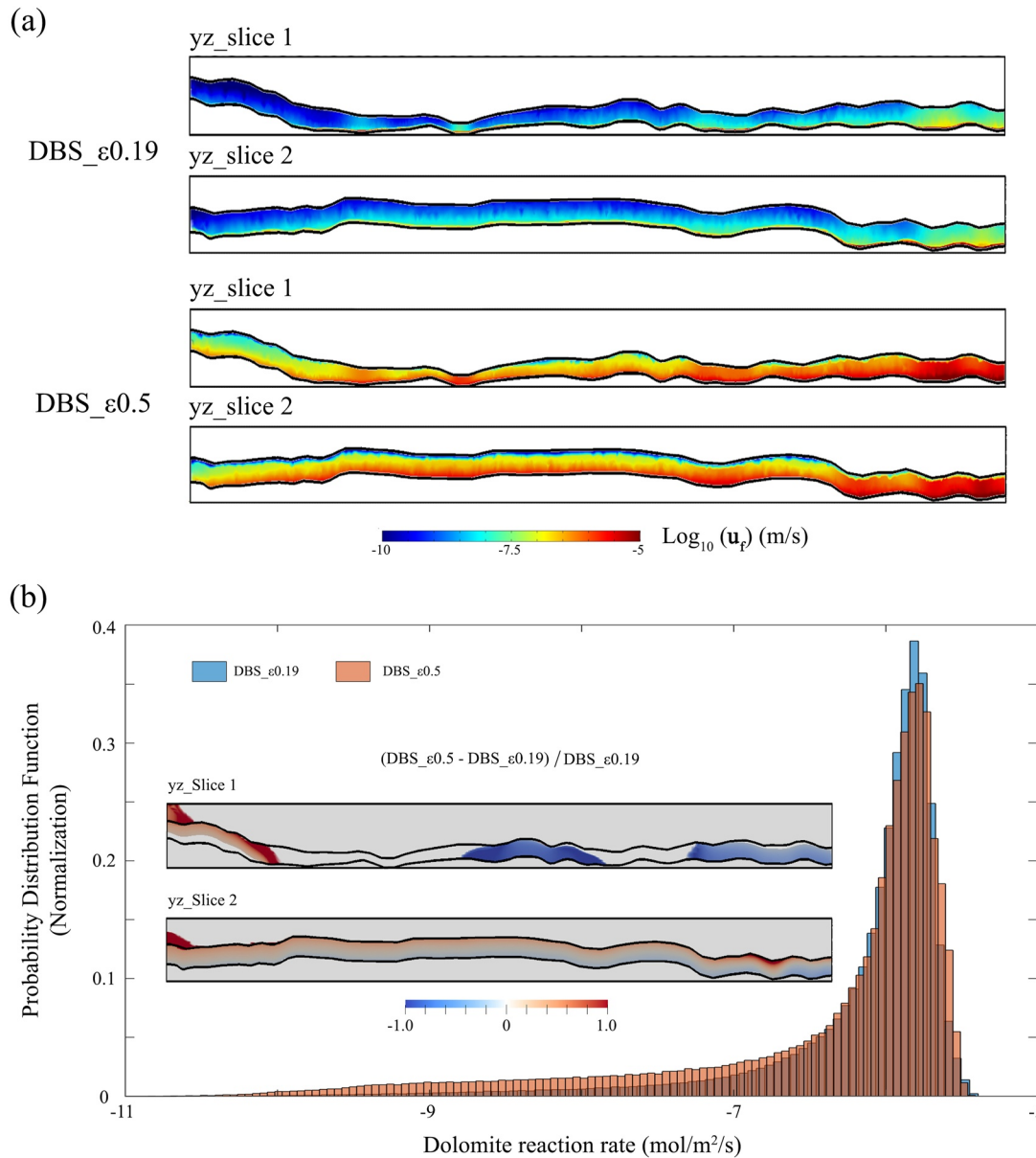


Figure 6. Comparisons of (a) spatial distributions of velocity (u_t) and (b) probability distribution functions and relative differences of dolomite reaction rates for DBS_ε0.19 and DBS_ε0.5 scenarios.

4.3. Spatial Differences in Reactions and Implications on Alteration

Because long-term alteration of the fractured media is dictated by local reaction rates whereas bulk chemistry is an integrated outcome of the system dynamics, here, we report a detailed comparison of different scenarios focusing on local dolomite reaction rates, to better understand the long-term implications of different AL processes.

4.3.1. Coupled Effects of Advective and Diffusive Transport on Spatial Dynamics

Comparison of the DBS_ε0.19 and DBS_ε0.5 scenarios highlight the coupled effects of advective and diffusive transport. Spatial distributions of velocity (Figure 6a) illustrate the non-uniform flow in the AL with higher velocities in regions close to the fracture surface for both DBS_ε0.19 and DBS_ε0.5 scenarios, especially where fracture apertures are smaller. The PDF of dolomite dissolution rates (Figure 6b) of the DBS_ε0.5 scenario shows a flatter and a slightly more uniform distribution, that is, there are more grid cells with high or low reaction rates.

The 2D profiles indicate that the increase in dolomite dissolution resulting from the increased advective and diffusive transport in the AL primarily occur in the regions with large apertures (e.g., yz_Slice 2 in Figure 6b). In the wider fracture region, dolomite reaction rate increase (more than 100% difference) is more evident close to the interface between the unreacted matrix and the AL, implying a stronger reaction front into the matrix, whereas dolomite reaction rate is reduced at the fracture surface. This means that the gradient in dolomite dissolution perpendicular to the fracture plane (as shown in Figure 4d) is more homogenized in these regions. In comparison, in regions with small apertures (yz_Slice 1 in Figure 6b), dolomite dissolution rate is mostly reduced except for zones close to the inlet before the channel flow is well developed. These observations imply that in rocks with higher fraction of fast-reacting minerals as in the DBS_ε0.5 scenario, the spatial heterogeneity of AL may be enhanced in the fracture plane, while staying more uniform vertically, which is an important assumption for using the analytical representation of diffusion limited reaction rate and as also assumed in the initial setup of the model here. The changes of aperture enlargement, however, will be more complex as even though the dissolution is dampened at the fracture surface, less remaining mineral is to be removed in this case.

4.3.2. Effects of Advective Transport on Spatial Dynamics

Comparisons of the N-S_ε0.19 and N-S_ε0.5 scenarios with the DBS_ε0.19 and DBS_ε0.5 scenarios provide a direct measure of the effect of advection within the AL. Spatial variations in the changes in dolomite dissolution rate are highlighted in Figure 7, and are especially evident (up to 20% difference in local reaction rate) in the case with large porosity in the AL as expected (i.e., DBS_ε0.5 and N-S_ε0.5). For the ε0.19 scenarios, advection in the AL increases dissolution of dolomite in the AL, mostly in the small aperture regions. In the ε0.50 simulations, advection increases dissolution of dolomite in the AL particularly in the large aperture regions; whereas the dissolution is enhanced in the upstream outside of the contact points and is reduced in the downstream in the small aperture regions.

These spatial variations even though limited, can still propagate throughout time. In a dynamic simulations in which the geometry/porosity is updated, spatial variations can be amplified following the positive feedback between reactions and transport, that is, the more reactive regions have more significant porosity increase and thus permeability and diffusivity enhancement, which in turn accelerate local reactions (Deng & Spycher, 2019; Fazeli et al., 2018; Garcia-Rios et al., 2017). As shown in Figure 4d, dolomite dissolution is faster in the wide aperture regions. The spatially varied enhancement in dolomite dissolution rate is likely to compound the spatial heterogeneity.

4.3.3. Effects of Diffusive Transport on Spatial Dynamics

Comparisons between the N-S_ε0.19 and N-S_ε0.5 scenarios (i.e., Group 1), and between the DBS_ε0.19 and DBS_ε0.5 (i.e., Group 2) isolate the effects of diffusivity in the AL. In both groups, the diffusivity differs by a factor of 5. Figure 8 shows the probability and spatial distribution differences of dolomite reaction rates from two groups. Overall, lower diffusivity in the AL increased dolomite dissolution rates (the relative difference can be more than 100%) close to the fracture- AL interface and reduced the rates close to the interface between the AL and the unreacted matrix. As a result, dolomite dissolution rate has a larger gradient across the AL and the high dissolution rates are rearranged into a narrower zone.

In addition, the regions with slow velocities in the fracture (yz_Slice 1) are strongly affected by the diffusivity compared to the region of fast flow paths (yz_Slice 2). This is especially evident when advection is considered (i.e., Group 2), as also highlighted by the red circle in Figure 8b. Without advection, local reaction rate is solely determined by the interplay between diffusion and surface reaction, whereas with advection, local reaction rate may be controlled by advection and surface reaction interactions instead, as also discussed in Section 4.1. As such, those regions with higher advection are not affected by the diffusivity reduction, and the reduced reactions caused by stronger diffusion limitations in other places can preserve higher thermodynamic driving force for surface reaction and thus higher local reaction rate in these regions.

Overall, the PDF of the reaction rate in the case without considering advection in the AL is narrower with the lower diffusivity. In contrast, with advection the PDF spreads over a wider range and is flatter, indicating a more heterogeneous spatial distribution when the diffusivity is lower. While the lower diffusivity in the AL highly impedes the alteration of unreacted matrix induced by calcite dissolution as observed in previous experiments (Al-Khulaifi et al., 2019; Deng et al., 2017; Ellis et al., 2013), the increase in dolomite dissolution close to the

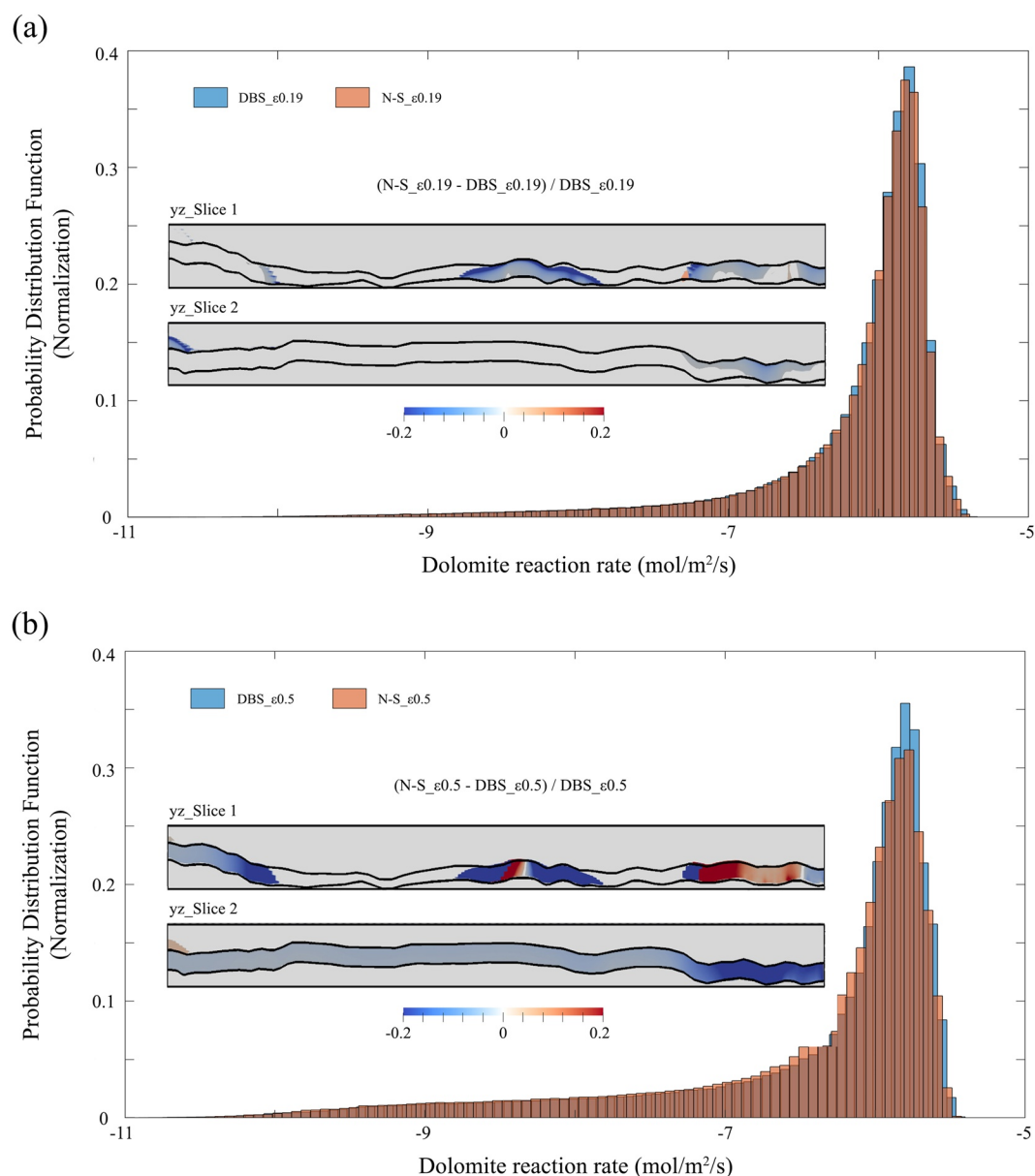


Figure 7. Comparison of dolomite reaction rates by probability distribution functions and relative differences for (a) DBS_ $\epsilon_{0.19}$ and N-S_ $\epsilon_{0.19}$ and (b) DBS_ $\epsilon_{0.5}$ and N-S_ $\epsilon_{0.5}$.

fracture-AL interface—the level of which depends on whether advection is accounted for—will enhance fracture enlargement.

4.3.4. Effects of Reactivity on Spatial Dynamics

The DBS_ A_{high} scenario with a higher reactive surface area of dolomite was performed to examine the effects of surface reaction. However, the increase in surface area and that in the extent of dolomite dissolution are not proportional, confirming the mixture of transport-controlled reaction and surface-controlled reaction (Section 4.1). The lower peak of PDF for DBS_ A_{high} compared to DBS_ $\epsilon_{0.19}$ is caused by the presence of zero dolomite dissolution rate in more grid cells. Relative differences (Figure 9) in dolomite dissolution between DBS_ A_{high} and DBS_ $\epsilon_{0.19}$ provide an indicator of the rate-limiting regimes. The red color corresponds to regions with significant rate increase where the reactions are more likely surface-controlled, whereas the blue indicates the regions of stronger transport limitation. Comparison of different slices highlights that close to larger fracture apertures,

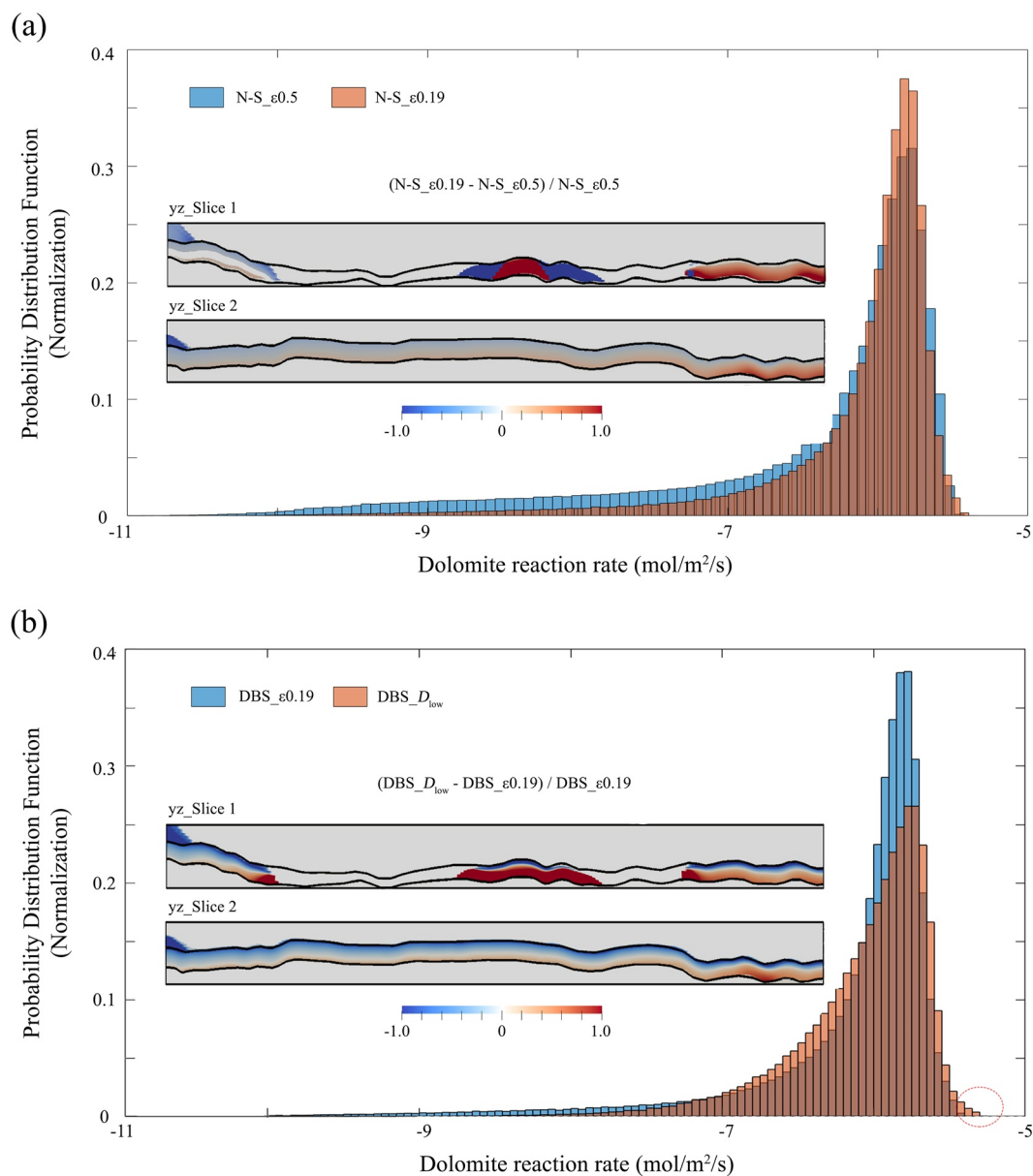


Figure 8. Comparison of dolomite reaction rates by probability distribution functions and relative differences for (a) $N-S_{\epsilon 0.19}$ and $N-S_{\epsilon 0.5}$ and (b) $DBS_{\epsilon 0.19}$ and $DBS_{D_{low}}$. The red color circle highlights the larger reaction rates from $DBS_{D_{low}}$ scenario.

reactant delivery and mixing are more effective and the reactions are primarily controlled by surface reactions. It also shows that reactions mainly happen close to the inlet and decrease along the flow path, which gives rise to a non-uniform evolution pattern of fractured media.

5. Conclusions and Implications

In this study, a DBS-based micro-continuum reactive transport model was developed by coupling the COMSOL multiphysics code and CrunchFlow. Using this modeling framework, we investigated the reactive transport processes in a fractured domain including a pore-scale component and a Darcy-scale component. The domain was reconstructed from a fractured dolostone after reacting with CO_2 -acidified fluid, which resulted in a porous AL bordering the fracture where calcite is depleted and the dolomite remains. In our simulations, flow, diffusion, and reactions were explicitly considered in the open fracture, the bordering porous AL, and the unreacted matrix,

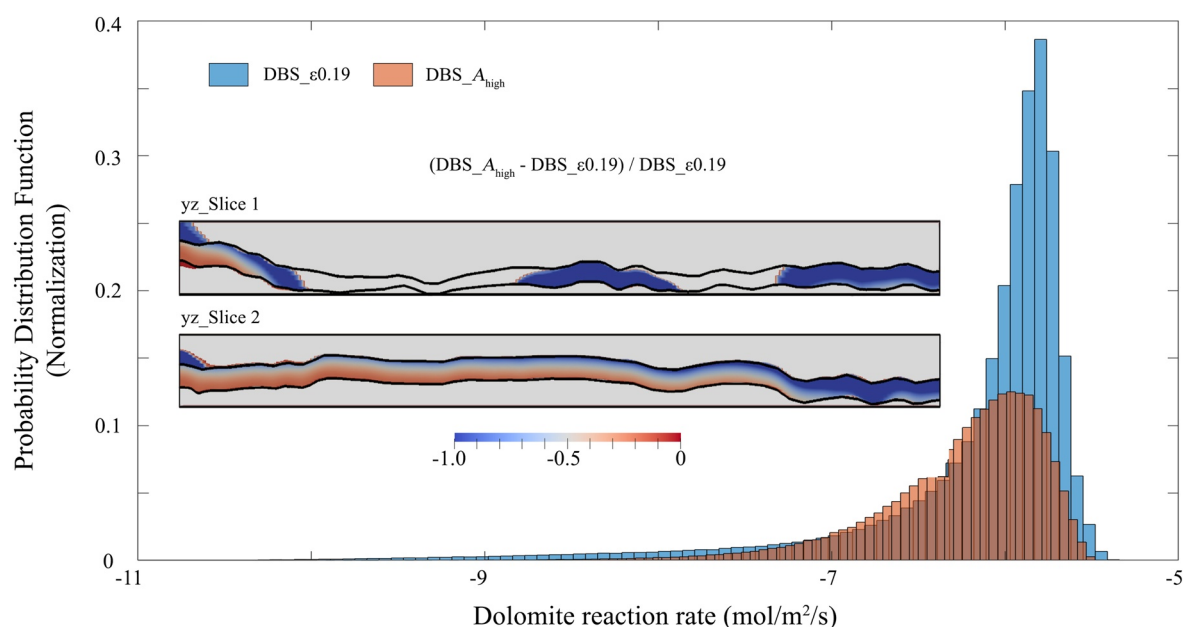


Figure 9. Comparison of dolomite reaction rates by probability distribution functions and relative differences for $\text{DBS}_{\epsilon 0.19}$ and $\text{DBS}_{A_{\text{high}}}$.

allowing accurate depiction of mass transfer within and across domains of contrast properties. By examining the bulk and local reactions at steady state for different simulation scenarios, we were able to, for the first time, investigate rigorously the impacts of the physical-chemical properties within the AL on reactive transport in the fractured media.

Our simulation results show good agreement with previous experimental and modeling studies of the same system. Calcite dissolution occurs primarily at the interface between the unreacted matrix and the AL, and its rate is strongly diffusion limited. Significant reduction in overall calcite dissolution (73%) was observed when the diffusivity is reduced by a factor of 5. In contrast, it is not significantly affected by advection in the AL. In this system, the reaction rate of calcite (i.e., the so-called fast-reacting minerals) can be slower than that of dolomite (i.e., the nominally slow-reacting minerals), indicating a critical role of the remaining mineral in the subsequent alteration of fractured media. Dolomite dissolution exhibits complex spatial variations in response to the changes in advective flow, diffusion, and reactive surface area in the AL. Advective flow provides an additional transport mechanism and tends to enhance dolomite dissolution within the AL in certain regions (by ~20%), especially when the porosity and thus permeability of the AL is high. The differences are also more evident in the small aperture regions. A 7% increase in the overall dolomite dissolution rate was predicted when advective flow is considered in an AL with porosity of 50%. The difference in the spatial distribution of dolomite reaction rate due to presence of the advective flow suggests that the evolution of the fracture aperture and AL may differ from that which ignores advective flow, as advective flow tends to increase spatial heterogeneity. In addition, when advective flow in the AL is considered, dolomite dissolution responds differently to the changes in the diffusivity of the AL. Overall dolomite dissolution increases as diffusivity decreases when advection in the AL is not considered, whereas the overall reaction decreases as diffusivity decreases when advection is considered. For a lower diffusivity in the AL, the dissolution rate of dolomite is enhanced at the fracture surface, which means enhanced fracture enlargement, and reduced at the matrix-AL interface, and the extent is more evident when advection is considered.

While bulk chemistry provides important insights on the overall behavior, spatial variations in local reactions are key for controlling long-term dynamics in heterogeneous fractured media. Overall, our simulations showed that even though advective flow in the open fracture is dominant, advective flow in the AL can still be important in controlling reactive transport in the system and thus the evolution of the fracture and the AL, especially when the AL is permeable or the fracture apertures are small. In these cases, explicit consideration of continuity of fluid fluxes and pressures at the interface between fractures and the bordering porous media is needed for accurate prediction. Moreover, reactions of the nominally slow-reacting minerals should be resolved in reactive transport

simulations as they directly affect the long-term alteration of fractures. While not considered in this study, time-evolving domains resulting from geochemical reactions can also be simulated in the COMSOL-CrunchFlow framework. Our future work will focus on capturing the dynamics in geometries and hydraulic properties of complex fractured porous media and the feedback with geochemical reactions. Such modeling efforts will be especially useful for co-design of simulations and experiments in 3D to fill gaps in theoretical understanding and observations of intimately coupled processes that control the evolution of fractured porous media.

Data Availability Statement

Input files for the multiscale modeling of the fractured dolostone, DOI: <https://doi.org/10.5281/zenodo.4594959>.

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The Effect of Pore-Scale Two-Phase Flow on Mineral Reaction Rates

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In various natural and engineered systems, mineral–fluid interactions take place in the presence of multiple fluid phases. While there is evidence that the interplay between multiphase flow processes and reactions controls the evolution of these systems, investigation of the dynamics that shape this interplay at the pore scale has received little attention. Specifically, continuum scale models rarely consider the effect of multiphase flow parameters on mineral reaction rates or apply simple corrections as a function of the reactive surface area or saturation of the aqueous phase, without developing a mechanistic understanding of the pore-scale dynamics. In this study, we developed a framework that couples the two-phase flow simulator of OpenFOAM (open field operation and manipulation) with the geochemical reaction capability of CrunchTope to examine pore-scale dynamics of two phase flow and their impacts on mineral reaction rates. For our investigations, flat 2D channels and single sine wave channels were used to represent smooth and rough geometries. Calcite dissolution in these channels was quantified with single phase flow and two phase flow at a range of velocities. We observed that the bulk calcite dissolution rates were not only affected by the loss of reactive surface area as it becomes occupied by the non-reactive non-aqueous phase, but also largely influenced by the changes in local velocity profiles, e.g., recirculation zones, due to the presence of the non-aqueous phase. The extent of the changes in reaction rates in the two-phase systems compared to the corresponding single phase system is dependent on the flow rate (i.e., capillary number) and channel geometry, and follows a non-monotonic relationship with respect to aqueous saturation. The pore-scale simulation results highlight the importance of interfacial dynamics in controlling mineral reactions and can be used to better constrain reaction rate descriptions in multiphase continuum scale models. These results also emphasize the need for experimental studies that underpin the development of mechanistic models for multiphase flow in reactive systems.

Keywords: multiphase reactive transport, pore-scale, reaction rate, gas bubble, roughness

INTRODUCTION

Interactions between multiphase flow, geochemical reactions, and solute transport in fractured porous media are ubiquitous in Earth's critical zone and subsurface systems, and affect the dynamics of many environmental and energy engineering applications. Examples include supercritical CO₂ (scCO₂) injection, light non-aqueous phase fluid (LNAPL) contamination, and Enhanced Oil Recovery (EOR). In geologic carbon storage systems, the injected scCO₂ displaces the native brine and can be trapped in small pores or can dissolve into the brine and react with the host rocks.

The resulting mineral dissolution and precipitation can modify the porosity and permeability of the reservoirs and caprocks, affecting injectivity of the reservoirs and long-term storage security (Johnson et al., 2001, 2004; Xu et al., 2011b). In EOR, low-salinity water is injected to the oil reservoir to improve sweeping efficiency by modifying wettability via surface complexation reactions (Zhang et al., 2006; Kumar et al., 2011). In the shallow subsurface, LNAPL contamination of groundwater is a widespread environmental problem. Light non-aqueous phase fluid is insoluble in water and typically fluctuates with the water table. As a result, LNAPL is spread vertically following local drainage-imbibition cycles, which in turn affects the degradation and thus the fate of the contaminants (Ngien et al., 2012; Pan et al., 2016; Govindarajan et al., 2018). The interactions between two-phase flow and electrochemical reactions have also been identified as a research priority for the design of effective fuel cells, as gas bubbles (of e.g., H_2 and O_2) can be generated by side reactions during charging and affect power generation (Chen et al., 2017; Grunewald et al., 2021).

Continuum-scale numerical models are widely used to investigate and predict the evolution of fractured porous media caused by multiphase reactive transport processes. Models such as TOUGHREACT, OpenGeoSys, and PFLOTTRAN are versatile tools that have been used to investigate $scCO_2$ migration in fractured formations (Xu et al., 2011a, 2019; Xiao et al., 2020), dissolution, transport and biodegradation of non-aqueous phase fluid (NAPL) in shallow aquifers (Pruess, 2004; Popp et al., 2015; Sookhak Lari et al., 2019), and heat extraction using CO_2 as a working fluid (Lichtner and Karra, 2014). These models however rely heavily on constitutive relations such as the Brooks and Corey equation and the van Genuchten equation that relates the water saturation with capillary pressure and relative permeability.

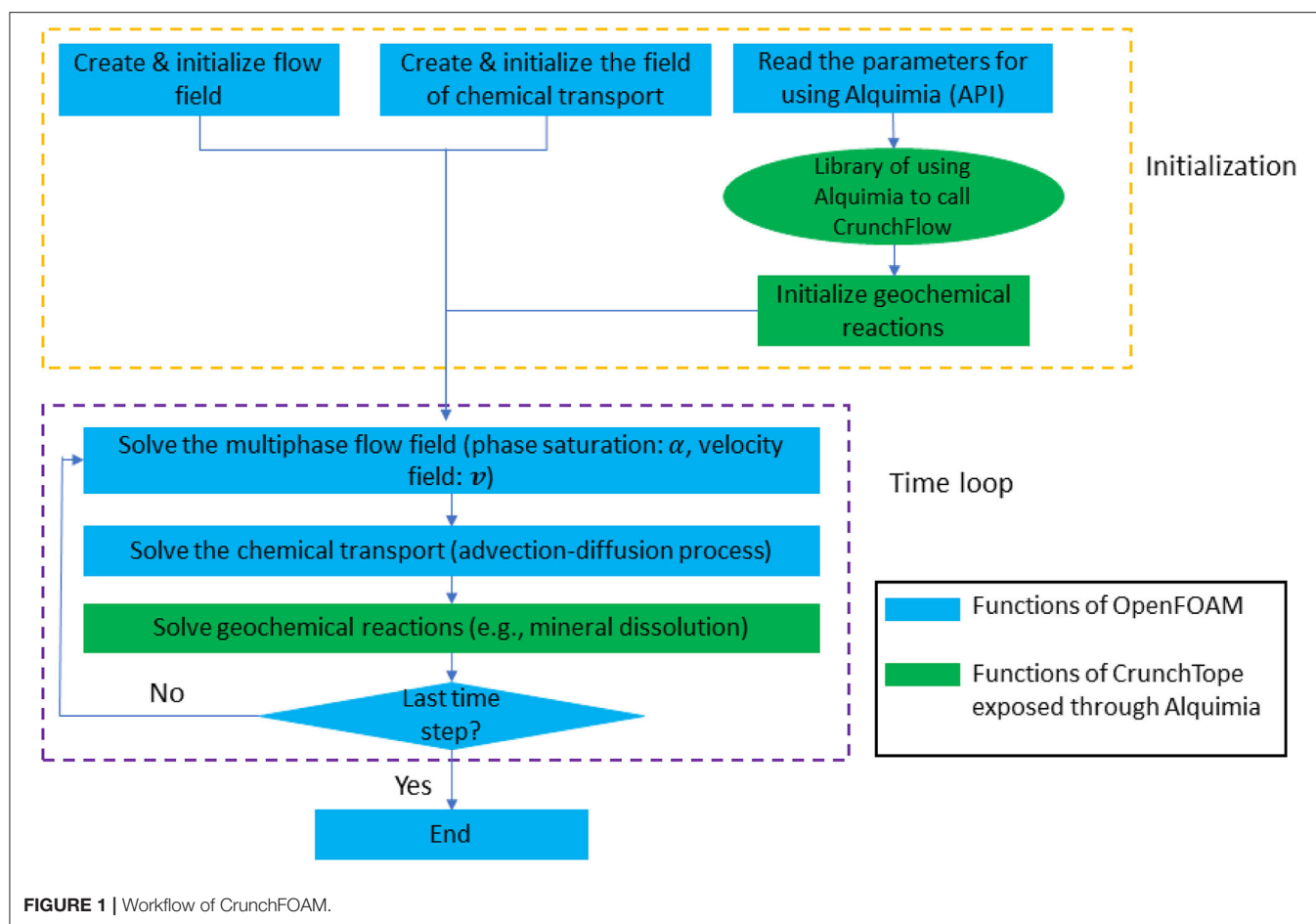
Pore-scale structural heterogeneity and processes that are not considered explicitly in the continuum description, however, could be important. For instance, microfluidic experiments from Karadimitriou et al. (2016, 2017) examined non-Fickian transport in a water-Fluorinert immiscible fluid system. The results demonstrated that the mass transfer rate between mobile-immobile zones under the two-phase flow condition is not constant, indicating that the conventional treatment of the mass transfer rate in the continuum scale Mobile-Immobile (MIM) model could introduce significant errors in the simulation. Pore-scale experiments from Jiménez-Martínez et al. (2015) using 2D microfluidic cells with homogeneous pore structures have also demonstrated that solute mixing can be significantly enhanced under multiphase flow conditions because of ramified finger flow structure, non-Fickian dispersion, and non-wetting phase clusters that limit the finger flow merging. Recent advancement in experimental techniques have also enabled direct observations of reactive transport in multiphase systems at the pore scale. Using biogenically calcite-functionalized micromodels, Song et al. (2018) observed a new microscale mechanism that affects reactive transport in the CO_2 -brine-calcite system. The CO_2 gas phase can accumulate on the mineral surfaces following calcite dissolution, which protects calcite from further dissolution. Jiménez-Martínez et al. (2020) developed a high-pressure geomaterial microfluidic device and

investigated mineral reactions in etched channels during co-injection of CO_2 -saturated brine and $scCO_2$. They observed that the presence of $scCO_2$ bubbles significantly changed both the flow dynamics and the reaction patterns, compared to the single phase flow experiments. In the competing channels with different widths, mineral dissolution was more homogenized and carbonate precipitation was enhanced in the low-velocity regions formed as a result of the presence of the bubbles, which was also confirmed by pore-scale Lattice Boltzmann Methods (LBM) simulations.

Pore-scale models provide an invaluable tool to further our understanding of pore-scale dynamics. Pore-network models (PNMs) are a computationally efficient option for simulating the reactive transport processes under two-phase flow conditions, but require simplifications of the geometric solid-fluid interface and assumptions about uniform aqueous concentrations within a pore (Xiong et al., 2016). Lattice Boltzmann Methods (Chen et al., 2013, 2015) and direct numerical simulations (DNS) (Haroun et al., 2010b; Marschall et al., 2012; Maes and Soulaïne, 2018; Soulaïne et al., 2018, 2021) provide alternatives that relax these restrictions. The LBM was able to reproduce experimental observations on wettability alteration during Low Salinity Water Flooding (Akai et al., 2020). It was also shown by Maes and Geiger (2018) using a DNS approach that the alternation is driven by the concentration of the potential determining ions (PDI) instead of pH. Moreover, a series of micro-continuum simulations showed that the production of CO_2 bubbles resulting from carbonate dissolution may limit the subsequent dissolution and prevent the emergence of wormholes (Soulaïne et al., 2018).

In spite of the growing use of fully-resolved pore-scale models (LBM or DNS) in multiphase reactive applications, they have not been used to understand how reaction rates are influenced by flow processes in the same way as it has been done in single-phase systems (e.g., Molins et al., 2012, 2014; Deng et al., 2018). In single phase systems, reactive surface area has been typically varied to account for flow dynamics and transport limitations. Pore-scale simulations have highlighted the complex dependence of the correction factor on pore-scale geometry and flow regimes (Deng et al., 2018). In multiphase systems, however, this information does not exist. As a result, multiphase continuum scale models rarely account for the effect of the flow dynamics on reaction rates. Further, reaction rates are commonly assumed to be independent of phase saturations (e.g., Xu et al., 2011a; Lichtner et al., 2015; Águila et al., 2020; Wu et al., 2021). Only in some rare instances, the reactive surface area is allowed to vary with liquid saturation. For example, in the active fracture module implemented in TOUGHREACT, the reactive surface area and thus reaction rate follows a power law relation with respect to the water saturation and the exponent is an empirical variable (Sonnenthal et al., 2005). However, there is still a lack of pore-scale studies and mechanistic understanding that support the development of this type of constitutive relationship for considering the impacts of multiphase flow on reaction rates.

Our study aims to bridge this gap, by performing a series of well-designed pore-scale multiphase reactive transport simulations. To this end, a pore-scale multiphase reactive transport modeling framework was developed by coupling



OpenFOAM and CrunchTope. A set of simulations were performed with co-injection of air and CO₂-acidified water into 2D calcite channels to examine calcite dissolution rate under a range of flow conditions. A sine wave geometry was used to introduce different levels of roughness in the channels. This simple geometry has been widely used to provide an idealized representation of surface roughness to investigate its impacts on fluid flow and chemical transport (Kitanidis and Dykaar, 1997; Bolster et al., 2009; Sund et al., 2015; Deng et al., 2018). Section Methodology details the modeling framework, the mathematical principles, and the simulation setups. The results of the numerical simulations, including analyses of the flow field and reaction rates, and observed relations between reaction rate and parameters such as saturation are presented in section Results. We discuss the broader implications of our study in section Discussion and conclude in section Summary and Conclusions.

METHODOLOGY

Our approach entails the simulation of two-phase flow and reactive transport in a series of synthetic geometries. For this purpose, we develop a modeling framework by coupling two widely used and thoroughly validated codes. In this section, we

present this modeling framework and we describe the setup of the simulations.

Modeling Framework

This pore-scale multiphase reactive transport modeling framework couples the geochemical reaction solver CrunchTope (Steefel et al., 2015) with the open source software package, OpenFOAM (open field operation and manipulation, OpenFOAM-v1812) and the related open source libraries, using Alquimia. Alquimia is a generic interface, which allows any flow and transport simulator to access geochemical reaction functionalities of existing, thoroughly validated codes such as CrunchTope (Andre et al., 2013). The modeling framework is referred to as CrunchFOAM for short.

Figure 1 illustrates the workflow of the modeling framework. The geochemical conditions and reaction kinetics are specified in the CrunchTope input files. The initial and boundary conditions for the flow and the transport of the primary chemical species are specified in OpenFOAM. Two-phase flow, transport, and geochemical reactions are solved sequentially following the operator splitting approach. The time stepping is controlled by the flow solver in OpenFOAM. Within each time step, the flow field from the two-phase flow solver is passed to the transport solver to calculate the concentrations of the primary

species by solving the advection-diffusion equation. Afterwards, CrunchTope is called in each cell to calculate aqueous speciation and mineral reactions, and to update the concentration fields for transport in the next time step.

Flow

Two phase flow is solved using interFoam, the standard OpenFOAM solver for transient incompressible isothermal flow of two immiscible fluids. The solver implements a modified version of the Volume of Fluid (VoF) method, and its performance has been confirmed for capillary numbers larger than 10^{-5} (Deshpande et al., 2012; Shuard et al., 2016).

The VoF approach treats the two fluid phases as an effective single phase. The velocity and pressure fields are solved by the single-field incompressible Navier Stokes Equation and continuity equation (Hirt and Nichols, 1981).

$$\begin{aligned} \frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) &= -\nabla p + [\nabla \cdot (\mu(\nabla \mathbf{u} + \nabla \mathbf{u}^T))] \\ &+ \rho \mathbf{g} + \mathbf{F}_{st} \quad (1) \\ \nabla \cdot \mathbf{u} &= 0 \quad (2) \end{aligned}$$

where $\mathbf{u}$ is the velocity. Fluid density ρ , and viscosity μ are weighted averages of the two fluid phases based on the volume fraction (α) of a designated fluid, which is usually the wetting fluid

$$\rho = \rho_1 \alpha + \rho_2 (1 - \alpha) \quad (3)$$

$$\mu = \mu_1 \alpha + \mu_2 (1 - \alpha) \quad (4)$$

$\mathbf{F}_{st}$ is the surface tension force and defined as

$$\mathbf{F}_{st} = \gamma \kappa \mathbf{n} \hat{\delta} \quad (5)$$

where γ is the interfacial tension, $\kappa = \nabla \cdot \mathbf{n}$ is the interface curvature, $\mathbf{n}$ is the unit vector normal to the interface given by $\frac{\nabla \alpha}{\|\nabla \alpha\|}$, $\hat{\delta}$ is a Dirac function located on the interface.

The phase volume fraction α is solved by the following transport equation

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \mathbf{u}) + \nabla \cdot (\alpha(1 - \alpha) \mathbf{u}_r) = 0 \quad (6)$$

where $\mathbf{u}_r$ is the relative velocity between the two fluids/phases. It is typically defined as the compression velocity $\mathbf{u}_c$ to ensure a sharp interface, and its amplitude is determined by the maximum of the single-field velocity

$$\mathbf{u}_r \equiv \mathbf{u}_c \equiv \mathbf{n} \left[\min(c_\alpha \frac{|\Phi|}{A_f}, \max\left(\frac{|\Phi|}{A_f}\right)) \right] \quad (7)$$

where Φ is the volumetric flux, A_f is the cell surface area, $0 \leq c_\alpha \leq 1$ limits the compression velocity below the maximum face flux velocity $\frac{|\Phi|}{A_f}$ and is a user-specified coefficient ($c_\alpha = 1$ in our simulations). This formulation helps minimize numerical diffusion (Rusche, 2002).

The contact angle (θ) is defined at the solid boundary and the following equation needs to be satisfied (Aziz et al., 2018):

$$\mathbf{n} \cdot \mathbf{n}_s = \cos \theta \quad (8)$$

where $\mathbf{n}_s$ is the normal vector to the solid wall.

Transport

For transport under multiphase flow conditions, the continuous species transfer (CST) method has been developed and implemented as a third-party solver in OpenFOAM (Haroun et al., 2010a; Marschall et al., 2012; Deising et al., 2016). The C-CST (compressive-CST) algorithm developed by (Maes and Soulaire, 2018) was implemented in this work as it minimizes numerical diffusion near the interface and ensures that the description of advection is fully consistent with the phase evolution equation (VoF equation). In this method, the transport of a species j dissolved in both phases is described by

$$\begin{aligned} \frac{\partial C_j}{\partial t} + \nabla \cdot (\mathbf{u} C_j) &= -\nabla \cdot \left(\frac{(1 - H_j) C_j}{\alpha + (1 - \alpha) H_j} \alpha (1 - \alpha) \mathbf{u}_r \right) \\ &+ \nabla \cdot (\hat{D}_j \nabla C_j + \Psi_j) + R_j \quad (9) \end{aligned}$$

where $\hat{D}_j$ is the interpolation of the diffusion coefficient of the chemical species in the two phases

$$\hat{D}_j = \alpha D_{j,w} + (1 - \alpha) D_{j,nw} \quad (10)$$

where $D_{j,w}$ and $D_{j,nw}$ is the diffusion coefficient of chemical j in the wetting and non-wetting fluid, respectively. Ψ_j describes the concentration jump at the interface

$$\Psi_j = -\hat{D}_j \frac{1 - H_j}{\alpha + (1 - \alpha) H_j} C_j \nabla \alpha \quad (11)$$

where H_j is Henry's law constant. In this study, mass transfer across the interface is not considered, and H_j is set to be a small value (1×10^{-12}) to avoid zero denominators in Equation (9) when $\alpha = 0$. Because only reactions in the aqueous phase are considered (see below), neglecting mass transfer across the interface implies that concentrations in the non-aqueous phase remain equal to the initial condition (which is zero).

Geochemical Reactions

R_j in Equation (9) accounts for the contribution of mineral reactions to the changes in the mass of chemical species j , and is described by the transition state theory rate law

$$R_j = k_{rxn} A (1 - IAP/K_{eq}) \quad (12)$$

where k_{rxn} is the kinetic coefficient ($\text{mol}/\text{m}^2 \cdot \text{s}$), A is the surface area of the mineral phase (m^2), and is determined directly from the geometry in OpenFOAM, and the chemical affinity term is calculated from the ion activity product (IAP) and the equilibrium constant of the mineral reaction (K_{eq}). The aqueous reactions are assumed to reach equilibrium instantaneously and speciation is calculated based on the law of mass action and the concentrations of the primary species.

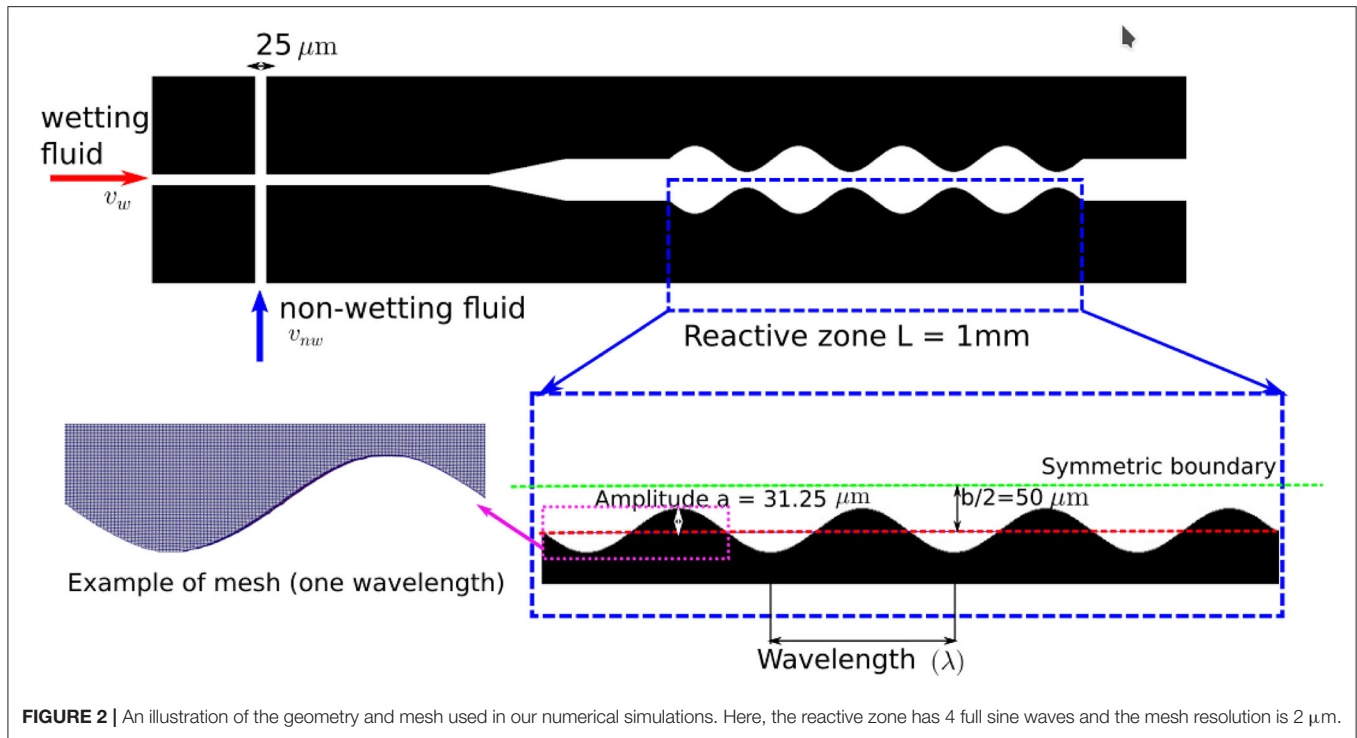


FIGURE 2 | An illustration of the geometry and mesh used in our numerical simulations. Here, the reactive zone has 4 full sine waves and the mesh resolution is 2 μm .

TABLE 1 | Summary of the geometries used in the simulations.

	Geometry a	Geometry b	Geometry c	Geometry d	Geometry e	Geometry f
Wavelength (λ)	N/A	1L	1/4L	1/8L	1/16L	1/32L
SRF	1	1.08	1.14	1.46	2.31	4.19

Simulation Setup

We investigate multiphase flow and reactive transport in a domain that seeks to represent a microcrack with an arbitrarily rough geometry, which can also be conceptualized as a sequence of pores. **Figure 2** provides an illustration of the geometry used in our simulations. To save computational time, we assumed a symmetric geometry and simulated half of the domain. A T-junction structure is used at the inlet for the injection of the wetting and non-wetting phase. The width of the inlets is 25 μm . The reactive zone, where the reactive mineral is located and highlighted in the blue box in **Figure 2**, is 1mm long and has an average width (b) of 100 μm . The dimensions are comparable to previous modeling and micromodel studies of geomaterials (Deng et al., 2018; Song et al., 2018; Jiménez-Martínez et al., 2020) and to the fiber diameter in batteries (Chen et al., 2017). In addition to the reference flat channel, a single sine wave was used to represent pore scale roughness in the reactive zone, following (Deng et al., 2018).

$$b(x) = \frac{\bar{b}}{2} + a \cdot \sin\left(\frac{2\pi x}{\lambda}\right) \quad (13)$$

where λ is the wavelength, and a is the amplitude. In order to explore different levels of roughness, a number of simulations

were performed each with a different wavelength (**Table 1**). In all cases, an amplitude of 31.25 μm was used.

The roughness in these sine wave geometries is measured by the surface roughness factor (SRF). It is defined as the ratio between the total surface area (A_{total}), which is calculated by summing the patch area defined as the mineral wall in the mesh generated by OpenFOAM, and the nominal surface area, which is equivalent of the surface area of the flat geometry (A_{flat}).

$$SRF = \frac{A_{total}}{A_{flat}} \quad (14)$$

The geometries were first generated by a Python script and Blender, and the STL files were then imported into OpenFOAM to generate the meshes using snappyHexMesh. The average mesh size was set to 2 μm . Although interFoam does not show convergence with decreasing mesh size (Pavuluri et al., 2018), this mesh size is comparable with the resolution used in previous studies that showed good results using interFoam in complicated pore structures (Yin et al., 2019; Carrillo et al., 2020). This mesh size also ensures that the single phase simulation results are not affected by further refinement.

Initially, the simulation domain is fully saturated with water except for the vertical branch of the T-junction, which is occupied

TABLE 2 | Physical properties of fluids (ρ_w , water density; ρ_a , air density; μ_w , dynamic viscosity of water; μ_a , dynamic viscosity of air; γ_{wa} , interfacial tension between water and air; θ , contact angle).

ρ_w (kg/m ³)	ρ_a (kg/m ³)	μ_w (Pa · s)	μ_a (Pa · s)	γ_{wa} (N/m)	θ
1000.0	1.0	1.05×10^{-3}	1.55×10^{-5}	0.072	30°

TABLE 3 | Boundary conditions for the two-phase flow simulations.

	Injected wetting fluid—inlet	Injected non-wetting fluid—inlet	Outlet
Flow	$v_{inw} = 0.1 \text{ m/s}$		Zero-gradient
	$v_{inw} = 0.04 \text{ m/s}$	$1.v_{inrw} = 0.25v_{inw}$	Zero-gradient
	$v_{inw} = 0.4 \text{ m/s}$	$2.v_{inrw} = 0.5v_{inw}$	Zero-gradient
Transport	Constant concentration	Constant concentration	Zero-gradient

by the non-wetting phase (i.e., air). The physical properties of the fluids are summarized in **Table 2**.

A constant velocity boundary condition is applied at the inlets. For each geometry, three velocities for the wetting phase were simulated, which are 0.04, 0.1, and 0.4 m/s. The velocity of the non-wetting phase is a fraction of the wetting phase, and two ratios, 0.25 and 0.5, were used to control the frequency of the bubbles generated by the co-injection. This results in a Reynolds number (Re) of 1–10 and a capillary number (Ca) of $\sim 10^{-4}$ – 10^{-3} . The Ca-values are within the range that is relevant for typical reservoirs (Satter and Iqbal, 2016) and battery systems (Grunewald et al., 2021). At the outlet, the zero gradient boundary condition is applied for both flow and transport (**Table 3**).

Geochemical reactions are assumed to take place in the aqueous phase only. The solid phase in the reactive zone is composed of a single mineral, calcite. It dissolves in water, which has a NaCl concentration of 0.01 M/L and a pH of 5 due to dissolution of atmospheric CO₂. The kinetic coefficients for the three elementary reaction pathways reported in Chou et al. (1989) for calcite dissolution are summarized in **Table 4** and used in the simulations. The aqueous reactions and their equilibrium constants are summarized in **Table 5**. The activity coefficients used to convert concentrations to activities (a_{species}) are calculated using the extended Debye-Hückle equation.

$$k_{rxn} = k_1 a_{H^+} + k_2 a_{H_2CO_3} + k_3 \quad (15)$$

In this study, the geometry is not updated and we focus on steady state behavior. Given the time scale that is needed to reach steady state (within seconds as shown in section Results), the mineral reaction is not expected to cause any geometric change. The simulations were run until both the flow and reaction rate reached a steady state. For the analyses of calcite dissolution rate, the absolute average instantaneous reaction rate at a time point t ($\overline{R_m^t}$) was calculated for both single-phase ($m = s$) and

TABLE 4 | Calcite dissolution reactions, the equilibrium constant, and the three reaction pathways with the kinetic coefficients.

Dissolution reaction	LogK _{eq}
$\text{CaCO}_3(\text{s}) + 2\text{H}^+ = \text{Ca}^{2+} + \text{H}_2\text{CO}_3^*(\text{aq})$	8.16
Elementary reaction	Logk(mol/m ² ·s)
$\text{CaCO}_3(\text{s}) + \text{H}^+ \rightleftharpoons \text{Ca}^{2+} + \text{HCO}_3^-$	$k_1 = -0.05$
$\text{CaCO}_3(\text{s}) + \text{H}_2\text{CO}_3^* \rightleftharpoons \text{Ca}^{2+} + 2\text{HCO}_3^-$	$k_2 = -3.3$
$\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CO}_3^{2-} + \text{Ca}^{2+}$	$k_3 = -6.19$

TABLE 5 | Aqueous reactions and the equilibrium constants.

Aqueous complexation reaction	LogK _{eq}
$\text{OH}^- + \text{H}^+ \rightleftharpoons \text{H}_2\text{O}$	13.99
$\text{CO}_3^{2-} + 2\text{H}^+ \rightleftharpoons \text{H}_2\text{CO}_3^*(\text{aq})$	16.67
$\text{HCO}_3^- + \text{H}^+ \rightleftharpoons \text{H}_2\text{CO}_3^*(\text{aq})$	6.34
$\text{CaCO}_3(\text{aq}) + 2\text{H}^+ \rightleftharpoons \text{Ca}^{2+} + \text{H}_2\text{CO}_3^*(\text{aq})$	13.35
$\text{CaHCO}_3^- + \text{H}^+ \rightleftharpoons \text{H}_2\text{CO}_3^*(\text{aq}) + \text{Ca}^{2+}$	5.30
$\text{CaOH}^+ + \text{H}^+ \rightleftharpoons \text{Ca}^{2+} + \text{H}_2\text{O}$	12.85
$\text{CaCl}^+ \rightleftharpoons \text{Ca}^{2+} + \text{Cl}^-$	0.7
$\text{CaCl}_2(\text{aq}) \rightleftharpoons \text{Ca}^{2+} + 2\text{Cl}^-$	0.65
$\text{NaCO}_3^- + 2\text{H}^+ \rightleftharpoons \text{H}_2\text{CO}_3^*(\text{aq}) + \text{Na}^+$	16.16
$\text{NaCl}(\text{aq}) \rightleftharpoons \text{Na}^+ + \text{Cl}^-$	0.78
$\text{NaHCO}_3^-(\text{aq}) + \text{H}^+ \rightleftharpoons \text{H}_2\text{CO}_3^*(\text{aq}) + \text{Na}^+$	6.18

two-phase ($m = t$) flow systems as

$$\left| \overline{R_m^t} \right| = \frac{\sum_{n=1} |R_{m,n}^t| V_n \alpha_n^t}{\sum_{n=1} A_n \beta_n^t} \quad (16)$$

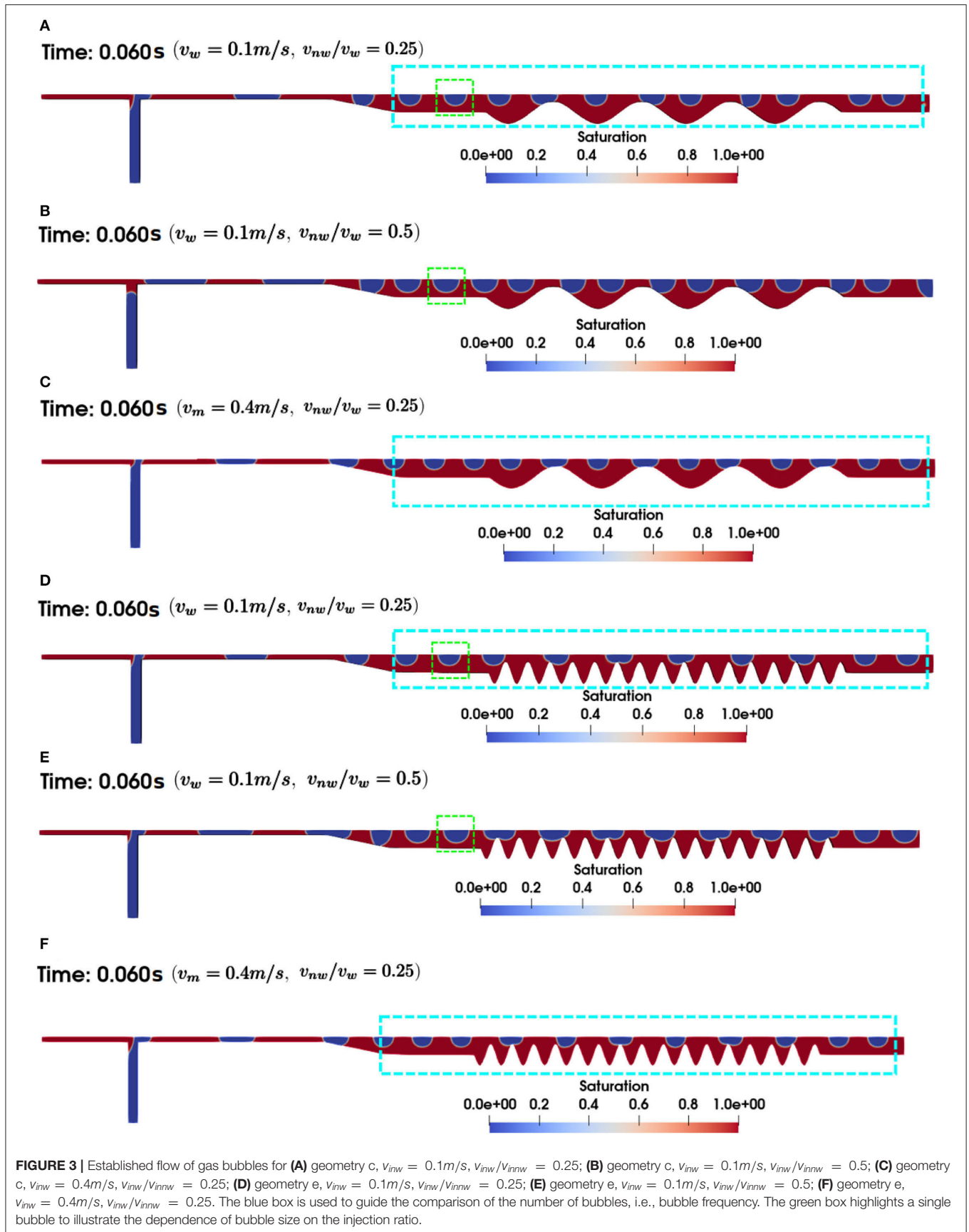
$|R_{m,n}^t|$ is the instantaneous reaction rate in the wall grid cell n with a volume of V_n and wall surface area of A_n at time point t , α_n^t is the volume fraction of the wetting phase, i.e., saturation, in the grid cell, and β_n^t is the wetted surface area ratio within the local wall grid cell, which is approximated by α_n^t .

In addition to the multiphase simulations described above, a single-phase simulation is performed for each roughness level (**Table 1**) and flow condition (**Table 3**), with the same total flux for comparison. These single-phase results are used to elucidate the multiphase effects on reaction rates. All the simulations were performed with the reactive transport solver described in Section Methodology and the saturation (α) is set to one for single-phase simulations.

RESULTS

Reaction Rates

The co-injection of the wetting and non-wetting phases produces a series of gas bubbles that migrate through the reactive zone (**Figure 3**). The size and frequency of the gas bubbles are primarily controlled by the injection rate and the ratio between the wetting and non-wetting phase. Consistent with previous studies (van Steijn et al., 2010; Malekzadeh and Roohi, 2015; Mi et al., 2019), we observed that the bubble size is controlled by the ratio (v_{nw}/v_w) and the frequency is determined by magnitudes



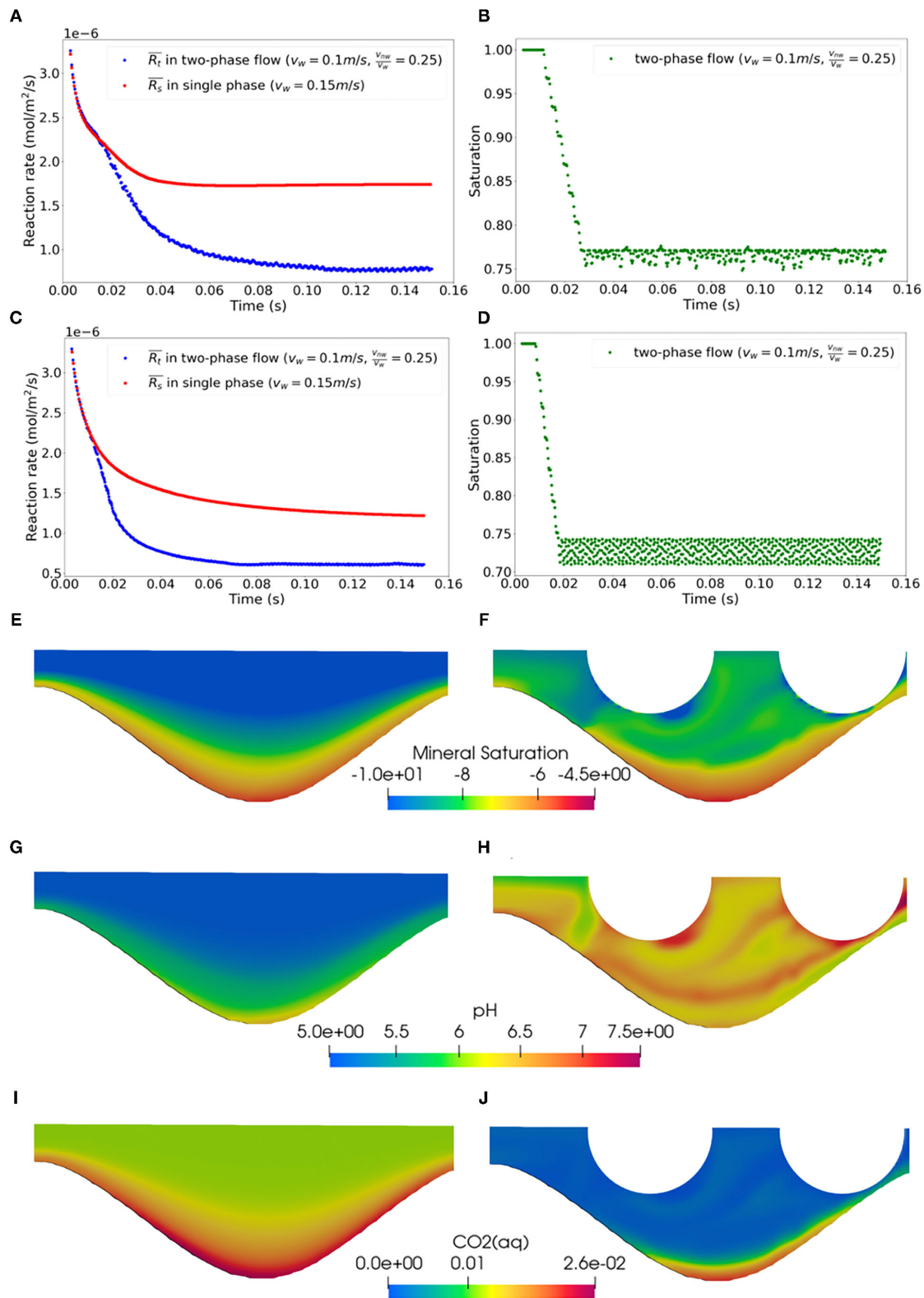


FIGURE 4 | Temporal profiles of the average instantaneous reaction rates (A,C) and average saturation (B,D) in the reactive zone for geometry c (A,B) and e (C,D), $v_{inw} = 0.1 \text{ m/s}$, $v_{inw}/v_w = 0.25$. Close-up spatial profiles of calcite saturation index ($\log(IAP/K_{eq})$) (E,F), pH (G,H), and total $\text{CO}_2(\text{aq})$ concentration (I,J) in geometry c for the (E,G,I) single and (F,H,J) two phase simulations at time 0.06s.

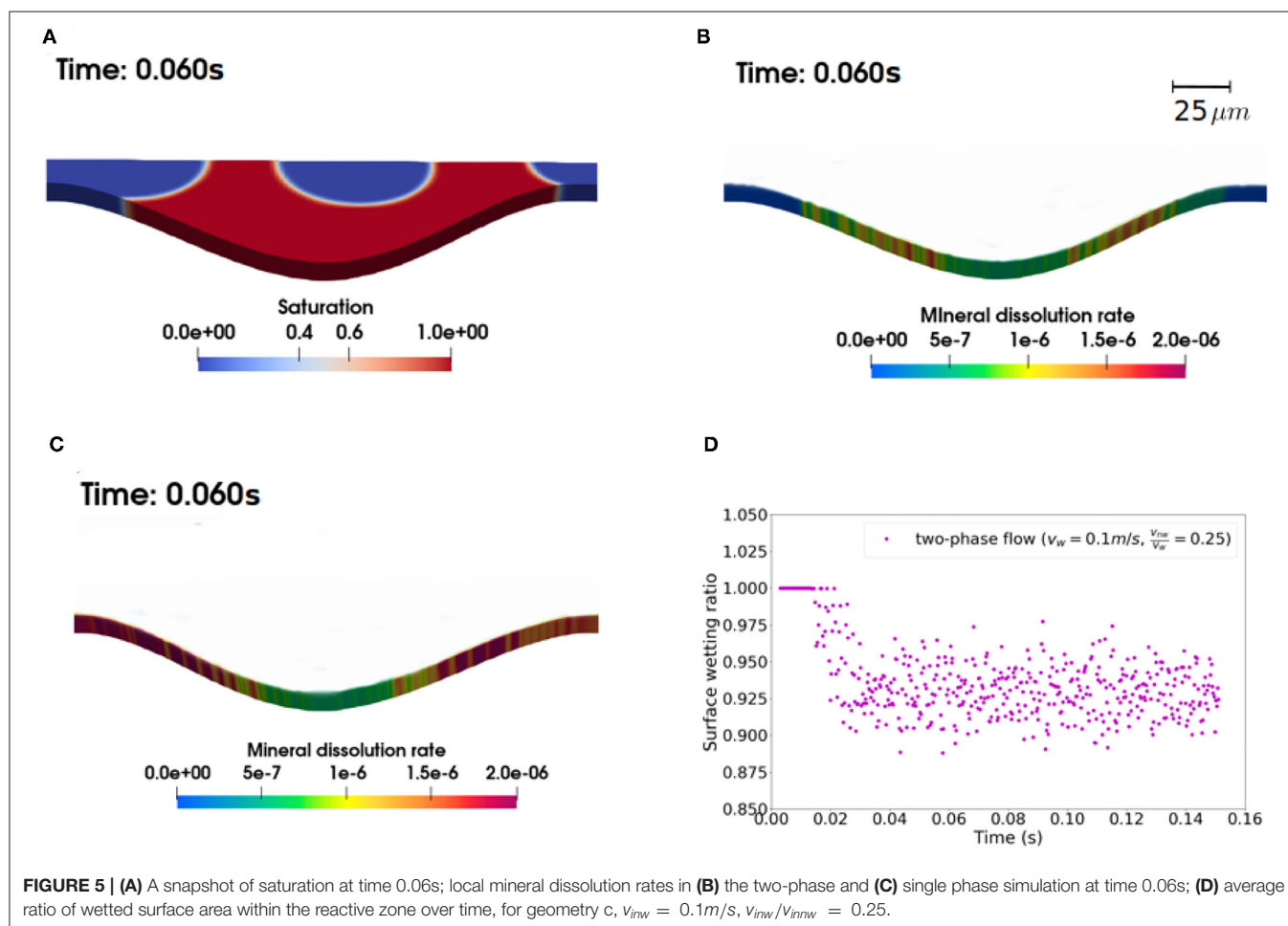
of v_{inw} and v_{innw} . The diameter of the bubbles is 64 and 74 μm for v_{nw}/v_w of 0.25 and 0.5, respectively. In cases of large roughness, the shape of the gas bubbles changes temporarily at the narrow throats and can be recovered after entering the wide channel locations, i.e., sine wave troughs. Once the gas bubble flow is established, the hydrodynamics of the system reaches the steady state, as indicated by the average saturation, which is the mean of the phase field (α) in the reactive zone. The average saturation displays small oscillations around the steady state value when the bubbles enter or exit the reactive domain (**Figures 4B,D**). **Figure 4A** shows the temporal profiles of the average instantaneous reaction rates for the single-phase and two-phase cases for geometry **c** at $v_w = 0.1\text{m/s}$ with $v_w/v_{nw} = 0.25$. The reaction rate for the single phase flow simulation decreases initially as the calcite saturation state starts to increase following the dissolution reaction, and stabilizes at about $1.8 \times 10^{-6} \text{ mol/m}^2\text{s}$ after $\sim 0.04\text{s}$. This dissolution rate is comparable with the value reported in previous single phase reactive transport simulations in a similar system using the pore-scale code ChomboCrunch (Deng et al., 2018). Given the relatively high velocity and large Reynolds number, overall the system is far from equilibrium with respect to calcite and the reaction rate is relatively high. The concentration of the reactive

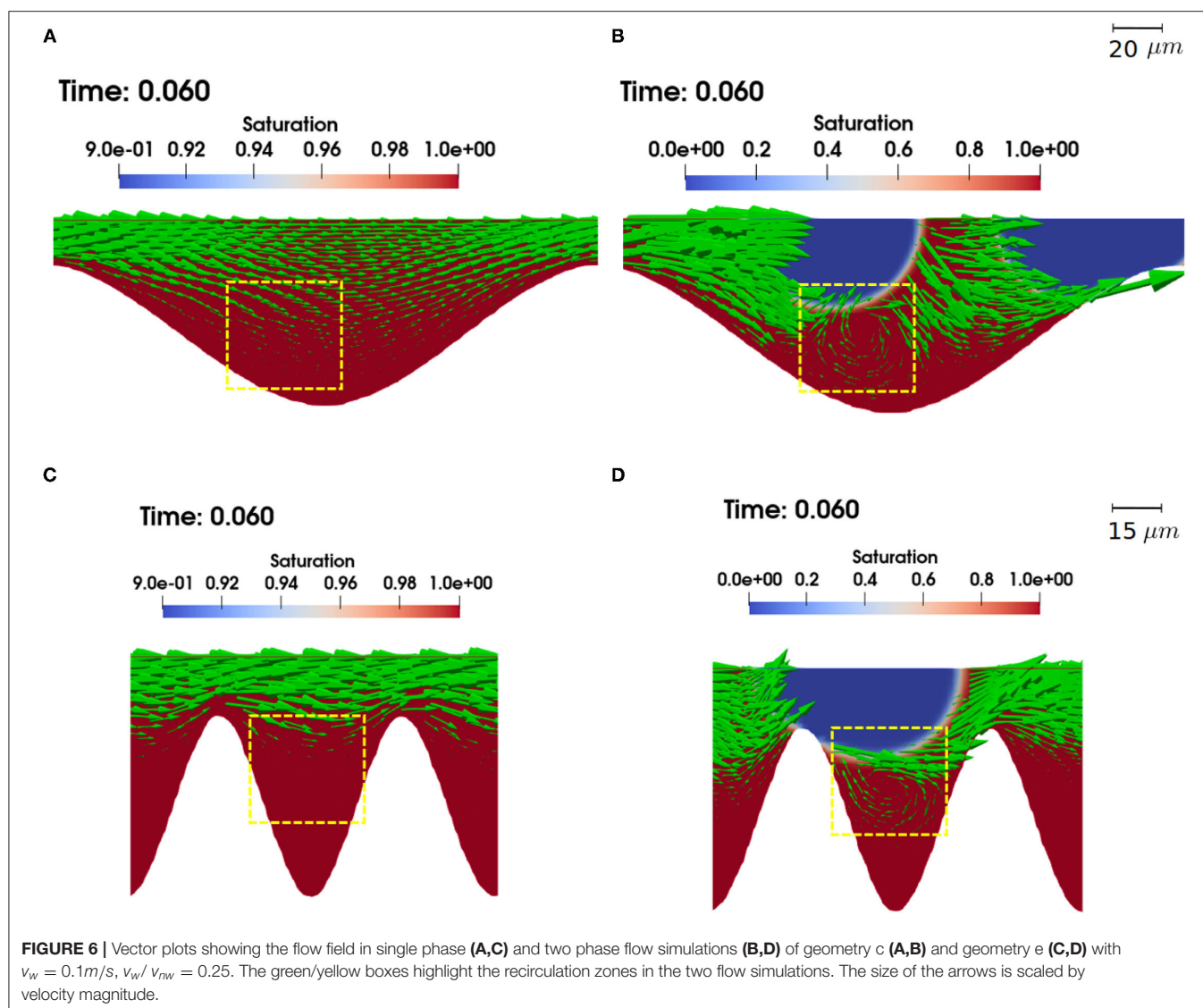
solutes and thus calcite saturation index shows a thin boundary layer at the fluid-solid interface and the concentration gradient across the flow direction is large (**Figures 4E,F**). This is consistent with the observations in Deng et al. (2018) that the effect of transverse transport could be important in these cases.

The reaction rate for the two phase flow simulation decreases over time as well. The initial decreasing trend overlaps with the single phase simulation as the gas bubbles have not entered the reactive zone. The decreasing trend diverges as the gas bubbles enter the reactive zone. $|\overline{R_m^t}|$ reaches the steady state at $\sim 0.8 \times 10^{-6} \text{ mol/m}^2\text{s}$ after $\sim 0.12\text{s}$. The steady state instantaneous reaction rate of the two phase flow case is significantly lower than that of the single phase flow case. This observation is also consistent across all geometries and flow conditions simulated. **Figures 4C,D** provide another example with similar results for geometry **e** under the same flow condition.

Mechanisms for Reaction Rate Modification in Two Phase Systems

Analysis of local reaction rate confirms that the lower reaction rate in the two phase flow case is partially attributed to the changes in accessibility of rock surface area. **Figure 5A** shows the



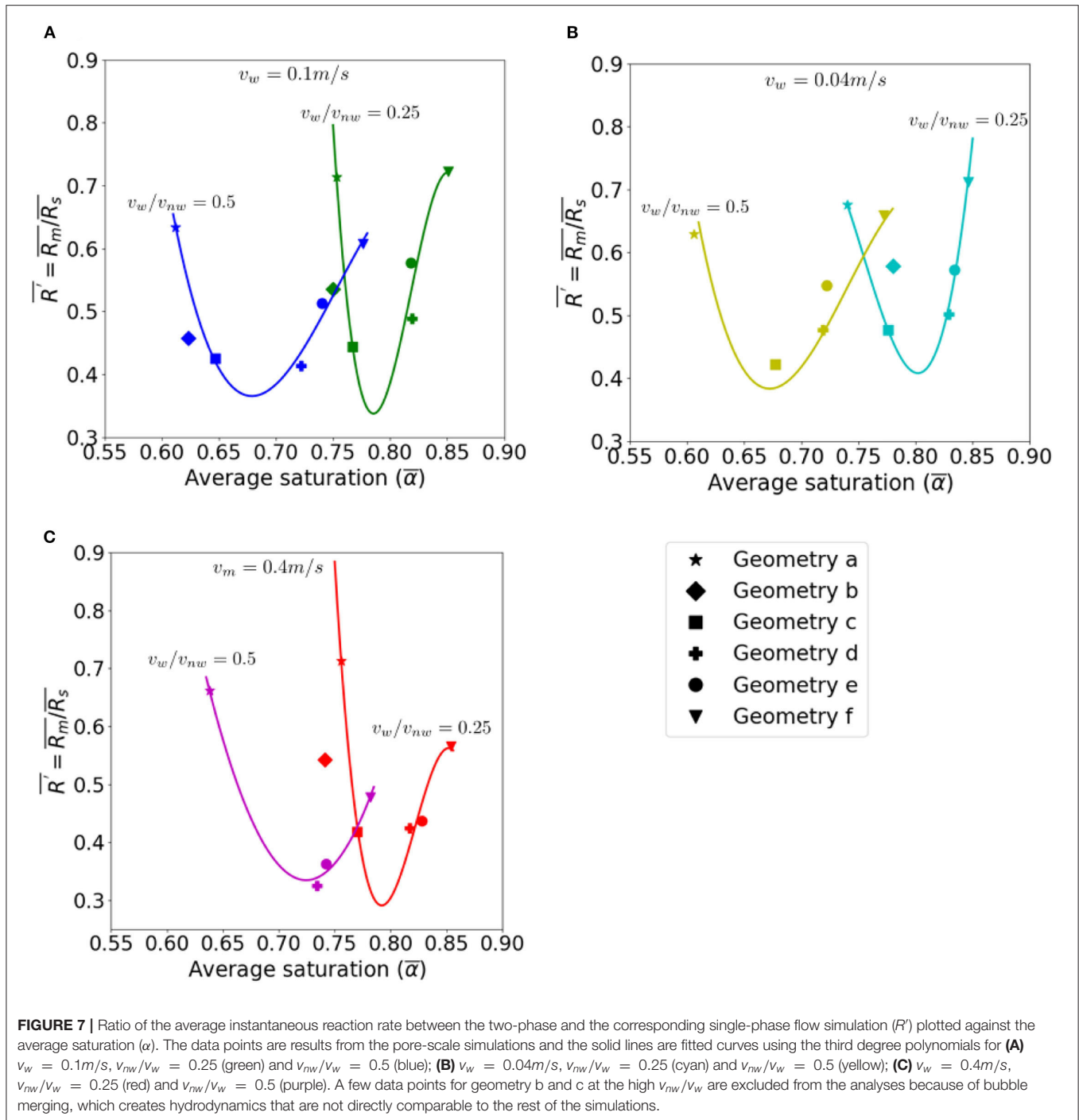


phase field for geometry **c** at $v_w = 0.1\text{m/s}$ and $v_w/v_{nw} = 0.25$, and highlights that when the gas bubbles migrate through the narrow throats, the local reactive surface area becomes inaccessible to water-rock interactions (Figure 5B). In the experimental study of Song et al. (2018), CO_2 gas bubbles generated by calcite dissolution were observed to block access of the reactive fluid phase to the mineral grain surfaces and thus local calcite dissolution is suppressed. Furthermore, in the corresponding single phase simulation (Figure 5C), the local reaction rates on the walls in the narrow throats ($\sim 2 \times 10^{-6} \text{ mol/m}^2\text{s}$) tend to be higher than those in the troughs. As such, even though the wetted surface area in the reactive zone—which provides a measure of accessible mineral surface area—is only reduced by <10% (Figure 5D), the reduction in the average reaction rate is much more significant (>50%).

As shown in Figures 5B,C, the local reaction rate outside of the narrow throats that are occupied by the non-aqueous

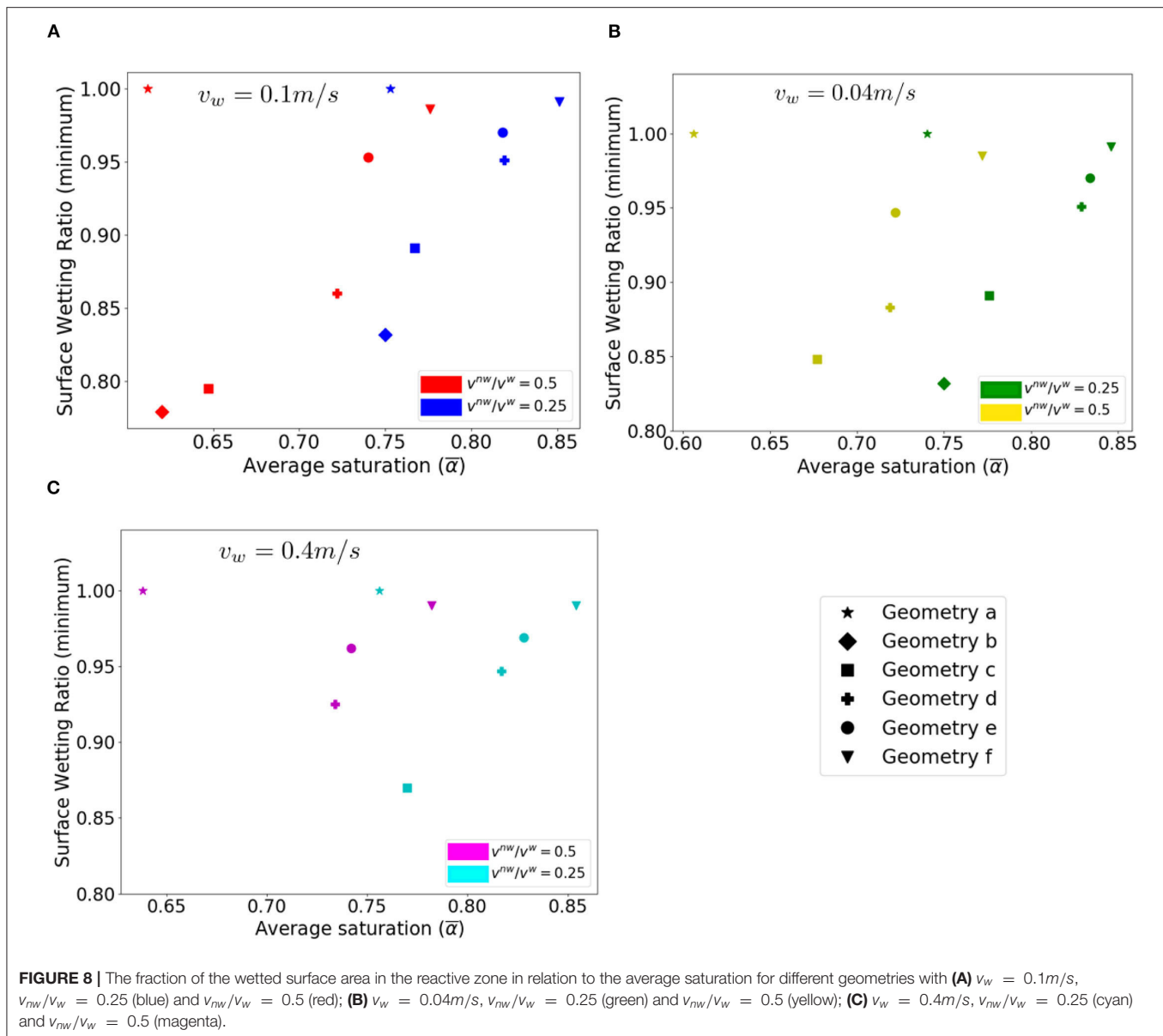
phase is also lower in the two-phase case than that in the single phase case. This indicates the presence of a stronger local transport limitation, which is also confirmed by the flow fields.

Figures 6A,B compare the velocity vectors in the single and two-phase flow simulations for geometry **c** at $v_w = 0.1\text{m/s}$ and $v_w/v_{nw} = 0.25$. In the two phase case, as the gas bubbles migrate through the troughs, the width of the wetting phase is compressed and recirculation zones formed locally. In contrast, for the same geometry and flow rate, no recirculation zones were observed in the single phase simulation. The recirculating phenomenon has been observed experimentally and numerically in two-phase systems across a wide range of flow conditions with Ca between 1×10^{-7} – 1×10^{-2} , as a result of the shear stress exerted by the fluid phase that migrates faster on the other fluid phase that is less mobile or immobile (Blois et al., 2015; Roman et al., 2016; Heshmati and Piri, 2018; Maes and Soulaine, 2018; Mohammadi Alamooti et al., 2020). Previous studies have highlighted that



recirculation zones can be an important mechanism that traps the solutes (Bolster et al., 2014; Sund et al., 2015; Deng et al., 2018; Yoon and Kang, 2021), reducing local thermodynamic driving force (Figures 4E,F), i.e., the chemical affinity term in Equation (12). In addition, the pH is higher and total concentration of $\text{CO}_2(\text{aq})$ is lower in the two phase case (Figures 4G–J), both would lead to a lower kinetic rate as given in Equation (15).

Figures 5C,D show the flow fields for geometry e at the same flow conditions as Figures 5A,B. Similar patterns were observed. The recirculation zone is also more predominant in the troughs or the pore-body with the rougher geometry, accounting for a larger portion of the troughs. The dependence of the recirculation zone on pore morphology has also been observed in previous experimental studies (Heshmati and Piri, 2018).



Correlation Between Reaction Rate Modification and Liquid Saturation

In order to gain some insights regarding constitutive relations that can be used to upscale the impacts of two-phase flow dynamics on reaction rate, **Figure 7** summarizes the steady state reaction rates and saturations from the pore scale simulations. The ratio between the reaction rate of the two phase simulation and that of the corresponding single phase simulation is used to evaluate the effect of two-phase flow dynamics, specifically gas bubble migrations in 2D rough channels, on reaction rates. Liquid saturation is an indicator of the two-phase flow dynamics as is the case in many multiphase continuum models.

The impacts on reaction rate do not change monotonically with respect to the steady state saturation. For a given flow condition, the reaction rate ratio decreases as the steady state

saturation increases for the less rough geometries (i.e., geometry a–c), whereas it increases with the saturation for the rougher geometries (i.e., geometry d–f). **Figure 8** shows that the steady state saturation increases with roughness, and so does the wetted surface area ratio (except for the flat reference geometry). This indicates that if the surface area accessibility is the dominant mechanism of the reaction rate reduction in the two phase case, a higher saturation corresponding to a rougher geometry should result in a lower reduction in reaction rate in the two phase case. However, as shown in **Figure 6**, the other mechanism, i.e., the transport limitation of the recirculation zones, is stronger in the rougher geometry. This implies that reaction rate reduction due to transport limitation arising from the presence of the gas bubbles is more significant in the rougher geometries which also have higher saturations. The tradeoff between the

TABLE 6 | Coefficients of polynomial equations and *R*-squared for the fitting curves in **Figure 7**.

	v_w/v_{nw}	A	B	C	D	R-squared
$v_w = 0.1\text{m/s}$	0.25	-2740.50	6726.12	-5493.92	1493.94	0.927
	0.5	-208.78	471.67	-351.75	87.10	0.976
$v_w = 0.04\text{m/s}$	0.25	845.11	-1911.40	1435.12	-357.09	0.993
	0.5	-256.41	569.40	-417.94	101.97	0.999
$v_w = 0.4\text{m/s}$	0.25	-2543.00	6272.58	-5149.03	-1407.47	0.969
	0.5	0	43.82	-63.48	23.33	0.994

two mechanisms can explain the inverted bell shape, which is observed for all flow conditions.

This non-monotonic trend indicates that a power-law relationship will not apply. The guiding lines in **Figure 7** were fitted using third degree polynomials, $\bar{R}' = A\alpha^3 + B\alpha^2 + C\alpha + D$ (**Table 6**). While the polynomial relationship provides a reasonable fit of the pore-scale modeling data statistically—the goodness of fitting as measured by R^2 is larger than 0.95 in most cases (**Table 6**)—they are not meant to be directly implemented or at least caution should be exercised.

Our observation is analogous to previously observed non-monotonic dependence on fluid saturation for other processes in the sense that competing mechanisms—because of their opposite “dependence” on saturation—are in play. For example, Jiménez-Martínez et al. (2017) reported that when saturation is above a threshold, mixing increases as saturation decreases because of stretching, whereas mixing decreases with saturation below the threshold as molecular diffusion becomes dominant. Our observations also reiterate the fact that saturation is a result of the multiphase dynamics and it alone does not provide a full description of the hydrodynamics in the system. For example, for a given flow condition in our simulations, saturation is determined by the roughness of the geometry, which can be expected for other more complex geometries. Given that both liquid saturation and reaction rate are dependent on the flow conditions and the geometries, future studies may focus on developing constitutive relations that are informed by the underlying physics or that explicitly integrate these controlling factors.

DISCUSSION

In our simulations, the recirculation zone is the dominant hydrodynamic feature affecting the mineral dissolution rate. The development of recirculation zones have been observed in single phase systems at high velocity when the contribution of inertial effect becomes significant; and roughness allows recirculation zones to form under lower velocities (Deng et al., 2018). Our results illustrate that in two phase systems, momentum transfer across the fluid–fluid interface further extends the conditions under which recirculation zones form as also reported in previous studies (Heshmati and Piri, 2018). Other multiphase flow dynamics are also observed in our results albeit the simplified setup considered. For instance,

in geometry **b**, a few simulations at the high and low flow velocities especially with large v_w/v_{nw} showed coalescence of the gas bubbles, which resulted in significantly lower saturation in the reactive zone. Bubble coalescence and breakup are widely observed phenomena that are dependent on fluid properties, velocities, and geometries (Jo and Revankar, 2009; Paulsen et al., 2014; Chen et al., 2017; Mahabadi et al., 2018; Ren et al., 2020; Grunewald et al., 2021). These processes are accompanied by the re-organization of the fluid–fluid interface and can introduce perturbations in the velocity field. The simulations with gas bubble coalescence were excluded from the analyses in section Correlation Between Reaction Rate Modification and Liquid Saturation as the new hydrodynamics is not directly comparable with the other simulations. Nonetheless, this observation highlights that compared to single phase systems in which a lower velocity typically transfers to a stronger transport limitation, the two phase systems require consideration of additional hydrodynamics that may arise at a different velocity (Blois et al., 2015).

A variety of complex hydrodynamics can arise from the migration of fluid–fluid interfaces depending on the velocity (e.g., capillary numbers) and pore morphology (e.g., Berg et al., 2013; Spurin et al., 2019; Li et al., 2021; Wang et al., 2021).

From a macroscopic perspective, as capillary number (Ca) increases, interface migration transitions from the capillary fingering regime with more random local movement to the viscous fingering with more stable displacement (Toussaint et al., 2012; Li et al., 2019; Grunewald et al., 2021). However, Ca alone does not provide a good description of the fluid–fluid interface dynamics (Armstrong et al., 2015), which is more sensitive to pore-scale roughness/morphology at low capillary numbers (Toussaint et al., 2012).

From a microscopic perspective, both the magnitude and direction of flow were observed to fluctuate before the arrival of the invading front (Roman et al., 2016; Li et al., 2017). Strong instabilities of the interface can also lead to pore-scale burst events such as Haines jumps and can increase local velocities by one–two orders of magnitude (Blois et al., 2015; Li et al., 2017, 2021). These hydrodynamics are reported to promote mixing and thus reactions in the liquid phases (Jiménez-Martínez et al., 2015, 2016, 2017), and may also increase local mineral reaction rates. In the study of Jiménez-Martínez et al. (2020), calcite dissolution rate in the two phase experiment with $Ca = \sim 10^{-5}$ and $v_w/v_{nw} = 1.0$ is 68% of the dissolution rate in the single phase experiment. This is comparable to the simulation results in the smooth channel, e.g., the reaction rate ratio is ~ 64 and $\sim 72\%$ at $v_w = 0.1\text{ m/s}$ (i.e., $Ca = \sim 3.5 \times 10^{-4}$) for v_w/v_{nw} of 0.5 and 0.25, respectively (**Figure 7**). Their reaction rate ratio is slightly higher than what would be expected based on our simulations, which is likely because of the oscillation of gas bubbles in the presence of a system of channels and thus higher local velocity at the fluid–fluid interface, in addition to the continuous supply of CO_2 from the co-injected scCO_2 phase. The competition of different fluid pathways can result in more dynamic bubble migration and flow field rearrangement, which influences self-organization of the system, which is also illustrated in the column experiment of Ott and Oedai (2015) and

the pore-scale numerical simulations of Soulaire et al. (2018). Geometry units such as pore-doublets can be used to capture such dynamics and thus offer additional insights (Mohammadi Alamooti et al., 2020; Alizadeh and Fatemi, 2021). More realistic geometries may be needed to fully examine the complexity of real systems, as flow instabilities are primarily driven by geometry when morphological heterogeneity is large (Li et al., 2017; Heshmati and Piri, 2018).

Water film is also an important contributor to the fluid–fluid interface (Li et al., 2019). While for the given Ca and channel width used in our study, water film is not expected to be well developed and thus contribute to transport significantly or affect our analyses (Roman et al., 2017), well-developed water film may become important in maintaining water-rock contact and transport pathways between isolated water parcels. For instance, water film has been observed during the drainage process in various experiments (Rücker et al., 2015; Schlüter et al., 2016; Roman et al., 2017; Moura et al., 2019). It enhances the connectivity between residual water and may cause the “snap-off” phenomenon, affecting the contact between water and the solid phase (e.g., rock) and the transport processes.

Overall, this study represents an early-stage effort in highlighting the importance of a dynamic coupling between reaction rates and multiphase flow dynamics. As such, it focuses on a specific geometric setup and a relatively limited number of flow scenarios. However, compared to the experimental studies that typically report temporally and/or spatially integrated reaction rates, our model provides information on local and instantaneous reaction rate along with detailed velocity field and solute transport, which helps to better explore microscopic mechanisms. The two mechanisms identified in our pore-scale two-phase reactive transport simulations—the surface area effect and local hydrodynamics—support recent observations from micromodel experiments (Song et al., 2018; Jiménez-Martínez et al., 2020). Moreover, the pore-scale perspective in the model enables insights into the impacts of multiphase flow dynamics on mineral reaction rate that have broader implications. Namely, results indicate that in addition to the fluid–solid interface, a direct measure of the reactive surface area, the fluid–fluid interface also plays an important role in controlling solid-phase reactions by modifying local hydrodynamics. As discussed above, local hydrodynamics that can affect mineral reaction rate is largely influenced by fluid–fluid interfaces, our study suggests that adding interfacial area in the formulation of reactive surface area in two-phase flow conditions in addition to saturation may be a logical step. In fact, it has been proposed to use the fluid–fluid interface as a measure of flow topology and thus an additional parameter in constitutive relations of relative permeability (Picchi and Battiato, 2018). Such consideration may be of particular interest in e.g., scCO_2 -brine systems where mass transfer across the fluid–fluid interface also affect fluid chemistry and thus mineral reaction rate.

In order to develop constitutive relations that faithfully reflect the coupling between mineral reaction rates and multiphase flow dynamics and are broadly applicable in natural and engineered fractured porous materials, systematic studies that explore a broader range of multiphase flow and geometric conditions are

needed. While the modeling framework developed in this work can be adapted to consider these processes and thus provides a suitable modeling tool in a wide range of applications for pore-scale mechanistic investigations, it needs to be acknowledged that these investigations hinge upon further development of modeling capabilities for multiphase flow dynamics (Aboukhedr et al., 2018; Qin et al., 2020). For instance, there is still a lack of comprehensive comparison and benchmarking of pore-scale models for low Ca flow systems (Zhao et al., 2019). It has also been shown that because of the sensitivity of multiphase flow to local perturbations, a deterministic reproduction of the dynamics as observed in experiments with relatively large porous domains using numerical models is challenging, while statistical behaviors can still be captured (Ferrari et al., 2015). The consideration of reactions, however, may require higher spatial accuracy because the spatial variations (in mineral distribution and reaction rate) can be important. The impacts of multiphase hydrodynamics on aqueous reactions and mineral reactions may also need to be considered separately. Taking the recirculation zone as an example, it increases interfacial mass transfer (Maes and Soulaire, 2018), but reduces mineral reactions (Deng et al., 2018). These research needs further emphasize the need for experimental studies that underpin the development of mechanistic models for multiphase flow in reactive systems, and studies that expand our investigations from quasi-2D micromodels to 3D systems. The development of constitutive relations will also require the investigations be extended to larger scales, which calls for development of a complementary approach that leverages modeling and experiments at different scales (Blunt et al., 2013).

SUMMARY AND CONCLUSIONS

A pore-scale multiphase reactive transport model was established and used to investigate the dependence of mineral reaction rate on pore-scale two-phase flow dynamics. A series of numerical simulations were performed, in which CO_2 -acidified water and air were co-injected at a range of velocities into 2D calcite channels with different levels of roughness as defined by a single sine wave. The simulation results showed that gas bubbles migrating through the reactive zone as a result of the two-phase co-injection caused the reaction rate to be lower than that of the single phase flow simulation with the same total injection rate. Our analyses revealed that the decrease of the mineral reaction rate is caused by a combination of two mechanisms: the reduction in the wetted surface area and the transport limitations that arise from the two phase flow. In the rough geometries, the narrow throats are reaction hotspots in the single phase flow simulations, whereas these locations are occupied by gas bubbles periodically and thus not accessible for reactions. Meanwhile, the flow field showed clear “vortices,” i.e., recirculation zones, as the gas bubbles migrate through the trough of the sine wave. As a result, more reaction products are trapped in the trough and the local thermodynamic driving force and kinetic rates are reduced. Additionally, the correlations between fluid saturation and the extent of reaction rate reduction—which

is measured by the ratio between the reaction rate from the two-phase flow simulation and that from the corresponding single phase simulation—were analyzed to provide insights for continuum-scale modeling. A non-monotonic relationship was observed, which is because the contribution from wetted surface area reduction and local transport limitation show opposite dependence on the roughness of the channel, which controls the saturation for a given flow condition. Through these numerical simulations, we highlighted the complexity of reactive transport in multiphase flow systems and identified two important mechanisms through which mineral reaction rates are affected. These results highlight the need for consideration of interfacial dynamics on mineral reaction rates in multiphase flow systems, and also emphasize the need for experimental studies that underpin the development of mechanistic models for multiphase flow in reactive systems.

DATA AVAILABILITY STATEMENT

The raw data supporting the conclusions of this article will be made available by the authors upon reasonable request, without undue reservation.

AUTHOR CONTRIBUTIONS

PL contributed to developing the code, designing the study, performing the simulations and analyses, and writing the

manuscript. HD designed the study, contributed to code development, simulation results analyses, and writing the manuscript. SM contributed to code development and writing the manuscript. All authors contributed to the article and approved the submitted version.

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